LANDSCAPE DASHBOARD — TIER-1 & TOP-N GENOMIC REPORT DIGEST

Cohort Subject: Daniel Ehrle (`Daniel_Ehrle`) · **Mean Called Depth:** `50.11x` (Median 50.0x) · **Date:** October 12, 2026

Pre-expanded static digest of the **Tier-1 / ClinVar Pathogenic genes** and **Top-N highest-scoring genes** across all 8 Dashboard Report views (Visual Explorer, ClinVar, AlphaGenome Atlas, CADD, SpliceAI, AlphaMissense, REVEL, and Rare Variants ≤1%).

Total Cohort Catalog: 1407 Variants / 1002 Genes
Output Formats: .HTML · .MHTML (RFC 2557) · .PDF

1. Visual Explorer — Tier-1 & Top Clinical Impact Genes

Master genomic catalog filtered strictly to genes carrying Tier-1 / ClinVar Pathogenic variants and highest multi-engine clinical impact scores.

Selected Scope: 45 Genes (45 Tier-1/Pathogenic)
Filtered from 1002 matching genes in view

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 35.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000304611.13
cDNA (HGVS):	c.2362G>A
Protein (HGVS):	p.Val788Met
VAF / Read Depth:	VAF 0.58 · DP —
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#556244 ↗
gnomAD v4 AF:	0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus

30%
28%
25%
12%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

COL23A1 COL23A1 chr5:178,262,215 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 33.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 COL23A1 is a member of the transmembrane collagens, a subfamily of the nonfibrillar collagens that contain a single pass hydrophobic transmembrane domain (Banyard et al., 2003 [PubMed 12644459]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs2973744 c.675+2T>C	SPL	Heterozygous	Unphased	Not reviewed	Tier1	33.0	0.91	26.3 (Splicing)	– / –	chr5:178262215 AF: 0.4032%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

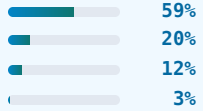
Transcript: ENST00000390654.8
cDNA (HGVS): c.675+2T>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.48 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.02 DL:0.91
ClinVar Review: criteria provided
ClinVar VCV: VCV#4314750 ↗
gnomAD v4 AF: 0.4032%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs5873597 c.1297-277_1297-276del	INT	Heterozygous	Unknown	Not reviewed	Tier3	–	–	–	– / –	chr5:178246735 AF: 0.0000%

CBLIF CBLIF chr11:59,845,374 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the cobalamin transport protein family. It encodes a glycoprotein secreted by parietal cells of the gastric mucosa and is required for adequate absorption of vitamin B12. Vitamin B12 is necessary for erythrocyte maturation and mutations in this gene may lead to congenital pernicious anemia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147785187 c.79+1G>A	SPL	Heterozygous	Maternal	Pathogenic	Tier1	32.0	0.99	33.9 (Splicing)	– / –	chr11:59845374 AF: 0.0256%

1. INHERITANCE & OMIM PATHOLOGY

Intrinsic Factor Deficiency / Cobalamin Malabsorption
Autosomal Recessive OMIM #261000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

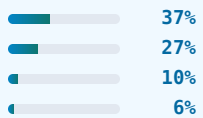
Transcript: ENST00000257248.3
cDNA (HGVS): c.79+1G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.47 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.04 DL:0.99
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#439755 ↗
gnomAD v4 AF: 0.0256%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.9 (Raw: 1.892 · Top 0.04%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000233813.5
cDNA (HGVS):	c.578G>A
Protein (HGVS):	p.Arg193His
VAF / Read Depth:	VAF 0.38 · DP —
Phased GT:	0/1 (Unphased)

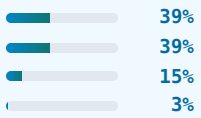
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.11 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0008%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PTDSS2 PTDSS2 chr11:488,272 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 31.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene catalyzes the conversion of phosphatidylethanolamine to phosphatidylserine, a structural membrane phospholipid that functions in cell signaling, blood coagulation, and apoptosis. The encoded enzyme also has a high affinity for docosahexaenoic acid (DHA) and can use it to make DHA-containing phosphatidylserine. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs908582716 p.Glu232Gly	MIS, p.Glu232Gly	Heterozygous	Unphased	Not reviewed	Tier1	31.0	—	25.8 (AlphaMissense)	0.77 / 0.78	chr11:488272 AF: 0.0096%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000308020.6
cDNA (HGVS):	c.695A>G
Protein (HGVS):	p.Glu232Gly
VAF / Read Depth:	VAF 0.68 · DP —
Phased GT:	0/1 (Unphased)

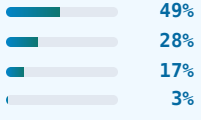
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0096%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.251 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs142613557 c.367+914G>A	INT	Heterozygous	Unphased	Not reviewed	Tier2	0.9	0.24	0.4 (Cactus)	— / —	chr11:474891 AF: 0.6829%

BTN3A2 BTN3A2 chr6:26,370,605 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 28.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the immunoglobulin superfamily, which resides in the juxta-telomeric region of the major histocompatability class 1 locus and is clustered with the other family members on chromosome 6. The encoded protein may be involved in the adaptive immune response. Alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Jun 2013]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs58367598 c.715+2T>G	SPL	Heterozygous	Unphased	Benign	Tier1	28.7	0.98	25.4 (Splicing)	— / —	chr6:26370605 AF: 3.8454%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377708.7
cDNA (HGVS):	c.715+2T>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.64 · DP —
Phased GT:	0/1 (Unphased)

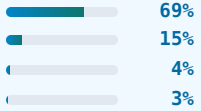
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.01 DL:0.98
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#1238754 ↗
gnomAD v4 AF:	3.8454%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.177 · Top 0.29%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.9 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	—	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367797.9
cDNA (HGVS):	c.883A>G
Protein (HGVS):	p.Thr295Ala
VAF / Read Depth:	VAF 0.34 · DP —
Phased GT:	0|1 (Maternal)

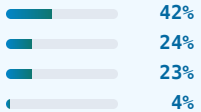
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2884751 ↗
gnomAD v4 AF:	0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs6025 p.Arg534Gln	MIS, p.Arg534Gln	Heterozygous	Unphased	drug response	Tier2	27.9	—	23.6 (Cactus)	— / 0.20	chr1:169549811 AF: 2.1432%

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.7 Showing 3 of 3 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000357077.9
cDNA (HGVS): c.11716C>T
Protein (HGVS): p.Arg3906Trp
VAF / Read Depth: VAF 0.52 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#18059 ↗
gnomAD v4 AF: 0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs11929834 c.85-22668G>A	INT	Heterozygous	Paternal	Not reviewed	Tier2	8.0	—	2.8 (PhastCons 470)	— / —	chr4:113151748 AF: 32.8900%
rs183491249 c.85-40398G>A	INT	Heterozygous	Maternal	Not reviewed	Tier3	5.2	—	4.2 (Cactus)	— / —	chr4:113134018 AF: 0.4297%

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	—	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

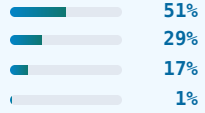
Transcript: ENST00000377801.8
cDNA (HGVS): c.734C>G
Protein (HGVS): p.Ser245Cys
VAF / Read Depth: VAF 0.49 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#3214611 ↗
gnomAD v4 AF: 0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ELOVL7 ELOVL7 chr5:60,771,922 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables fatty acid elongase activity. Involved in fatty acid elongation, polyunsaturated fatty acid; fatty acid elongation, saturated fatty acid; and very long-chain fatty acid biosynthetic process. Located in endoplasmic reticulum. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs368309905 p.Ser79Tyr	MIS, p.Ser79Tyr	Heterozygous	Maternal	Not reviewed	Tier1	27.5	–	31.4 (AlphaMissense)	0.69 / 0.98	chr5:60771922 AF: 0.0210%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000508821.6
cDNA (HGVS):	c.236C>A
Protein (HGVS):	p.Ser79Tyr
VAF / Read Depth:	VAF 0.60 · DP –
Phased GT:	0|1 (Maternal)

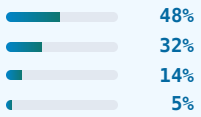
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0210%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 31.4 (Raw: 1.638 · Top 0.07%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	–	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000355112.8
cDNA (HGVS):	c.2171T>C
Protein (HGVS):	p.Val724Ala
VAF / Read Depth:	VAF 0.44 · DP –
Phased GT:	0/1 (Unphased)

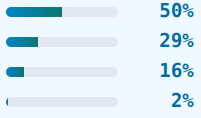
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#567134 ↗
gnomAD v4 AF:	0.0054%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.1 (Raw: 1.282 · Top 0.25%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SLC9A2 SLC9A2 chr2:102,701,098 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 26.3 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the sodium-hydrogen exchanger (NHE) protein family. These proteins are involved in sodium-ion transport by exchanging intracellular hydrogen ions to external sodium ions and help in the regulation of cell pH and volume. The encoded protein is localized to the apical membrane and is involved in apical absorption of sodium. [provided by RefSeq, Jun 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371605244 p.Arg539Trp	MIS, p.Arg539Trp	Heterozygous	Maternal	Not reviewed	Tier1	26.3	–	22.8 (AlphaMissense)	0.46 / 0.54	chr2:102701098 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000233969.3
cDNA (HGVS):	c.1615C>T
Protein (HGVS):	p.Arg539Trp
VAF / Read Depth:	VAF 0.51 · DP –
Phased GT:	0|1 (Maternal)

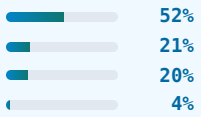
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.8 (Raw: 0.787 · Top 0.52%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs4499446 c.289+5344G>C	INT	Homozygous	N/A	Not reviewed	Tier3	0.9	–	6.3 (DNASE-seq)	– / –	chr2:102625481 AF: 54.2955%

IQCA1 IQCA1 chr2:236,338,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 26.2 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ATPases Associated with diverse cellular Activities (AAA) superfamily. Members of this superfamily, found in all organisms, participate in a large number of cellular processes and contain the ATPase module consisting of an alpha-beta-alpha core domain and the Walker A and B motifs of the P-loop NTPases. Alternative splicing results in multiple transcript variants that encode different protein isoforms. [provided by RefSeq, Jul 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs115960780 p.Cys693Arg	MIS, p.Cys693Arg	Heterozygous	Paternal	Not reviewed	Tier1	26.2	–	24.0 (AlphaMissense)	0.75 / 0.49	chr2:236338262 AF: 0.6424%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000409907.8
cDNA (HGVS):	c.2077T>C
Protein (HGVS):	p.Cys693Arg
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	1|0 (Paternal)

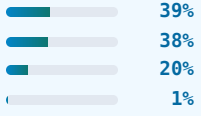
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.6424%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.0 (Raw: 0.948 · Top 0.40%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs1553608365 c.1870-2866_1870-2865insTATATATATAGA	INT	Homozygous	N/A	Not reviewed	Tier3	–	–	–	– / –	chr2:236347509 AF: 0.4781%

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 26.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	–	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases
Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder
Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

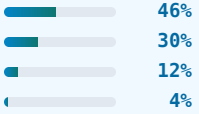
Transcript:	ENST00000373366.3
cDNA (HGVS):	c.223C>T
Protein (HGVS):	p.Arg75Cys
VAF / Read Depth:	VAF 0.37 · DP –
Phased GT:	0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#297191 ↗
gnomAD v4 AF:	0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder
Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

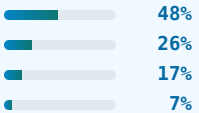
Transcript:	ENST00000377403.7
cDNA (HGVS):	c.1004C>T
Protein (HGVS):	p.Pro335Leu
VAF / Read Depth:	VAF 0.43 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.12 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#773105 ↗
gnomAD v4 AF:	0.2420%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.6 (Raw: 1.050 · Top 0.35%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	–	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000361439.9
cDNA (HGVS):	c.692G>A
Protein (HGVS):	p.Cys231Tyr
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

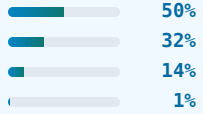
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2358669 ↗
gnomAD v4 AF:	0.0064%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.6 (Raw: 1.597 · Top 0.09%)

Primary Driver: AlphaMissense



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ACACB ACACB chr12:109,260,564 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acetyl-CoA carboxylase (ACC) is a complex multifunctional enzyme system. ACC is a biotin-containing enzyme which catalyzes the carboxylation of acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid synthesis. ACC-beta is thought to control fatty acid oxidation by means of the ability of malonyl-CoA to inhibit carnitine-palmitoyl-CoA transferase I, the rate-limiting step in fatty acid uptake and oxidation by mitochondria. ACC-beta may be involved in the regulation of fatty acid oxidation, rather than fatty acid biosynthesis. [provided by RefSeq, Oct 2022]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139082246 p.Pro2194Leu	MIS, p.Pro2194Leu	Heterozygous	Paternal	Not reviewed	Tier1	25.5	–	24.2 (Cactus)	0.70 / 0.36	chr12:109260564 AF: 0.0037%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000338432.12
cDNA (HGVS):	c.6581C>T
Protein (HGVS):	p.Pro2194Leu
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

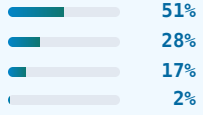
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0037%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.2 (Raw: 0.986 · Top 0.38%)

Primary Driver: Cactus



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	–	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia

Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

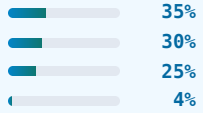
Transcript: **ENST00000268097.10**
cDNA (HGVS): **c.739C>T**
Protein (HGVS): **p.Arg247Trp**
VAF / Read Depth: **VAF 0.56 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#3922 ↗
gnomAD v4 AF: **0.2782%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **21.0** (Raw: 0.629 · Top 0.80%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PAK3 PAK3 chrX:111,163,700 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a serine-threonine kinase and forms an activated complex with GTP-bound RAS-like (P21), CDC2 and RAC1. This protein may be necessary for dendritic development and for the rapid cytoskeletal reorganization in dendritic spines associated with synaptic plasticity. Defects in this gene are the cause of a non-syndromic form of X-linked intellectual disability. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
chrX:111163700 p.Thr247Pro	MIS, p.Thr247Pro	Heterozygous	Unphased	Not reviewed	Tier1	25.1	–	24.9 (AlphaMissense)	0.48 / 0.72	chrX:111163700 AF: 0.0000%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

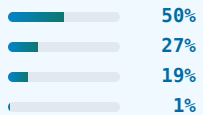
Transcript: **ENST00000372007.10**
cDNA (HGVS): **c.739A>C**
Protein (HGVS): **p.Thr247Pro**
VAF / Read Depth: **VAF 0.27 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0000%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.9** (Raw: 1.106 · Top 0.32%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	–	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

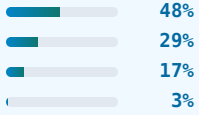
Transcript: **ENST00000294304.12**
cDNA (HGVS): **c.698T>A**
Protein (HGVS): **p.Val233Glu**
VAF / Read Depth: **VAF 0.55 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#866038 ↗
gnomAD v4 AF: **0.0007%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.4** (Raw: 1.189 · Top 0.29%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PARP3 PARP3 chr3:51,944,496 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PARP family. These enzymes modify nuclear proteins by poly-ADP-ribosylation, which is required for DNA repair, regulation of apoptosis, and maintenance of genomic stability. This gene encodes the poly(ADP-ribosyl)transferase 3, which is preferentially localized to the daughter centriole throughout the cell cycle. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs761611493 p.Arg140Gln	MIS, p.Arg140Gln	Heterozygous	Unphased	Not reviewed	Tier1	24.8	–	15.5 (Cactus)	0.40 / 0.14	chr3:51944496 AF: 0.0061%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

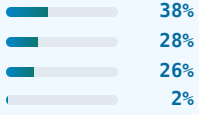
Transcript: **ENST00000398755.8**
cDNA (HGVS): **c.419G>A**
Protein (HGVS): **p.Arg140Gln**
VAF / Read Depth: **VAF 0.56 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0061%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **15.5** (Raw: 0.340 · Top 2.83%)
Primary Driver: **Cactus**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

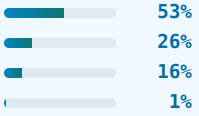
Transcript:	ENST00000435607.3
cDNA (HGVS):	c.1796A>G
Protein (HGVS):	p.His599Arg
VAF / Read Depth:	VAF 0.59 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#324537 ↗
gnomAD v4 AF:	0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	–	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4
Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

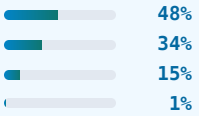
Transcript:	ENST00000323670.14
cDNA (HGVS):	c.124G>A
Protein (HGVS):	p.Gly42Arg
VAF / Read Depth:	VAF 0.40 · DP –
Phased GT:	0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#183298 ↗
gnomAD v4 AF:	0.0040%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.4 (Raw: 1.532 · Top 0.12%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.3 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	—	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000278550.12
cDNA (HGVS):	c.2804C>G
Protein (HGVS):	p.Pro935Arg
VAF / Read Depth:	VAF 0.44 · DP —
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#708690 ↗
gnomAD v4 AF:	0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)

Primary Driver: AlphaMissense

40%
37%
13%
4%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs191549326 p.Pro485Ser	MIS, p.Pro485Ser	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier2	24.3	—	24.2 (Cactus)	0.36 / 0.36	chr11:78855981 AF: 0.4409%
rs181299543 c.-264-22173G>C	INT	Heterozygous	Paternal	Not reviewed	Tier3	17.0	—	13.3 (Cactus)	— / —	chr11:79238082 AF: 0.1990%

TTN TTN chr2:178,621,751 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene Tier1 · CADD 23.4 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a large abundant protein of striated muscle. The product of this gene is divided into two regions, a N-terminal I-band and a C-terminal A-band. The I-band, which is the elastic part of the molecule, contains two regions of tandem immunoglobulin domains on either side of a PEVK region that is rich in proline, glutamate, valine and lysine. The A-band, which is thought to act as a protein-ruler, contains a mixture of immunoglobulin and fibronectin repeats, and possesses kinase activity. An N-terminal Z-disc region and a C-terminal M-line region bind to the Z-line and M-line of the sarcomere, respectively, so that a single titin molecule spans half the length of a sarcomere. Titin also contains binding sites for muscle associated proteins so it serves as an adhesion template for the assembly of contractile machinery in muscle cells. It has also been identified as a structural protein for chromosomes. Alternative splicing of this gene results in multiple transcript variants. Considerable variability exists in the I-band, the M-line and the Z-disc regions of titin. Variability in the I-band region contributes to the differences in elasticity of different titin isoforms and, therefore, to the differences in elasticity of different muscle types. Mutations in this gene are associated with familial hypertrophic cardiomyopathy 9, and autoantibodies to titin are produced in patients with the autoimmune disease scleroderma. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72677222 c.45083-10A>G	INT	Heterozygous	Unphased	Benign/Likely benign	Tier1	17.8	1.00	26.0 (Splicing)	– / –	chr2:178621751 AF: 1.0145%

1. INHERITANCE & OMIM PATHOLOGY

Dilated cardiomyopathy 1G
Complex / Multifactorial

Autosomal recessive limb-girdle muscular dystrophy type 2J
Autosomal Recessive

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000589042.5
cDNA (HGVS):	c.45083-10A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.49 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:1.00 AL:0.78 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#46980 ↗
gnomAD v4 AF:	1.0145%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.0 (Raw: 1.269 · Top 0.25%)

Primary Driver: Splicing

84%
14%
1%
0%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs33917087 p.Val2823Phe	MIS, p.Val2823Phe	Heterozygous	Maternal	Benign/Likely benign	Tier2	23.0	0.21	22.9 (AlphaMissense)	0.42 / 0.46	chr2:178770234 AF: 1.8566%
rs146983095 p.Glu5963Gly	MIS, p.Glu5963Gly	Heterozygous	Paternal	Benign/Likely benign	Tier2	22.8	–	20.6 (Cactus)	0.11 / 0.10	chr2:178730645 AF: 0.9710%

GRIA4 GRIA4 chr11:105,611,080 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 23.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Glutamate receptors are the predominant excitatory neurotransmitter receptors in the mammalian brain and are activated in a variety of normal neurophysiologic processes. These receptors are heteromeric protein complexes composed of multiple subunits, arranged to form ligand-gated ion channels. The classification of glutamate receptors is based on their activation by different pharmacologic agonists. The subunit encoded by this gene belongs to a family of AMPA (alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate)-sensitive glutamate receptors, and is subject to RNA editing (AGA->GGA; R->G). Alternative splicing of this gene results in transcript variants encoding different isoforms, which may vary in their signal transduction properties. Some haplotypes of this gene show a positive association with schizophrenia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs150577695 p.Gln28Arg	MIS, p.Gln28Arg	Heterozygous	Paternal	Not reviewed	Tier1	23.0	–	25.8 (AlphaMissense)	0.30 / 0.78	chr11:105611080 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000282499.10
cDNA (HGVS):	c.83A>G
Protein (HGVS):	p.Gln28Arg
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	1|0 (Paternal)

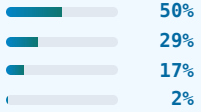
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.245 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB2 GJB2 chr13:20,189,481 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 20.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the gap junction protein family (gap junction beta 2) but is more commonly known as connexin 26. The gap junctions were first characterized by electron microscopy as regionally specialized structures on plasma membranes of contacting adherent cells. These structures were shown to consist of cell-to-cell channels that facilitate the transfer of ions and small molecules between cells. Connexins form hexameric channels in the plasma membrane which regulate passage of ions and small molecules between the cell and its environment or, when joined with another cell, form a dodecameric intercellular gap junction channel. The connexin proteins are grouped into alpha, beta, and gamma subfamilies. Mutations in this gene are responsible for as much as 50% of pre-lingual, recessive deafness. [provided by RefSeq, Mar 2026]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs35887622 p.Met34Thr	MIS, p.Met34Thr	Heterozygous	Unphased	Pathogenic	Tier1	20.9	–	23.6 (Cactus)	0.70 / 0.43	chr13:20189481 AF: 1.2041%

1. INHERITANCE & OMIM PATHOLOGY

Autosomal Recessive Deafness 1A / Dominant 3A

Autosomal Recessive Autosomal Dominant

OMIM #121011 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000382848.5
cDNA (HGVS):	c.101T>C
Protein (HGVS):	p.Met34Thr
VAF / Read Depth:	VAF 0.63 · DP –
Phased GT:	0/1 (Unphased)

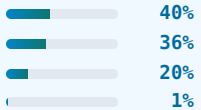
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	reviewed by expert panel
ClinVar VCV:	VCV#17000 ↗
gnomAD v4 AF:	1.2041%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.6 (Raw: 0.892 · Top 0.43%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TRAK1 TRAK1 chr3:42,115,352 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 17.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable GABA receptor binding activity and myosin binding activity. Involved in endosome to lysosome transport. Located in early endosome and mitochondrion. Implicated in developmental and epileptic encephalopathy 68. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs17220321 c.92-10068T>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	17.7	0.89	28.6 (Splicing)	— / —	chr3:42115352 AF: 1.5740%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

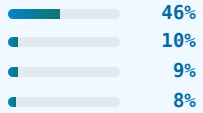
Transcript: **ENST00000327628.10**
cDNA (HGVS): **c.92-10068T>A**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.43 · DP —**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.89 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **1.5740%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **28.6** (Raw: 1.480 · Top 0.14%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

JKAMP JKAMP chr14:59,502,424 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 13.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables ubiquitin protein ligase binding activity. Involved in ERAD pathway. Predicted to be located in endoplasmic reticulum membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs113112586 c.717+1157A>G	INT	Heterozygous	Paternal	Not reviewed	Tier1	13.9	0.65	13.5 (Splicing)	— / —	chr14:59502424 AF: 4.0038%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

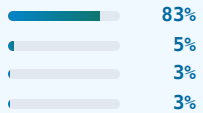
Transcript: **ENST00000616435.5**
cDNA (HGVS): **c.717+1157A>G**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.63 · DP —**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.40 AL:0.00 DG:0.65 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **4.0038%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **13.5** (Raw: 0.265 · Top 4.50%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

DNAH7 DNAH7 chr2:196,024,488 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 12.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 DNAH7 is a component of the inner dynein arm of ciliary axonemes (Zhang et al., 2002 [PubMed 11877439]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs555558470 p.Lys228=	SYN, p.Lys228=	Heterozygous	Paternal	Not reviewed	Tier1	12.9	0.74	23.0 (Splicing)	– / –	chr2:196024488 AF: 0.0021%

1. INHERITANCE & OMIM PATHOLOGY

Primary Ciliary Dyskinesia / Respiratory Clearance

Autosomal Recessive OMIM #610061 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

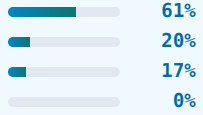
Transcript: ENST00000312428.11
cDNA (HGVS): c.684A>G
Protein (HGVS): p.Lys228=
VAF / Read Depth: VAF 0.65 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.74 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0021%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.0 (Raw: 0.808 · Top 0.50%)
Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NEK10 NEK10 chr3:27,344,024 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 12.8 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein kinase activity. Involved in several processes, including mucociliary clearance; positive regulation of protein phosphorylation; and regulation of ERK1 and ERK2 cascade. Part of protein kinase complex. Implicated in primary ciliary dyskinesia 44. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs113156482 c.362+248G>A	INT	Heterozygous	Maternal	Not reviewed	Tier1	12.8	0.56	20.4 (Splicing)	– / –	chr3:27344024 AF: 1.7310%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

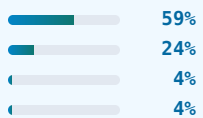
Transcript: ENST00000691995.1
cDNA (HGVS): c.362+248G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.41 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.56 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 1.7310%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.4 (Raw: 0.584 · Top 0.92%)
Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1267845859 c.2971-433A>G	INT	Heterozygous	Paternal	Not reviewed	Tier2	0.8	0.47	14.6 (Splicing)	– / –	chr3:27132423 AF: 0.0007%

CNTN1 CNTN1 chr12:40,943,448 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 10.2 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the immunoglobulin superfamily. It is a glycosylphosphatidylinositol (GPI)-anchored neuronal membrane protein that functions as a cell adhesion molecule. It may play a role in the formation of axon connections in the developing nervous system. Multiple alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139915282 c.1380-149C>T	INT	Heterozygous	Paternal	Likely benign	Tier1	10.2	0.53	12.1 (Splicing)	— / —	chr12:40943448 AF: 1.8289%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

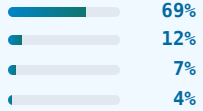
Transcript: **ENST00000551295.7**
cDNA (HGVS): **c.1380-149C>T**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.52 · DP —**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.53 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#680275 ↗
gnomAD v4 AF: **1.8289%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **12.1** (Raw: 0.221 · Top 6.13%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs146742800 c.2420-1480G>A	INT	Heterozygous	Maternal	Not reviewed	Tier2	6.3	0.22	9.9 (Splicing)	— / —	chr12:41018857 AF: 1.5444%

NPM1 NPM1 chr5:171,388,520 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 8.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is involved in several cellular processes, including centrosome duplication, protein chaperoning, and cell proliferation. The encoded phosphoprotein shuttles between the nucleolus, nucleus, and cytoplasm, chaperoning ribosomal proteins and core histones from the nucleus to the cytoplasm. This protein is also known to sequester the tumor suppressor ARF in the nucleolus, protecting it from degradation until it is needed. Mutations in this gene are associated with acute myeloid leukemia. Dozens of pseudogenes of this gene have been identified. [provided by RefSeq, Aug 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1770393339 c.58+514T>G	INT	Heterozygous	Unphased	Not reviewed	Tier1	8.7	0.82	19.6 (Splicing)	— / —	chr5:171388520 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

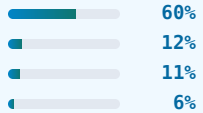
Transcript: **ENST00000296930.10**
cDNA (HGVS): **c.58+514T>G**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.49 · DP —**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.01 AL:0.00 DG:0.82 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0007%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **19.6** (Raw: 0.536 · Top 1.10%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

INPP5K INPP5K chr17:1,508,258 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 7.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with 5-phosphatase activity toward polyphosphate inositol. The protein localizes to the cytosol in regions lacking actin stress fibers. It is thought that this protein may negatively regulate the actin cytoskeleton. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs74437970 c.555-32A>G	INT	Heterozygous	Maternal	Not reviewed	Tier1	7.5	0.63	13.0 (Splicing)	— / —	chr17:1508258 AF: 2.2461%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

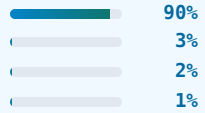
Transcript: **ENST00000421807.7**
cDNA (HGVS): **c.555-32A>G**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.50 · DP —**
Phased GT: **0|1 (Maternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.63 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4417749 ↗
gnomAD v4 AF: **2.2461%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.0 (Raw: 0.249 · Top 5.02%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 6.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs562989479 c.99+808G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	6.9	0.57	19.2 (Splicing)	— / —	chr1:165828740 AF: 0.3704%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

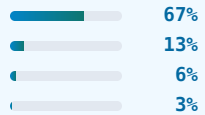
Transcript: **ENST00000367879.9**
cDNA (HGVS): **c.99+808G>T**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.47 · DP —**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.57 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.3704%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

RAD51 RAD51 chr15:40,695,628 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 5.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the RAD51 protein family. RAD51 family members are highly similar to bacterial RecA and Saccharomyces cerevisiae Rad51, and are known to be involved in the homologous recombination and repair of DNA. This protein can interact with the ssDNA-binding protein RPA and RAD52, and it is thought to play roles in homologous pairing and strand transfer of DNA. This protein is also found to interact with BRCA1 and BRCA2, which may be important for the cellular response to DNA damage. BRCA2 is shown to regulate both the intracellular localization and DNA-binding ability of this protein. Loss of these controls following BRCA2 inactivation may be a key event leading to genomic instability and tumorigenesis. Multiple transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Aug 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs5030791 c.-3+203G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	5.5	0.81	19.1 (Splicing)	– / –	chr15:40695628 AF: 1.3014%

1. INHERITANCE & OMIM PATHOLOGY

Breast-Ovarian Cancer Predisposition
Autosomal Dominant OMIM #179570 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

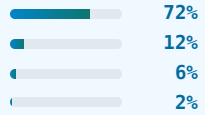
Transcript: ENST00000267868.8
cDNA (HGVS): c.-3+203G>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.60 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.81 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4401961 ↗
gnomAD v4 AF: 1.3014%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.1 (Raw: 0.510 · Top 1.23%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLB1 PLB1 chr2:28,461,598 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 3.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a membrane-associated phospholipase that displays lysophospholipase and phospholipase A2 activities through removal of sn-1 and sn-2 fatty acids of glycerophospholipids. In addition, it displays lipase and retinyl ester hydrolase activities. The encoded protein is highly conserved and is composed of a large, glycosylated extracellular domain composed of four tandem homologous domains, followed by a hydrophobic segment that anchors the enzyme to the membrane and a short C-terminal cytoplasmic tail. This gene has been identified as a candidate rheumatoid arthritis risk gene. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs56116810 c.-114+4401G>A	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.5	–	3.6 (Cactus)	– / –	chr2:28461598 AF: 0.0012%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

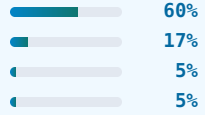
Transcript: ENST00000416713.5
cDNA (HGVS): c.-114+4401G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0012%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 3.6 (Raw: 0.009 · Top 43.90%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PKD1L3 PKD1L3 chr16:71,979,984 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 3.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polycystin protein family. The encoded protein contains 11 transmembrane domains, a latrophilin/CL-1-like GPCR proteolytic site (GPS) domain, and a polycystin-1, lipoxygenase, alpha-toxin (PLAT) domain. This protein may function as a component of cation channel pores.[provided by RefSeq, Apr 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72803747 c.1271+23G>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.4	0.84	10.4 (Splicing)	– / –	chr16:71979984 AF: 0.9916%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

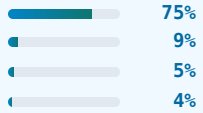
Transcript: ENST00000620267.2
cDNA (HGVS): c.1271+23G>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.39 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.84 DL:0.02
ClinVar Review: criteria provided
ClinVar VCV: VCV#4415201 ↗
gnomAD v4 AF: 0.9916%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 10.4 (Raw: 0.168 · Top 9.03%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CYP2E1 CYP2E1 chr10:133,520,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 2.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the cytochrome P450 superfamily of enzymes. The cytochrome P450 proteins are monooxygenases which catalyze many reactions involved in drug metabolism and synthesis of cholesterol, steroids and other lipids. This protein localizes to the endoplasmic reticulum and is induced by ethanol, the diabetic state, and starvation. The enzyme metabolizes both endogenous substrates, such as ethanol, acetone, and acetal, as well as exogenous substrates including benzene, carbon tetrachloride, ethylene glycol, and nitrosamines which are premutagens found in cigarette smoke. Due to its many substrates, this enzyme may be involved in such varied processes as gluconeogenesis, hepatic cirrhosis, diabetes, and cancer. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141625332 c.-118+195C>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	2.5	0.51	1.0 (Cactus)	– / –	chr10:133520755 AF: 1.0596%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

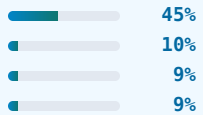
Transcript: ENST00000463117.6
cDNA (HGVS): c.-118+195C>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.43 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.51 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 1.0596%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.0 (Raw: -0.042 · Top 80.14%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLA2G6 PLA2G6 chr22:38,115,172 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 1.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is an A2 phospholipase, a class of enzyme that catalyzes the release of fatty acids from phospholipids. The encoded protein may play a role in phospholipid remodelling, arachidonic acid release, leukotriene and prostaglandin synthesis, fas-mediated apoptosis, and transmembrane ion flux in glucose-stimulated B-cells. Several transcript variants encoding multiple isoforms have been described, but the full-length nature of only three of them have been determined to date. [provided by RefSeq, Dec 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147795054 c.2034+355G>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	1.7	0.95	25.0 (Splicing)	– / –	chr22:38115172 AF: 0.5672%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000332509.8
cDNA (HGVS):	c.2034+355G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.55 · DP –
Phased GT:	0/1 (Unphased)

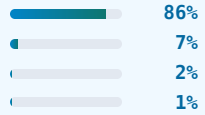
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.95 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.5672%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.0 (Raw: 1.119 · Top 0.32%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

EGFL7 EGFL7 chr9:136,670,809 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 0.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a secreted endothelial cell protein that contains two epidermal growth factor-like domains. The encoded protein may play a role in regulating vasculogenesis. This protein may be involved in the growth and proliferation of tumor cells. Alternate splicing results in multiple transcript variants. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs920684955 c.572-141C>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	0.6	0.73	16.7 (Splicing)	– / –	chr9:136670809 AF: 0.0147%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000308874.12
cDNA (HGVS):	c.572-141C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.58 · DP –
Phased GT:	0|1 (Maternal)

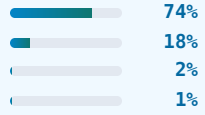
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.73 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0147%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 16.7 (Raw: 0.391 · Top 2.16%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

AZIN2 AZIN2 chr1:33,081,925 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 0.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the antizyme inhibitor family, which plays a role in cell growth and proliferation by maintaining polyamine homeostasis within the cell. Antizyme inhibitors are homologs of ornithine decarboxylase (ODC, the key enzyme in polyamine biosynthesis) that have lost the ability to decarboxylase ornithine; however, retain the ability to bind to antizymes. Antizymes negatively regulate intracellular polyamine levels by binding to ODC and targeting it for degradation, as well as by inhibiting polyamine uptake. Antizyme inhibitors function as positive regulators of polyamine levels by sequestering antizymes and neutralizing their effect. This gene encodes antizyme inhibitor 2, the second member of this gene family. Like antizyme inhibitor 1, antizyme inhibitor 2 interacts with all 3 antizymes and stimulates ODC activity and polyamine uptake. However, unlike antizyme inhibitor 1, which is ubiquitously expressed and localized in the nucleus and cytoplasm, antizyme inhibitor 2 is predominantly expressed in the brain and testis and localized in the endoplasmic reticulum-golgi intermediate compartment. Recent studies indicate that antizyme inhibitor 2 is also expressed in specific cell types in ovaries, adrenal glands and pancreas, and in mast cells. The exact function of this gene is not known, however, available data suggest its role in cell growth, spermiogenesis, vesicular trafficking and secretion. Accumulation of antizyme inhibitor 2 has also been observed in brains of patients with Alzheimer's disease. There has been confusion in literature and databases over the nomenclature of this gene, stemming from an earlier report that a human cDNA clone (identical to ODCp/AZIN2) had arginine decarboxylase (ADC) activity (PMID:14738999). Subsequent studies in human and mouse showed that antizyme inhibitor 2 was devoid of arginine decarboxylase activity (PMID:19956990). Alternatively spliced transcript variants have been described for this gene. [provided by RefSeq, Sep 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs190309518 c.-73+113G>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	0.4	0.67	13.2 (Splicing)	– / –	chr1:33081925 AF: 0.0490%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

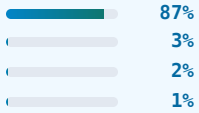
Transcript: **ENST00000294517.11**
cDNA (HGVS): **c.-73+113G>T**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.40 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.06 AL:0.00 DG:0.67 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4417896 ↗
gnomAD v4 AF: **0.0490%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **13.2** (Raw: 0.256 · Top 4.79%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TMEM270 TMEM270 chr7:73,865,329 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 0.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs73142829 p.Gly137Ser	MIS, p.Gly137Ser	Heterozygous	Paternal	Not reviewed	Tier1	0.0	0.72	20.1 (Splicing)	0.02 / 0.08	chr7:73865329 AF: 4.6362%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

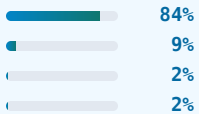
Transcript: **ENST00000320531.3**
cDNA (HGVS): **c.409G>A**
Protein (HGVS): **p.Gly137Ser**
VAF / Read Depth: **VAF 0.47 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.72 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4339331 ↗
gnomAD v4 AF: **4.6362%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **20.1** (Raw: 0.569 · Top 0.97%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VDR VDR chr12:47,906,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier3 · CADD 5.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes vitamin D3 receptor, which is a member of the nuclear hormone receptor superfamily of ligand-inducible transcription factors. This receptor also functions as a receptor for the secondary bile acid, lithocholic acid. Downstream targets of vitamin D3 receptor are principally involved in mineral metabolism, though this receptor regulates a variety of other metabolic pathways, such as those involved in immune response and cancer. Mutations in this gene are associated with type II vitamin D-resistant rickets. A single nucleotide polymorphism in the initiation codon results in an alternate translation start site three codons downstream. Alternatively spliced transcript variants encoding different isoforms have been described for this gene. A recent study provided evidence for translational readthrough in this gene, and expression of an additional C-terminally extended isoform via the use of an alternative in-frame translation termination codon. [provided by RefSeq, Jun 2018]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs4516035 c.-1172A>G	2KU	Homozygous	N/A	Likely pathogenic	Tier3	5.4	–	2.1 (DNASE-seq)	– / –	chr12:47906043 AF: 30.5142%

1. INHERITANCE & OMIM PATHOLOGY

Chronic obstructive pulmonary disease

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000549336.6
cDNA (HGVS):	c.-1172A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 1.00 · DP –
Phased GT:	1/1 (N/A)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	no assertion criteria provided
ClinVar VCV:	VCV#3336650 ↗
gnomAD v4 AF:	30.5142%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 2.1 (Raw: -0.015 · Top 61.95%)

Primary Driver: DNASE-seq



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CDKN2B CDKN2B chr9:22,003,368 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier3 · CADD 0.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene lies adjacent to the tumor suppressor gene CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. This gene encodes a cyclin-dependent kinase inhibitor, which forms a complex with CDK4 or CDK6, and prevents the activation of the CDK kinases, thus the encoded protein functions as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be dramatically induced by TGF beta, which suggested its role in the TGF beta induced growth inhibition. Two alternatively spliced transcript variants of this gene, which encode distinct proteins, have been reported. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1063192 c.*2619C>T	UT3	Heterozygous	Unphased	Likely pathogenic|protective	Tier3	0.2	–	1.3 (Cactus)	– / –	chr9:22003368 AF: 68.9409%

1. INHERITANCE & OMIM PATHOLOGY

Coronary Artery Disease & Breast Carcinoma

Complex / Polygenic Protective Allele OMIM #600431 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000276925.7
cDNA (HGVS):	c.*2619C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.50 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	no assertion criteria provided
ClinVar VCV:	VCV#812639 ↗
gnomAD v4 AF:	68.9409%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.3 (Raw: -0.032 · Top 74.07%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

2. ClinVar Pathogenicity & Clinical Curation — Tier-1 & Actionable Genes

Genes carrying ClinVar-curated Pathogenic, Likely Pathogenic, Risk Factor, Drug Response, or Protective alleles.

Selected Scope: 20 Genes (20 Tier-1/Pathogenic)
Filtered from 303 matching genes in view

CBLIF CBLIF chr11:59,845,374 Multisystem 🔥 Tier-1 / Pathogenic Gene

Pathogenic · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the cobalamin transport protein family. It encodes a glycoprotein secreted by parietal cells of the gastric mucosa and is required for adequate absorption of vitamin B12. Vitamin B12 is necessary for erythrocyte maturation and mutations in this gene may lead to congenital pernicious anemia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147785187 c.79+1G>A	SPL	Heterozygous	Maternal	Pathogenic	Tier1	32.0	0.99	33.9 (Splicing)	– / –	chr11:59845374 AF: 0.0256%

1. INHERITANCE & OMIM PATHOLOGY

Intrinsic Factor Deficiency / Cobalamin Malabsorption
Autosomal Recessive OMIM #261000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

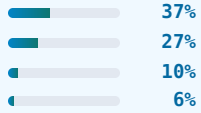
Transcript: ENST00000257248.3
cDNA (HGVS): c.79+1G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.47 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.04 DL:0.99
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#439755 ↗
gnomAD v4 AF: 0.0256%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.9 (Raw: 1.892 · Top 0.04%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB2 GJB2 chr13:20,189,481 Multisystem 🔥 Tier-1 / Pathogenic Gene

Pathogenic · CADD 20.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the gap junction protein family (gap junction beta 2) but is more commonly known as connexin 26. The gap junctions were first characterized by electron microscopy as regionally specialized structures on plasma membranes of contacting adherent cells. These structures were shown to consist of cell-to-cell channels that facilitate the transfer of ions and small molecules between cells. Connexins form hexameric channels in the plasma membrane which regulate passage of ions and small molecules between the cell and its environment or, when joined with another cell, form a dodecameric intercellular gap junction channel. The connexin proteins are grouped into alpha, beta, and gamma subfamilies. Mutations in this gene are responsible for as much as 50% of pre-lingual, recessive deafness. [provided by RefSeq, Mar 2026]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs35887622 p.Met34Thr	MIS, p.Met34Thr	Heterozygous	Unphased	Pathogenic	Tier1	20.9	–	23.6 (Cactus)	0.70 / 0.43	chr13:20189481 AF: 1.2041%

1. INHERITANCE & OMIM PATHOLOGY

Autosomal Recessive Deafness 1A / Dominant 3A
Autosomal Recessive Autosomal Dominant
OMIM #121011 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

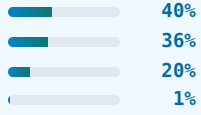
Transcript: ENST00000382848.5
cDNA (HGVS): c.101T>C
Protein (HGVS): p.Met34Thr
VAF / Read Depth: VAF 0.63 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: reviewed by expert panel
ClinVar VCV: VCV#17000 ↗
gnomAD v4 AF: 1.2041%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.6 (Raw: 0.892 · Top 0.43%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VDR VDR chr12:47,906,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely pathogenic · CADD 5.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes vitamin D3 receptor, which is a member of the nuclear hormone receptor superfamily of ligand-inducible transcription factors. This receptor also functions as a receptor for the secondary bile acid, lithocholic acid. Downstream targets of vitamin D3 receptor are principally involved in mineral metabolism, though this receptor regulates a variety of other metabolic pathways, such as those involved in immune response and cancer. Mutations in this gene are associated with type II vitamin D-resistant rickets. A single nucleotide polymorphism in the initiation codon results in an alternate translation start site three codons downstream. Alternatively spliced transcript variants encoding different isoforms have been described for this gene. A recent study provided evidence for translational readthrough in this gene, and expression of an additional C-terminally extended isoform via the use of an alternative in-frame translation termination codon. [provided by RefSeq, Jun 2018]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs4516035 c.-1172A>G	2KU	Homozygous	N/A	Likely pathogenic	Tier3	5.4	–	2.1 (DNASE-seq)	– / –	chr12:47906043 AF: 30.5142%

1. INHERITANCE & OMIM PATHOLOGY

Chronic obstructive pulmonary disease

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000549336.6
cDNA (HGVS):	c.-1172A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 1.00 · DP –
Phased GT:	1/1 (N/A)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	no assertion criteria provided
ClinVar VCV:	VCV#3336650 ↗
gnomAD v4 AF:	30.5142%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 2.1 (Raw: -0.015 · Top 61.95%)

Primary Driver: DNASE-seq



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

Uncertain significance · CADD 27.9 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	–	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367797.9
cDNA (HGVS):	c.883A>G
Protein (HGVS):	p.Thr295Ala
VAF / Read Depth:	VAF 0.34 · DP –
Phased GT:	0|1 (Maternal)

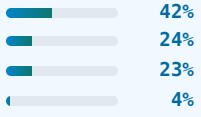
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2884751 ↗
gnomAD v4 AF:	0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs6025 p.Arg534Gln	MIS, p.Arg534Gln	Heterozygous	Unphased	drug response	Tier2	27.9	–	23.6 (Cactus)	– / 0.20	chr1:169549811 AF: 2.1432%

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

Uncertain significance · CADD 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	—	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377801.8
cDNA (HGVS):	c.734C>G
Protein (HGVS):	p.Ser245Cys
VAF / Read Depth:	VAF 0.49 · DP —
Phased GT:	0/1 (Unphased)

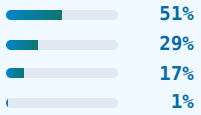
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#3214611 ↗
gnomAD v4 AF:	0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

Uncertain significance · CADD 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	—	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000355112.8
cDNA (HGVS):	c.2171T>C
Protein (HGVS):	p.Val724Ala
VAF / Read Depth:	VAF 0.44 · DP —
Phased GT:	0/1 (Unphased)

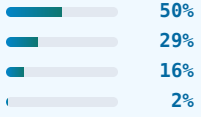
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#567134 ↗
gnomAD v4 AF:	0.0054%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.1 (Raw: 1.282 · Top 0.25%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

Uncertain significance · CADD 25.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	–	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000361439.9
cDNA (HGVS):	c.692G>A
Protein (HGVS):	p.Cys231Tyr
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

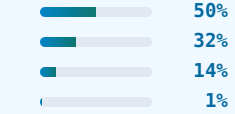
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2358669 ↗
gnomAD v4 AF:	0.0064%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.6 (Raw: 1.597 · Top 0.09%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Uncertain significance · CADD 24.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	–	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000294304.12
cDNA (HGVS):	c.698T>A
Protein (HGVS):	p.Val233Glu
VAF / Read Depth:	VAF 0.55 · DP –
Phased GT:	0/1 (Unphased)

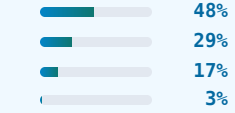
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#866038 ↗
gnomAD v4 AF:	0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.189 · Top 0.29%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Conflicting classifications of pathogenicity · CADD 35.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000304611.13**
cDNA (HGVS): **c.2362G>A**
Protein (HGVS): **p.Val788Met**
VAF / Read Depth: **VAF 0.58 · DP —**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.07 DL:0.76**
ClinVar Review: **criteria provided, conflicting classifications**
ClinVar VCV: VCV#556244 ↗
gnomAD v4 AF: **0.0007%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **32.3** (Raw: 1.716 · Top 0.06%)
Primary Driver: **Cactus**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BTN3A2 BTN3A2 chr6:26,370,605 Multisystem 🔥 Tier-1 / Pathogenic Gene

Benign · CADD 28.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the immunoglobulin superfamily, which resides in the juxta-telomeric region of the major histocompatability class 1 locus and is clustered with the other family members on chromosome 6. The encoded protein may be involved in the adaptive immune response. Alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Jun 2013]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs58367598 c.715+2T>G	SPL	Heterozygous	Unphased	Benign	Tier1	28.7	0.98	25.4 (Splicing)	— / —	chr6:26370605 AF: 3.8454%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

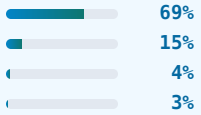
Transcript: **ENST00000377708.7**
cDNA (HGVS): **c.715+2T>G**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.64 · DP —**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.01 DL:0.98**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#1238754 ↗
gnomAD v4 AF: **3.8454%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.4** (Raw: 1.177 · Top 0.29%)
Primary Driver: **Splicing**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

Conflicting classifications of pathogenicity · CADD 27.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000357077.9
cDNA (HGVS): c.11716C>T
Protein (HGVS): p.Arg3906Trp
VAF / Read Depth: VAF 0.52 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#18059 ↗
gnomAD v4 AF: 0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)
Primary Driver: Splicing

36%
25%
20%
16%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Conflicting classifications of pathogenicity · CADD 26.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	—	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases
Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder
Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000373366.3
cDNA (HGVS): c.223C>T
Protein (HGVS): p.Arg75Cys
VAF / Read Depth: VAF 0.37 · DP —
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#297191 ↗
gnomAD v4 AF: 0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)
Primary Driver: AlphaMissense

46%
30%
12%
4%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely benign · CADD 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377403.7
cDNA (HGVS):	c.1004C>T
Protein (HGVS):	p.Pro335Leu
VAF / Read Depth:	VAF 0.43 · DP —
Phased GT:	0/1 (Unphased)

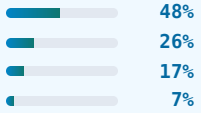
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.12 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#773105 ↗
gnomAD v4 AF:	0.2420%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.6 (Raw: 1.050 · Top 0.35%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

Benign/Likely benign|other · CADD 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	—	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia

Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000268097.10
cDNA (HGVS):	c.739C>T
Protein (HGVS):	p.Arg247Trp
VAF / Read Depth:	VAF 0.56 · DP —
Phased GT:	0/1 (Unphased)

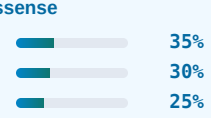
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#3922 ↗
gnomAD v4 AF:	0.2782%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.629 · Top 0.80%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Conflicting classifications of pathogenicity · CADD 24.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000435607.3
cDNA (HGVS):	c.1796A>G
Protein (HGVS):	p.His599Arg
VAF / Read Depth:	VAF 0.59 · DP –
Phased GT:	0/1 (Unphased)

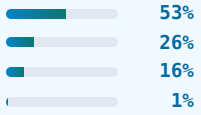
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#324537 ↗
gnomAD v4 AF:	0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

Conflicting classifications of pathogenicity · CADD 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	–	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4
Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000323670.14
cDNA (HGVS):	c.124G>A
Protein (HGVS):	p.Gly42Arg
VAF / Read Depth:	VAF 0.40 · DP –
Phased GT:	0|1 (Maternal)

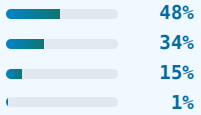
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#183298 ↗
gnomAD v4 AF:	0.0040%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.4 (Raw: 1.532 · Top 0.12%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

Conflicting classifications of pathogenicity · CADD 24.3 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	–	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000278550.12
cDNA (HGVS):	c.2804C>G
Protein (HGVS):	p.Pro935Arg
VAF / Read Depth:	VAF 0.44 · DP –
Phased GT:	0/1 (Unphased)

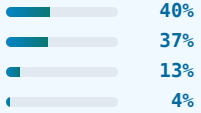
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#708690 ↗
gnomAD v4 AF:	0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs191549326 p.Pro485Ser	MIS, p.Pro485Ser	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier2	24.3	–	24.2 (Cactus)	0.36 / 0.36	chr11:78855981 AF: 0.4409%

TTN TTN chr2:178,621,751 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Benign/Likely benign · CADD 23.4 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a large abundant protein of striated muscle. The product of this gene is divided into two regions, a N-terminal I-band and a C-terminal A-band. The I-band, which is the elastic part of the molecule, contains two regions of tandem immunoglobulin domains on either side of a PEVK region that is rich in proline, glutamate, valine and lysine. The A-band, which is thought to act as a protein-ruler, contains a mixture of immunoglobulin and fibronectin repeats, and possesses kinase activity. An N-terminal Z-disc region and a C-terminal M-line region bind to the Z-line and M-line of the sarcomere, respectively, so that a single titin molecule spans half the length of a sarcomere. Titin also contains binding sites for muscle associated proteins so it serves as an adhesion template for the assembly of contractile machinery in muscle cells. It has also been identified as a structural protein for chromosomes. Alternative splicing of this gene results in multiple transcript variants. Considerable variability exists in the I-band, the M-line and the Z-disc regions of titin. Variability in the I-band region contributes to the differences in elasticity of different titin isoforms and, therefore, to the differences in elasticity of different muscle types. Mutations in this gene are associated with familial hypertrophic cardiomyopathy 9, and autoantibodies to titin are produced in patients with the autoimmune disease scleroderma. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72677222 c.45083-10A>G	INT	Heterozygous	Unphased	Benign/Likely benign	Tier1	17.8	1.00	26.0 (Splicing)	– / –	chr2:178621751 AF: 1.0145%
rs13021201 p.Ser21716Leu	MIS, p.Ser21716Leu	Heterozygous	Maternal	Benign/Likely benign	Tier3	23.4	–	22.8 (Cactus)	0.36 / 0.15	chr2:178584404 AF: 3.6610%
rs33917087 p.Val2823Phe	MIS, p.Val2823Phe	Heterozygous	Maternal	Benign/Likely benign	Tier2	23.0	0.21	22.9 (AlphaMissense)	0.42 / 0.46	chr2:178770234 AF: 1.8566%

1. INHERITANCE & OMIM PATHOLOGY

Dilated cardiomyopathy 1G
Complex / Multifactorial

Autosomal recessive limb-girdle muscular dystrophy type 2J
Autosomal Recessive

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

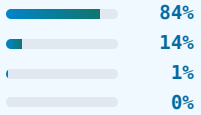
Transcript: **ENST00000589042.5**
cDNA (HGVS): **c.45083-10A>G**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.49 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:1.00 AL:0.78 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#46980 ↗
gnomAD v4 AF: **1.0145%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **26.0** (Raw: 1.269 · Top 0.25%)
Primary Driver: **Splicing**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CNTN1 CNTN1 chr12:40,943,448 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely benign · CADD 10.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the immunoglobulin superfamily. It is a glycosylphosphatidylinositol (GPI)-anchored neuronal membrane protein that functions as a cell adhesion molecule. It may play a role in the formation of axon connections in the developing nervous system. Multiple alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139915282 c.1380-149C>T	INT	Heterozygous	Paternal	Likely benign	Tier1	10.2	0.53	12.1 (Splicing)	– / –	chr12:40943448 AF: 1.8289%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

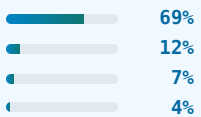
Transcript: **ENST00000551295.7**
cDNA (HGVS): **c.1380-149C>T**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.52 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.53 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#680275 ↗
gnomAD v4 AF: **1.8289%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **12.1** (Raw: 0.221 · Top 6.13%)
Primary Driver: **Splicing**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CDKN2B CDKN2B chr9:22,003,368 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely pathogenic|protective · CADD 0.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene lies adjacent to the tumor suppressor gene CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. This gene encodes a cyclin-dependent kinase inhibitor, which forms a complex with CDK4 or CDK6, and prevents the activation of the CDK kinases, thus the encoded protein functions as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be dramatically induced by TGF beta, which suggested its role in the TGF beta induced growth inhibition. Two alternatively spliced transcript variants of this gene, which encode distinct proteins, have been reported. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1063192 c.*2619C>T	UT3	Heterozygous	Unphased	Likely pathogenic|protective	Tier3	0.2	–	1.3 (Cactus)	– / –	chr9:22003368 AF: 68.9409%

1. INHERITANCE & OMIM PATHOLOGY

Coronary Artery Disease & Breast Carcinoma

Complex / Polygenic Protective Allele OMIM #600431 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000276925.7
cDNA (HGVS):	c.*2619C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.50 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	no assertion criteria provided
ClinVar VCV:	VCV#812639 ↗
gnomAD v4 AF:	68.9409%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.3 (Raw: -0.032 · Top 74.07%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

3. DeepMind AlphaGenome Atlas — Top Genes by AVI Phred & 18-Modality Drivers

Top genes ranked by DeepMind AlphaGenome Variant Impact (AVI) Phred score across 18 genomic and epigenomic modalities (RNA-seq, Splicing, ATAC/DNase, ChIP).

Selected Scope: 45 Genes (45 Tier-1/Pathogenic)
Filtered from 913 matching genes in view

CBLIF CBLIF chr11:59,845,374 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 33.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the cobalamin transport protein family. It encodes a glycoprotein secreted by parietal cells of the gastric mucosa and is required for adequate absorption of vitamin B12. Vitamin B12 is necessary for erythrocyte maturation and mutations in this gene may lead to congenital pernicious anemia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147785187 c.79+1G>A	SPL	Heterozygous	Maternal	Pathogenic	Tier1	32.0	0.99	33.9 (Splicing)	– / –	chr11:59845374 AF: 0.0256%

1. INHERITANCE & OMIM PATHOLOGY

Intrinsic Factor Deficiency / Cobalamin Malabsorption
Autosomal Recessive OMIM #261000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

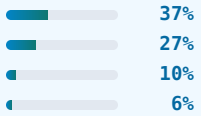
Transcript:	ENST00000257248.3
cDNA (HGVS):	c.79+1G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.47 · DP –
Phased GT:	0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.04 DL:0.99
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#439755 ↗
gnomAD v4 AF:	0.0256%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.9 (Raw: 1.892 · Top 0.04%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 32.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	–	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases
Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder
Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

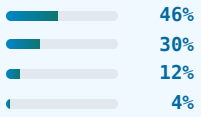
Transcript:	ENST00000373366.3
cDNA (HGVS):	c.223C>T
Protein (HGVS):	p.Arg75Cys
VAF / Read Depth:	VAF 0.37 · DP –
Phased GT:	0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#297191 ↗
gnomAD v4 AF:	0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 32.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000304611.13
cDNA (HGVS): c.2362G>A
Protein (HGVS): p.Val788Met
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#556244 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ELOVL7 ELOVL7 chr5:60,771,922 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 31.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables fatty acid elongase activity. Involved in fatty acid elongation, polyunsaturated fatty acid; fatty acid elongation, saturated fatty acid; and very long-chain fatty acid biosynthetic process. Located in endoplasmic reticulum. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs368309905 p.Ser79Tyr	MIS, p.Ser79Tyr	Heterozygous	Maternal	Not reviewed	Tier1	27.5	–	31.4 (AlphaMissense)	0.69 / 0.98	chr5:60771922 AF: 0.0210%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

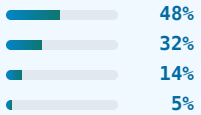
Transcript: ENST00000508821.6
cDNA (HGVS): c.236C>A
Protein (HGVS): p.Ser79Tyr
VAF / Read Depth: VAF 0.60 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0210%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 31.4 (Raw: 1.638 · Top 0.07%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 30.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	–	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

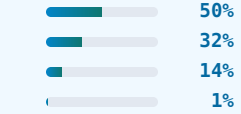
Transcript: **ENST00000361439.9**
cDNA (HGVS): **c.692G>A**
Protein (HGVS): **p.Cys231Tyr**
VAF / Read Depth: **VAF 0.53 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#2358669 ↗
gnomAD v4 AF: **0.0064%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **30.6** (Raw: 1.597 · Top 0.09%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 29.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	–	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4
Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

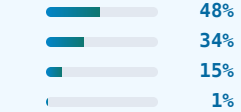
Transcript: **ENST00000323670.14**
cDNA (HGVS): **c.124G>A**
Protein (HGVS): **p.Gly42Arg**
VAF / Read Depth: **VAF 0.40 · DP –**
Phased GT: **0|1 (Maternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, conflicting classifications**
ClinVar VCV: VCV#183298 ↗
gnomAD v4 AF: **0.0040%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **29.4** (Raw: 1.532 · Top 0.12%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TRAK1 TRAK1 chr3:42,115,352 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 28.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable GABA receptor binding activity and myosin binding activity. Involved in endosome to lysosome transport. Located in early endosome and mitochondrion. Implicated in developmental and epileptic encephalopathy 68. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs17220321 c.92-10068T>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	17.7	0.89	28.6 (Splicing)	– / –	chr3:42115352 AF: 1.5740%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000327628.10
cDNA (HGVS):	c.92-10068T>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.43 · DP –
Phased GT:	0/1 (Unphased)

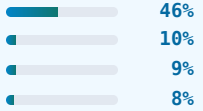
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.89 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	1.5740%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.6 (Raw: 1.480 · Top 0.14%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 27.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000435607.3
cDNA (HGVS):	c.1796A>G
Protein (HGVS):	p.His599Arg
VAF / Read Depth:	VAF 0.59 · DP –
Phased GT:	0/1 (Unphased)

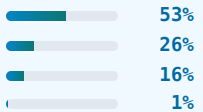
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#324537 ↗
gnomAD v4 AF:	0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.8 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	–	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

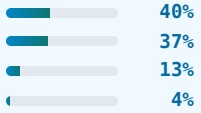
Transcript: ENST00000278550.12
cDNA (HGVS): c.2804C>G
Protein (HGVS): p.Pro935Arg
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#708690 ↗
gnomAD v4 AF: 0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs191549326 p.Pro485Ser	MIS, p.Pro485Ser	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier2	24.3	–	24.2 (Cactus)	0.36 / 0.36	chr11:78855981 AF: 0.4409%
rs181299543 c.-264-22173G>C	INT	Heterozygous	Paternal	Not reviewed	Tier3	17.0	–	13.3 (Cactus)	– / –	chr11:79238082 AF: 0.1990%

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000233813.5
cDNA (HGVS): c.578G>A
Protein (HGVS): p.Arg193His
VAF / Read Depth: VAF 0.38 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.11 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0008%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

COL23A1 COL23A1 chr5:178,262,215 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 COL23A1 is a member of the transmembrane collagens, a subfamily of the nonfibrillar collagens that contain a single pass hydrophobic transmembrane domain (Banyard et al., 2003 [PubMed 12644459]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs2973744 c.675+2T>C	SPL	Heterozygous	Unphased	Not reviewed	Tier1	33.0	0.91	26.3 (Splicing)	– / –	chr5:178262215 AF: 0.4032%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000390654.8
cDNA (HGVS):	c.675+2T>C
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.48 · DP –
Phased GT:	0/1 (Unphased)

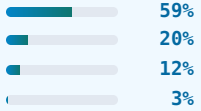
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.02 DL:0.91
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4314750 ↗
gnomAD v4 AF:	0.4032%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	–	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000355112.8
cDNA (HGVS):	c.2171T>C
Protein (HGVS):	p.Val724Ala
VAF / Read Depth:	VAF 0.44 · DP –
Phased GT:	0/1 (Unphased)

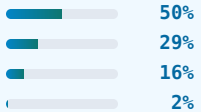
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#567134 ↗
gnomAD v4 AF:	0.0054%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.1 (Raw: 1.282 · Top 0.25%)

Primary Driver: AlphaMissense



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TTN TTN chr2:178,621,751 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene AlphaGenome AVI: 26.0 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a large abundant protein of striated muscle. The product of this gene is divided into two regions, a N-terminal I-band and a C-terminal A-band. The I-band, which is the elastic part of the molecule, contains two regions of tandem immunoglobulin domains on either side of a PEVK region that is rich in proline, glutamate, valine and lysine. The A-band, which is thought to act as a protein-ruler, contains a mixture of immunoglobulin and fibronectin repeats, and possesses kinase activity. An N-terminal Z-disc region and a C-terminal M-line region bind to the Z-line and M-line of the sarcomere, respectively, so that a single titin molecule spans half the length of a sarcomere. Titin also contains binding sites for muscle associated proteins so it serves as an adhesion template for the assembly of contractile machinery in muscle cells. It has also been identified as a structural protein for chromosomes. Alternative splicing of this gene results in multiple transcript variants. Considerable variability exists in the I-band, the M-line and the Z-disc regions of titin. Variability in the I-band region contributes to the differences in elasticity of different titin isoforms and, therefore, to the differences in elasticity of different muscle types. Mutations in this gene are associated with familial hypertrophic cardiomyopathy 9, and autoantibodies to titin are produced in patients with the autoimmune disease scleroderma. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72677222 c.45083-10A>G	INT	Heterozygous	Unphased	Benign/Likely benign	Tier1	17.8	1.00	26.0 (Splicing)	– / –	chr2:178621751 AF: 1.0145%

1. INHERITANCE & OMIM PATHOLOGY

Dilated cardiomyopathy 1G
Complex / Multifactorial

Autosomal recessive limb-girdle muscular dystrophy type 2J
Autosomal Recessive

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000589042.5
cDNA (HGVS):	c.45083-10A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.49 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:1.00 AL:0.78 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#46980 ↗
gnomAD v4 AF:	1.0145%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.0 (Raw: 1.269 · Top 0.25%)

Primary Driver: Splicing

84%
14%
1%
0%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs33917087 p.Val2823Phe	MIS, p.Val2823Phe	Heterozygous	Maternal	Benign/Likely benign	Tier2	23.0	0.21	22.9 (AlphaMissense)	0.42 / 0.46	chr2:178770234 AF: 1.8566%
rs13021201 p.Ser21716Leu	MIS, p.Ser21716Leu	Heterozygous	Maternal	Benign/Likely benign	Tier3	23.4	–	22.8 (Cactus)	0.36 / 0.15	chr2:178584404 AF: 3.6610%

PTDSS2 PTDSS2 chr11:488,272 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.8 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene catalyzes the conversion of phosphatidylethanolamine to phosphatidylserine, a structural membrane phospholipid that functions in cell signaling, blood coagulation, and apoptosis. The encoded enzyme also has a high affinity for docosahexaenoic acid (DHA) and can use it to make DHA-containing phosphatidylserine. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs908582716 p.Glu232Gly	MIS, p.Glu232Gly	Heterozygous	Unphased	Not reviewed	Tier1	31.0	—	25.8 (AlphaMissense)	0.77 / 0.78	chr11:488272 AF: 0.0096%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000308020.6
cDNA (HGVS):	c.695A>G
Protein (HGVS):	p.Glu232Gly
VAF / Read Depth:	VAF 0.68 · DP —
Phased GT:	0/1 (Unphased)

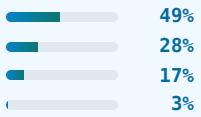
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0096%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.251 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs142613557 c.367+914G>A	INT	Heterozygous	Unphased	Not reviewed	Tier2	0.9	0.24	0.4 (Cactus)	— / —	chr11:474891 AF: 0.6829%

GRIA4 GRIA4 chr11:105,611,080 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Glutamate receptors are the predominant excitatory neurotransmitter receptors in the mammalian brain and are activated in a variety of normal neurophysiologic processes. These receptors are heteromeric protein complexes composed of multiple subunits, arranged to form ligand-gated ion channels. The classification of glutamate receptors is based on their activation by different pharmacologic agonists. The subunit encoded by this gene belongs to a family of AMPA (alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate)-sensitive glutamate receptors, and is subject to RNA editing (AGA->GGA; R->G). Alternative splicing of this gene results in transcript variants encoding different isoforms, which may vary in their signal transduction properties. Some haplotypes of this gene show a positive association with schizophrenia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs150577695 p.Gln28Arg	MIS, p.Gln28Arg	Heterozygous	Paternal	Not reviewed	Tier1	23.0	—	25.8 (AlphaMissense)	0.30 / 0.78	chr11:105611080 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000282499.10
cDNA (HGVS):	c.83A>G
Protein (HGVS):	p.Gln28Arg
VAF / Read Depth:	VAF 0.56 · DP —
Phased GT:	1|0 (Paternal)

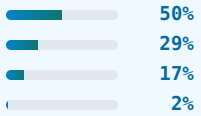
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.245 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	–	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

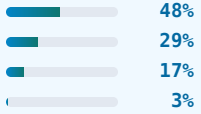
Transcript: ENST00000294304.12
cDNA (HGVS): c.698T>A
Protein (HGVS): p.Val233Glu
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#866038 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.189 · Top 0.29%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BTN3A2 BTN3A2 chr6:26,370,605 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the immunoglobulin superfamily, which resides in the juxta-telomeric region of the major histocompatability class 1 locus and is clustered with the other family members on chromosome 6. The encoded protein may be involved in the adaptive immune response. Alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Jun 2013]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs58367598 c.715+2T>G	SPL	Heterozygous	Unphased	Benign	Tier1	28.7	0.98	25.4 (Splicing)	– / –	chr6:26370605 AF: 3.8454%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

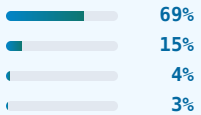
Transcript: ENST00000377708.7
cDNA (HGVS): c.715+2T>G
Protein (HGVS): –
VAF / Read Depth: VAF 0.64 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.01 DL:0.98
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#1238754 ↗
gnomAD v4 AF: 3.8454%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.177 · Top 0.29%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.2 Showing 3 of 3 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

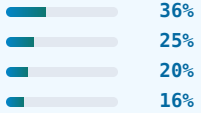
Transcript: ENST00000357077.9
cDNA (HGVS): c.11716C>T
Protein (HGVS): p.Arg3906Trp
VAF / Read Depth: VAF 0.52 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#18059 ↗
gnomAD v4 AF: 0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs183491249 c.85-40398G>A	INT	Heterozygous	Maternal	Not reviewed	Tier3	5.2	–	4.2 (Cactus)	– / –	chr4:113134018 AF: 0.4297%
rs11929834 c.85-22668G>A	INT	Heterozygous	Paternal	Not reviewed	Tier2	8.0	–	2.8 (PhastCons 470)	– / –	chr4:113151748 AF: 32.8900%

PLA2G6 PLA2G6 chr22:38,115,172 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is an A2 phospholipase, a class of enzyme that catalyzes the release of fatty acids from phospholipids. The encoded protein may play a role in phospholipid remodelling, arachidonic acid release, leukotriene and prostaglandin synthesis, fas-mediated apoptosis, and transmembrane ion flux in glucose-stimulated B-cells. Several transcript variants encoding multiple isoforms have been described, but the full-length nature of only three of them have been determined to date. [provided by RefSeq, Dec 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147795054 c.2034+355G>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	1.7	0.95	25.0 (Splicing)	– / –	chr22:38115172 AF: 0.5672%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

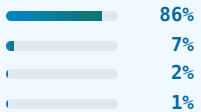
Transcript: ENST00000332509.8
cDNA (HGVS): c.2034+355G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.95 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.5672%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.0 (Raw: 1.119 · Top 0.32%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PAK3 PAK3 chrX:111,163,700 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a serine-threonine kinase and forms an activated complex with GTP-bound RAS-like (P21), CDC2 and RAC1. This protein may be necessary for dendritic development and for the rapid cytoskeletal reorganization in dendritic spines associated with synaptic plasticity. Defects in this gene are the cause of a non-syndromic form of X-linked intellectual disability. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
chrX:111163700 p.Thr247Pro	MIS, p.Thr247Pro	Heterozygous	Unphased	Not reviewed	Tier1	25.1	–	24.9 (AlphaMissense)	0.48 / 0.72	chrX:111163700 AF: 0.0000%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

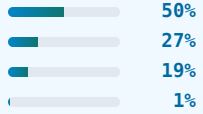
Transcript: **ENST00000372007.10**
cDNA (HGVS): **c.739A>C**
Protein (HGVS): **p.Thr247Pro**
VAF / Read Depth: **VAF 0.27 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0000%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.9** (Raw: 1.106 · Top 0.32%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder
Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

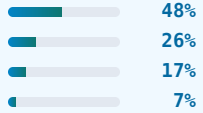
Transcript: **ENST00000377403.7**
cDNA (HGVS): **c.1004C>T**
Protein (HGVS): **p.Pro335Leu**
VAF / Read Depth: **VAF 0.43 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.12 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#773105 ↗
gnomAD v4 AF: **0.2420%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.6** (Raw: 1.050 · Top 0.35%)
Primary Driver: **Cactus**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ACACB ACACB chr12:109,260,564 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acetyl-CoA carboxylase (ACC) is a complex multifunctional enzyme system. ACC is a biotin-containing enzyme which catalyzes the carboxylation of acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid synthesis. ACC-beta is thought to control fatty acid oxidation by means of the ability of malonyl-CoA to inhibit carnitine-palmitoyl-CoA transferase I, the rate-limiting step in fatty acid uptake and oxidation by mitochondria. ACC-beta may be involved in the regulation of fatty acid oxidation, rather than fatty acid biosynthesis. [provided by RefSeq, Oct 2022]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139082246 p.Pro2194Leu	MIS, p.Pro2194Leu	Heterozygous	Paternal	Not reviewed	Tier1	25.5	–	24.2 (Cactus)	0.70 / 0.36	chr12:109260564 AF: 0.0037%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

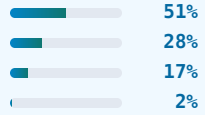
Transcript: ENST00000338432.12
cDNA (HGVS): c.6581C>T
Protein (HGVS): p.Pro2194Leu
VAF / Read Depth: VAF 0.53 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0037%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.2 (Raw: 0.986 · Top 0.38%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	–	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

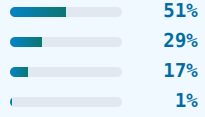
Transcript: ENST00000377801.8
cDNA (HGVS): c.734C>G
Protein (HGVS): p.Ser245Cys
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#3214611 ↗
gnomAD v4 AF: 0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IQCA1 IQCA1 chr2:236,338,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ATPases Associated with diverse cellular Activities (AAA) superfamily. Members of this superfamily, found in all organisms, participate in a large number of cellular processes and contain the ATPase module consisting of an alpha-beta-alpha core domain and the Walker A and B motifs of the P-loop NTPases. Alternative splicing results in multiple transcript variants that encode different protein isoforms. [provided by RefSeq, Jul 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs115960780 p.Cys693Arg	MIS, p.Cys693Arg	Heterozygous	Paternal	Not reviewed	Tier1	26.2	–	24.0 (AlphaMissense)	0.75 / 0.49	chr2:236338262 AF: 0.6424%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000409907.8
cDNA (HGVS): c.2077T>C
Protein (HGVS): p.Cys693Arg
VAF / Read Depth: VAF 0.56 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.6424%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.0 (Raw: 0.948 · Top 0.40%)
Primary Driver: AlphaMissense

39%
38%
20%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB2 GJB2 chr13:20,189,481 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 23.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the gap junction protein family (gap junction beta 2) but is more commonly known as connexin 26. The gap junctions were first characterized by electron microscopy as regionally specialized structures on plasma membranes of contacting adherent cells. These structures were shown to consist of cell-to-cell channels that facilitate the transfer of ions and small molecules between cells. Connexins form hexameric channels in the plasma membrane which regulate passage of ions and small molecules between the cell and its environment or, when joined with another cell, form a dodecameric intercellular gap junction channel. The connexin proteins are grouped into alpha, beta, and gamma subfamilies. Mutations in this gene are responsible for as much as 50% of pre-lingual, recessive deafness. [provided by RefSeq, Mar 2026]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs35887622 p.Met34Thr	MIS, p.Met34Thr	Heterozygous	Unphased	Pathogenic	Tier1	20.9	–	23.6 (Cactus)	0.70 / 0.43	chr13:20189481 AF: 1.2041%

1. INHERITANCE & OMIM PATHOLOGY

Autosomal Recessive Deafness 1A / Dominant 3A

Autosomal Recessive Autosomal Dominant

OMIM #121011 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000382848.5
cDNA (HGVS): c.101T>C
Protein (HGVS): p.Met34Thr
VAF / Read Depth: VAF 0.63 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: reviewed by expert panel
ClinVar VCV: VCV#17000 ↗
gnomAD v4 AF: 1.2041%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.6 (Raw: 0.892 · Top 0.43%)
Primary Driver: Cactus

40%
36%
20%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 23.6 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	–	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367797.9
cDNA (HGVS):	c.883A>G
Protein (HGVS):	p.Thr295Ala
VAF / Read Depth:	VAF 0.34 · DP –
Phased GT:	0|1 (Maternal)

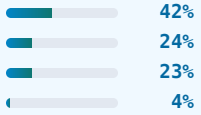
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2884751 ↗
gnomAD v4 AF:	0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs6025 p.Arg534Gln	MIS, p.Arg534Gln	Heterozygous	Unphased	drug response	Tier2	27.9	–	23.6 (Cactus)	– / 0.20	chr1:169549811 AF: 2.1432%

DNAH7 DNAH7 chr2:196,024,488 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 23.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 DNAH7 is a component of the inner dynein arm of ciliary axonemes (Zhang et al., 2002 [PubMed 11877439]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs555558470 p.Lys228=	SYN, p.Lys228=	Heterozygous	Paternal	Not reviewed	Tier1	12.9	0.74	23.0 (Splicing)	– / –	chr2:196024488 AF: 0.0021%

1. INHERITANCE & OMIM PATHOLOGY

Primary Ciliary Dyskinesia / Respiratory Clearance

Autosomal Recessive OMIM #610061 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000312428.11
cDNA (HGVS):	c.684A>G
Protein (HGVS):	p.Lys228=
VAF / Read Depth:	VAF 0.65 · DP –
Phased GT:	1|0 (Paternal)

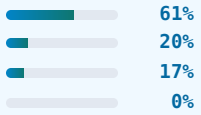
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.74 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0021%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.0 (Raw: 0.808 · Top 0.50%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SLC9A2 SLC9A2 chr2:102,701,098 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 22.8 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the sodium-hydrogen exchanger (NHE) protein family. These proteins are involved in sodium-ion transport by exchanging intracellular hydrogen ions to external sodium ions and help in the regulation of cell pH and volume. The encoded protein is localized to the apical membrane and is involved in apical absorption of sodium. [provided by RefSeq, Jun 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371605244 p.Arg539Trp	MIS, p.Arg539Trp	Heterozygous	Maternal	Not reviewed	Tier1	26.3	–	22.8 (AlphaMissense)	0.46 / 0.54	chr2:102701098 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

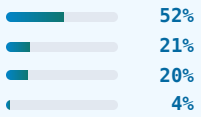
Transcript: ENST00000233969.3
cDNA (HGVS): c.1615C>T
Protein (HGVS): p.Arg539Trp
VAF / Read Depth: VAF 0.51 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.8 (Raw: 0.787 · Top 0.52%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs4499446 c.289+5344G>C	INT	Homozygous	N/A	Not reviewed	Tier3	0.9	–	6.3 (DNASE-seq)	– / –	chr2:102625481 AF: 54.2955%

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 21.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	–	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia
Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

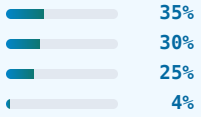
Transcript: ENST00000268097.10
cDNA (HGVS): c.739C>T
Protein (HGVS): p.Arg247Trp
VAF / Read Depth: VAF 0.56 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#3922 ↗
gnomAD v4 AF: 0.2782%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.629 · Top 0.80%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NEK10 NEK10 chr3:27,344,024 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 20.4 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein kinase activity. Involved in several processes, including mucociliary clearance; positive regulation of protein phosphorylation; and regulation of ERK1 and ERK2 cascade. Part of protein kinase complex. Implicated in primary ciliary dyskinesia 44. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs113156482 c.362+248G>A	INT	Heterozygous	Maternal	Not reviewed	Tier1	12.8	0.56	20.4 (Splicing)	– / –	chr3:27344024 AF: 1.7310%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

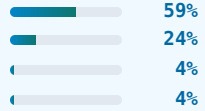
Transcript: ENST00000691995.1
cDNA (HGVS): c.362+248G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.41 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.56 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 1.7310%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.4 (Raw: 0.584 · Top 0.92%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs1267845859 c.2971-433A>G	INT	Heterozygous	Paternal	Not reviewed	Tier2	0.8	0.47	14.6 (Splicing)	– / –	chr3:27132423 AF: 0.0007%

TMEM270 TMEM270 chr7:73,865,329 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 20.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs73142829 p.Gly137Ser	MIS, p.Gly137Ser	Heterozygous	Paternal	Not reviewed	Tier1	0.0	0.72	20.1 (Splicing)	0.02 / 0.08	chr7:73865329 AF: 4.6362%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

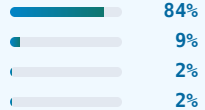
Transcript: ENST00000320531.3
cDNA (HGVS): c.409G>A
Protein (HGVS): p.Gly137Ser
VAF / Read Depth: VAF 0.47 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.72 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4339331 ↗
gnomAD v4 AF: 4.6362%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.1 (Raw: 0.569 · Top 0.97%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NPM1 NPM1 chr5:171,388,520 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 19.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is involved in several cellular processes, including centrosome duplication, protein chaperoning, and cell proliferation. The encoded phosphoprotein shuttles between the nucleolus, nucleus, and cytoplasm, chaperoning ribosomal proteins and core histones from the nucleus to the cytoplasm. This protein is also known to sequester the tumor suppressor ARF in the nucleolus, protecting it from degradation until it is needed. Mutations in this gene are associated with acute myeloid leukemia. Dozens of pseudogenes of this gene have been identified. [provided by RefSeq, Aug 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1770393339 c.58+514T>G	INT	Heterozygous	Unphased	Not reviewed	Tier1	8.7	0.82	19.6 (Splicing)	— / —	chr5:171388520 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000296930.10
cDNA (HGVS):	c.58+514T>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.49 · DP —
Phased GT:	0/1 (Unphased)

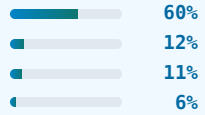
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.01 AL:0.00 DG:0.82 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.6 (Raw: 0.536 · Top 1.10%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 19.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs562989479 c.99+808G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	6.9	0.57	19.2 (Splicing)	— / —	chr1:165828740 AF: 0.3704%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367879.9
cDNA (HGVS):	c.99+808G>T
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.47 · DP —
Phased GT:	1|0 (Paternal)

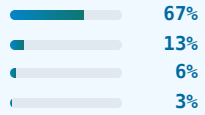
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.57 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.3704%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

RAD51 RAD51 chr15:40,695,628 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 19.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the RAD51 protein family. RAD51 family members are highly similar to bacterial RecA and Saccharomyces cerevisiae Rad51, and are known to be involved in the homologous recombination and repair of DNA. This protein can interact with the ssDNA-binding protein RPA and RAD52, and it is thought to play roles in homologous pairing and strand transfer of DNA. This protein is also found to interact with BRCA1 and BRCA2, which may be important for the cellular response to DNA damage. BRCA2 is shown to regulate both the intracellular localization and DNA-binding ability of this protein. Loss of these controls following BRCA2 inactivation may be a key event leading to genomic instability and tumorigenesis. Multiple transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Aug 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs5030791 c.-3+203G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	5.5	0.81	19.1 (Splicing)	– / –	chr15:40695628 AF: 1.3014%

1. INHERITANCE & OMIM PATHOLOGY

Breast-Ovarian Cancer Predisposition
Autosomal Dominant OMIM #179570 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

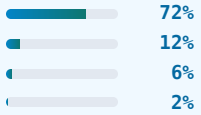
Transcript: ENST00000267868.8
cDNA (HGVS): c.-3+203G>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.60 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.81 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4401961 ↗
gnomAD v4 AF: 1.3014%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.1 (Raw: 0.510 · Top 1.23%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

EGFL7 EGFL7 chr9:136,670,809 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 16.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a secreted endothelial cell protein that contains two epidermal growth factor-like domains. The encoded protein may play a role in regulating vasculogenesis. This protein may be involved in the growth and proliferation of tumor cells. Alternate splicing results in multiple transcript variants. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs920684955 c.572-141C>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	0.6	0.73	16.7 (Splicing)	– / –	chr9:136670809 AF: 0.0147%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

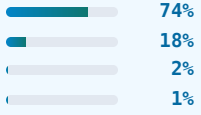
Transcript: ENST00000308874.12
cDNA (HGVS): c.572-141C>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.73 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0147%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 16.7 (Raw: 0.391 · Top 2.16%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PARP3 PARP3 chr3:51,944,496 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 15.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PARP family. These enzymes modify nuclear proteins by poly-ADP-ribosylation, which is required for DNA repair, regulation of apoptosis, and maintenance of genomic stability. This gene encodes the poly(ADP-ribosyl)transferase 3, which is preferentially localized to the daughter centriole throughout the cell cycle. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs761611493 p.Arg140Gln	MIS, p.Arg140Gln	Heterozygous	Unphased	Not reviewed	Tier1	24.8	–	15.5 (Cactus)	0.40 / 0.14	chr3:51944496 AF: 0.0061%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

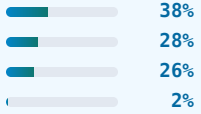
Transcript: **ENST00000398755.8**
cDNA (HGVS): **c.419G>A**
Protein (HGVS): **p.Arg140Gln**
VAF / Read Depth: **VAF 0.56 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0061%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **15.5** (Raw: 0.340 · Top 2.83%)
Primary Driver: **Cactus**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

JKAMP JKAMP chr14:59,502,424 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 13.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables ubiquitin protein ligase binding activity. Involved in ERAD pathway. Predicted to be located in endoplasmic reticulum membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs113112586 c.717+1157A>G	INT	Heterozygous	Paternal	Not reviewed	Tier1	13.9	0.65	13.5 (Splicing)	– / –	chr14:59502424 AF: 4.0038%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

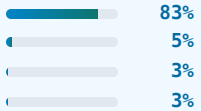
Transcript: **ENST00000616435.5**
cDNA (HGVS): **c.717+1157A>G**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.63 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.40 AL:0.00 DG:0.65 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **4.0038%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **13.5** (Raw: 0.265 · Top 4.50%)
Primary Driver: **Splicing**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

AZIN2 AZIN2 chr1:33,081,925 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 13.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the antizyme inhibitor family, which plays a role in cell growth and proliferation by maintaining polyamine homeostasis within the cell. Antizyme inhibitors are homologs of ornithine decarboxylase (ODC, the key enzyme in polyamine biosynthesis) that have lost the ability to decarboxylase ornithine; however, retain the ability to bind to antizymes. Antizymes negatively regulate intracellular polyamine levels by binding to ODC and targeting it for degradation, as well as by inhibiting polyamine uptake. Antizyme inhibitors function as positive regulators of polyamine levels by sequestering antizymes and neutralizing their effect. This gene encodes antizyme inhibitor 2, the second member of this gene family. Like antizyme inhibitor 1, antizyme inhibitor 2 interacts with all 3 antizymes and stimulates ODC activity and polyamine uptake. However, unlike antizyme inhibitor 1, which is ubiquitously expressed and localized in the nucleus and cytoplasm, antizyme inhibitor 2 is predominantly expressed in the brain and testis and localized in the endoplasmic reticulum-golgi intermediate compartment. Recent studies indicate that antizyme inhibitor 2 is also expressed in specific cell types in ovaries, adrenal glands and pancreas, and in mast cells. The exact function of this gene is not known, however, available data suggest its role in cell growth, spermiogenesis, vesicular trafficking and secretion. Accumulation of antizyme inhibitor 2 has also been observed in brains of patients with Alzheimer's disease. There has been confusion in literature and databases over the nomenclature of this gene, stemming from an earlier report that a human cDNA clone (identical to ODCp/AZIN2) had arginine decarboxylase (ADC) activity (PMID:14738999). Subsequent studies in human and mouse showed that antizyme inhibitor 2 was devoid of arginine decarboxylase activity (PMID:19956990). Alternatively spliced transcript variants have been described for this gene. [provided by RefSeq, Sep 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs190309518 c.-73+113G>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	0.4	0.67	13.2 (Splicing)	— / —	chr1:33081925 AF: 0.0490%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

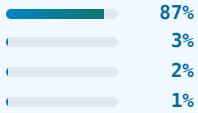
Transcript: ENST00000294517.11
cDNA (HGVS): c.-73+113G>T
Protein (HGVS): —
VAF / Read Depth: VAF 0.40 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.06 AL:0.00 DG:0.67 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4417896 ↗
gnomAD v4 AF: 0.0490%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.2 (Raw: 0.256 · Top 4.79%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

INPP5K INPP5K chr17:1,508,258 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 13.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with 5-phosphatase activity toward polyphosphate inositol. The protein localizes to the cytosol in regions lacking actin stress fibers. It is thought that this protein may negatively regulate the actin cytoskeleton. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs74437970 c.555-32A>G	INT	Heterozygous	Maternal	Not reviewed	Tier1	7.5	0.63	13.0 (Splicing)	— / —	chr17:1508258 AF: 2.2461%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

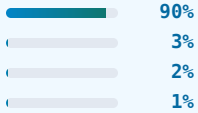
Transcript: ENST00000421807.7
cDNA (HGVS): c.555-32A>G
Protein (HGVS): —
VAF / Read Depth: VAF 0.50 · DP —
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.63 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4417749 ↗
gnomAD v4 AF: 2.2461%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.0 (Raw: 0.249 · Top 5.02%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CNTN1 CNTN1 chr12:40,943,448 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 12.1 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the immunoglobulin superfamily. It is a glycosylphosphatidylinositol (GPI)-anchored neuronal membrane protein that functions as a cell adhesion molecule. It may play a role in the formation of axon connections in the developing nervous system. Multiple alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139915282 c.1380-149C>T	INT	Heterozygous	Paternal	Likely benign	Tier1	10.2	0.53	12.1 (Splicing)	– / –	chr12:40943448 AF: 1.8289%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

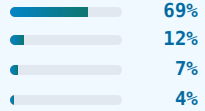
Transcript: **ENST00000551295.7**
cDNA (HGVS): **c.1380-149C>T**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.52 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.53 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#680275 ↗
gnomAD v4 AF: **1.8289%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 12.1 (Raw: 0.221 · Top 6.13%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs146742800 c.2420-1480G>A	INT	Heterozygous	Maternal	Not reviewed	Tier2	6.3	0.22	9.9 (Splicing)	– / –	chr12:41018857 AF: 1.5444%

PKD1L3 PKD1L3 chr16:71,979,984 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 10.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polycystin protein family. The encoded protein contains 11 transmembrane domains, a latrophilin/CL-1-like GPCR proteolytic site (GPS) domain, and a polycystin-1, lipoxygenase, alpha-toxin (PLAT) domain. This protein may function as a component of cation channel pores.[provided by RefSeq, Apr 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72803747 c.1271+23G>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.4	0.84	10.4 (Splicing)	– / –	chr16:71979984 AF: 0.9916%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

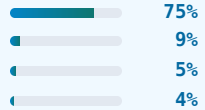
Transcript: **ENST00000620267.2**
cDNA (HGVS): **c.1271+23G>T**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.39 · DP –**
Phased GT: **0|1 (Maternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.84 DL:0.02**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4415201 ↗
gnomAD v4 AF: **0.9916%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 10.4 (Raw: 0.168 · Top 9.03%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLB1 PLB1 chr2:28,461,598 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 3.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a membrane-associated phospholipase that displays lysophospholipase and phospholipase A2 activities through removal of sn-1 and sn-2 fatty acids of glycerophospholipids. In addition, it displays lipase and retinyl ester hydrolase activities. The encoded protein is highly conserved and is composed of a large, glycosylated extracellular domain composed of four tandem homologous domains, followed by a hydrophobic segment that anchors the enzyme to the membrane and a short C-terminal cytoplasmic tail. This gene has been identified as a candidate rheumatoid arthritis risk gene. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs56116810 c.-114+4401G>A	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.5	–	3.6 (Cactus)	– / –	chr2:28461598 AF: 0.0012%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000416713.5
cDNA (HGVS):	c.-114+4401G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.50 · DP –
Phased GT:	0|1 (Maternal)

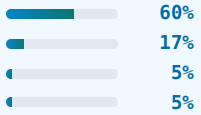
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0012%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 3.6 (Raw: 0.009 · Top 43.90%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VDR VDR chr12:47,906,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 2.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes vitamin D3 receptor, which is a member of the nuclear hormone receptor superfamily of ligand-inducible transcription factors. This receptor also functions as a receptor for the secondary bile acid, lithocholic acid. Downstream targets of vitamin D3 receptor are principally involved in mineral metabolism, though this receptor regulates a variety of other metabolic pathways, such as those involved in immune response and cancer. Mutations in this gene are associated with type II vitamin D-resistant rickets. A single nucleotide polymorphism in the initiation codon results in an alternate translation start site three codons downstream. Alternatively spliced transcript variants encoding different isoforms have been described for this gene. A recent study provided evidence for translational readthrough in this gene, and expression of an additional C-terminally extended isoform via the use of an alternative in-frame translation termination codon. [provided by RefSeq, Jun 2018]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs4516035 c.-1172A>G	2KU	Homozygous	N/A	Likely pathogenic	Tier3	5.4	–	2.1 (DNASE-seq)	– / –	chr12:47906043 AF: 30.5142%

1. INHERITANCE & OMIM PATHOLOGY

Chronic obstructive pulmonary disease

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000549336.6
cDNA (HGVS):	c.-1172A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 1.00 · DP –
Phased GT:	1/1 (N/A)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	no assertion criteria provided
ClinVar VCV:	VCV#3336650 ↗
gnomAD v4 AF:	30.5142%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 2.1 (Raw: -0.015 · Top 61.95%)

Primary Driver: DNASE-seq



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CDKN2B CDKN2B chr9:22,003,368 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 1.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene lies adjacent to the tumor suppressor gene CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. This gene encodes a cyclin-dependent kinase inhibitor, which forms a complex with CDK4 or CDK6, and prevents the activation of the CDK kinases, thus the encoded protein functions as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be dramatically induced by TGF beta, which suggested its role in the TGF beta induced growth inhibition. Two alternatively spliced transcript variants of this gene, which encode distinct proteins, have been reported. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1063192 c.*2619C>T	UT3	Heterozygous	Unphased	Likely pathogenic|protective	Tier3	0.2	–	1.3 (Cactus)	– / –	chr9:22003368 AF: 68.9409%

1. INHERITANCE & OMIM PATHOLOGY

Coronary Artery Disease & Breast Carcinoma

Complex / Polygenic Protective Allele OMIM #600431 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000276925.7
cDNA (HGVS):	c.*2619C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.50 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	no assertion criteria provided
ClinVar VCV:	VCV#812639 ↗
gnomAD v4 AF:	68.9409%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.3 (Raw: -0.032 · Top 74.07%)

Primary Driver: Cactus

22%
16%
14%
11%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CYP2E1 CYP2E1 chr10:133,520,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 1.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the cytochrome P450 superfamily of enzymes. The cytochrome P450 proteins are monooxygenases which catalyze many reactions involved in drug metabolism and synthesis of cholesterol, steroids and other lipids. This protein localizes to the endoplasmic reticulum and is induced by ethanol, the diabetic state, and starvation. The enzyme metabolizes both endogenous substrates, such as ethanol, acetone, and acetal, as well as exogenous substrates including benzene, carbon tetrachloride, ethylene glycol, and nitrosamines which are premutagens found in cigarette smoke. Due to its many substrates, this enzyme may be involved in such varied processes as gluconeogenesis, hepatic cirrhosis, diabetes, and cancer. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141625332 c.-118+195C>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	2.5	0.51	1.0 (Cactus)	– / –	chr10:133520755 AF: 1.0596%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000463117.6
cDNA (HGVS):	c.-118+195C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.43 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.51 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	1.0596%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.0 (Raw: -0.042 · Top 80.14%)

Primary Driver: Cactus

45%
10%
9%
9%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

4. CADD Deleteriousness Ranking — Tier-1 & Top Genes (Phred ≥ 15)

Top deleterious genes ranked by Combined Annotation Dependent Depletion (CADD v1.7) Phred-scaled score.

Selected Scope: 28 Genes (28 Tier-1/Pathogenic)
Filtered from 497 matching genes in view

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 35.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000304611.13
cDNA (HGVS): c.2362G>A
Protein (HGVS): p.Val788Met
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#556244 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

COL23A1 COL23A1 chr5:178,262,215 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 33.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 COL23A1 is a member of the transmembrane collagens, a subfamily of the nonfibrillar collagens that contain a single pass hydrophobic transmembrane domain (Banyard et al., 2003 [PubMed 12644459]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs2973744 c.675+2T>C	SPL	Heterozygous	Unphased	Not reviewed	Tier1	33.0	0.91	26.3 (Splicing)	– / –	chr5:178262215 AF: 0.4032%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

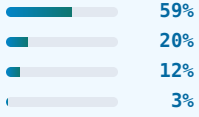
Transcript: ENST00000390654.8
cDNA (HGVS): c.675+2T>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.48 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.02 DL:0.91
ClinVar Review: criteria provided
ClinVar VCV: VCV#4314750 ↗
gnomAD v4 AF: 0.4032%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)
Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CBLIF CBLIF chr11:59,845,374 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the cobalamin transport protein family. It encodes a glycoprotein secreted by parietal cells of the gastric mucosa and is required for adequate absorption of vitamin B12. Vitamin B12 is necessary for erythrocyte maturation and mutations in this gene may lead to congenital pernicious anemia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147785187 c.79+1G>A	SPL	Heterozygous	Maternal	Pathogenic	Tier1	32.0	0.99	33.9 (Splicing)	– / –	chr11:59845374 AF: 0.0256%

1. INHERITANCE & OMIM PATHOLOGY

Intrinsic Factor Deficiency / Cobalamin Malabsorption

Autosomal Recessive OMIM #261000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000257248.3
cDNA (HGVS):	c.79+1G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.47 · DP –
Phased GT:	0|1 (Maternal)

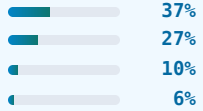
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.04 DL:0.99
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#439755 ↗
gnomAD v4 AF:	0.0256%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.9 (Raw: 1.892 · Top 0.04%)

Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000233813.5
cDNA (HGVS):	c.578G>A
Protein (HGVS):	p.Arg193His
VAF / Read Depth:	VAF 0.38 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.11 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0008%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)

Primary Driver: AlphaMissense



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PTDSS2 PTDSS2 chr11:488,272 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 31.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene catalyzes the conversion of phosphatidylethanolamine to phosphatidylserine, a structural membrane phospholipid that functions in cell signaling, blood coagulation, and apoptosis. The encoded enzyme also has a high affinity for docosahexaenoic acid (DHA) and can use it to make DHA-containing phosphatidylserine. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs908582716 p.Glu232Gly	MIS, p.Glu232Gly	Heterozygous	Unphased	Not reviewed	Tier1	31.0	—	25.8 (AlphaMissense)	0.77 / 0.78	chr11:488272 AF: 0.0096%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000308020.6
cDNA (HGVS):	c.695A>G
Protein (HGVS):	p.Glu232Gly
VAF / Read Depth:	VAF 0.68 · DP —
Phased GT:	0/1 (Unphased)

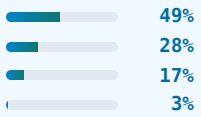
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0096%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.251 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BTN3A2 BTN3A2 chr6:26,370,605 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 28.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the immunoglobulin superfamily, which resides in the juxta-telomeric region of the major histocompatability class 1 locus and is clustered with the other family members on chromosome 6. The encoded protein may be involved in the adaptive immune response. Alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Jun 2013]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs58367598 c.715+2T>G	SPL	Heterozygous	Unphased	Benign	Tier1	28.7	0.98	25.4 (Splicing)	— / —	chr6:26370605 AF: 3.8454%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377708.7
cDNA (HGVS):	c.715+2T>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.64 · DP —
Phased GT:	0/1 (Unphased)

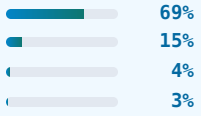
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.01 DL:0.98
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#1238754 ↗
gnomAD v4 AF:	3.8454%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.177 · Top 0.29%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.9 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	–	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency
Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000367797.9
cDNA (HGVS): c.883A>G
Protein (HGVS): p.Thr295Ala
VAF / Read Depth: VAF 0.34 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#2884751 ↗
gnomAD v4 AF: 0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)
Primary Driver: Cactus

42%
24%
23%
4%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs6025 p.Arg534Gln	MIS, p.Arg534Gln	Heterozygous	Unphased	drug response	Tier2	27.9	–	23.6 (Cactus)	– / 0.20	chr1:169549811 AF: 2.1432%

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000357077.9
cDNA (HGVS): c.11716C>T
Protein (HGVS): p.Arg3906Trp
VAF / Read Depth: VAF 0.52 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#18059 ↗
gnomAD v4 AF: 0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)
Primary Driver: Splicing

36%
25%
20%
16%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	–	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377801.8
cDNA (HGVS):	c.734C>G
Protein (HGVS):	p.Ser245Cys
VAF / Read Depth:	VAF 0.49 · DP –
Phased GT:	0/1 (Unphased)

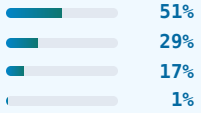
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#3214611 ↗
gnomAD v4 AF:	0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ELOVL7 ELOVL7 chr5:60,771,922 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables fatty acid elongase activity. Involved in fatty acid elongation, polyunsaturated fatty acid; fatty acid elongation, saturated fatty acid; and very long-chain fatty acid biosynthetic process. Located in endoplasmic reticulum. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs368309905 p.Ser79Tyr	MIS, p.Ser79Tyr	Heterozygous	Maternal	Not reviewed	Tier1	27.5	–	31.4 (AlphaMissense)	0.69 / 0.98	chr5:60771922 AF: 0.0210%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000508821.6
cDNA (HGVS):	c.236C>A
Protein (HGVS):	p.Ser79Tyr
VAF / Read Depth:	VAF 0.60 · DP –
Phased GT:	0|1 (Maternal)

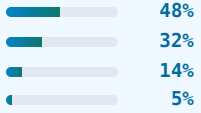
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0210%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 31.4 (Raw: 1.638 · Top 0.07%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	–	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

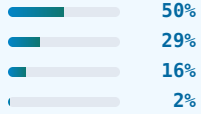
Transcript: ENST00000355112.8
cDNA (HGVS): c.2171T>C
Protein (HGVS): p.Val724Ala
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#567134 ↗
gnomAD v4 AF: 0.0054%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.1 (Raw: 1.282 · Top 0.25%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SLC9A2 SLC9A2 chr2:102,701,098 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the sodium-hydrogen exchanger (NHE) protein family. These proteins are involved in sodium-ion transport by exchanging intracellular hydrogen ions to external sodium ions and help in the regulation of cell pH and volume. The encoded protein is localized to the apical membrane and is involved in apical absorption of sodium. [provided by RefSeq, Jun 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371605244 p.Arg539Trp	MIS, p.Arg539Trp	Heterozygous	Maternal	Not reviewed	Tier1	26.3	–	22.8 (AlphaMissense)	0.46 / 0.54	chr2:102701098 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

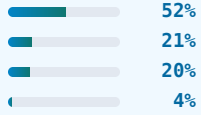
Transcript: ENST00000233969.3
cDNA (HGVS): c.1615C>T
Protein (HGVS): p.Arg539Trp
VAF / Read Depth: VAF 0.51 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.8 (Raw: 0.787 · Top 0.52%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IQCA1 IQCA1 chr2:236,338,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 26.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ATPases Associated with diverse cellular Activities (AAA) superfamily. Members of this superfamily, found in all organisms, participate in a large number of cellular processes and contain the ATPase module consisting of an alpha-beta-alpha core domain and the Walker A and B motifs of the P-loop NTPases. Alternative splicing results in multiple transcript variants that encode different protein isoforms. [provided by RefSeq, Jul 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs115960780 p.Cys693Arg	MIS, p.Cys693Arg	Heterozygous	Paternal	Not reviewed	Tier1	26.2	–	24.0 (AlphaMissense)	0.75 / 0.49	chr2:236338262 AF: 0.6424%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000409907.8
cDNA (HGVS):	c.2077T>C
Protein (HGVS):	p.Cys693Arg
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	1|0 (Paternal)

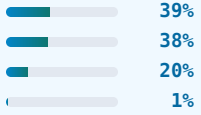
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.6424%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.0 (Raw: 0.948 · Top 0.40%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 26.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	–	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases

Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder

Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000373366.3
cDNA (HGVS):	c.223C>T
Protein (HGVS):	p.Arg75Cys
VAF / Read Depth:	VAF 0.37 · DP –
Phased GT:	0|1 (Maternal)

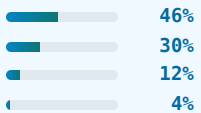
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#297191 ↗
gnomAD v4 AF:	0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

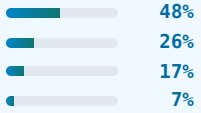
Transcript: **ENST00000377403.7**
cDNA (HGVS): **c.1004C>T**
Protein (HGVS): **p.Pro335Leu**
VAF / Read Depth: **VAF 0.43 · DP —**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.12 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#773105 ↗
gnomAD v4 AF: **0.2420%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.6** (Raw: 1.050 · Top 0.35%)
Primary Driver: **Cactus**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	—	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

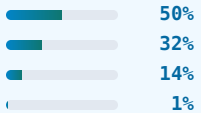
Transcript: **ENST00000361439.9**
cDNA (HGVS): **c.692G>A**
Protein (HGVS): **p.Cys231Tyr**
VAF / Read Depth: **VAF 0.53 · DP —**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#2358669 ↗
gnomAD v4 AF: **0.0064%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **30.6** (Raw: 1.597 · Top 0.09%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ACACB ACACB chr12:109,260,564 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acetyl-CoA carboxylase (ACC) is a complex multifunctional enzyme system. ACC is a biotin-containing enzyme which catalyzes the carboxylation of acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid synthesis. ACC-beta is thought to control fatty acid oxidation by means of the ability of malonyl-CoA to inhibit carnitine-palmitoyl-CoA transferase I, the rate-limiting step in fatty acid uptake and oxidation by mitochondria. ACC-beta may be involved in the regulation of fatty acid oxidation, rather than fatty acid biosynthesis. [provided by RefSeq, Oct 2022]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139082246 p.Pro2194Leu	MIS, p.Pro2194Leu	Heterozygous	Paternal	Not reviewed	Tier1	25.5	–	24.2 (Cactus)	0.70 / 0.36	chr12:109260564 AF: 0.0037%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

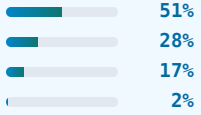
Transcript: **ENST00000338432.12**
cDNA (HGVS): **c.6581C>T**
Protein (HGVS): **p.Pro2194Leu**
VAF / Read Depth: **VAF 0.53 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0037%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.2** (Raw: 0.986 · Top 0.38%)
Primary Driver: **Cactus**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	–	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia
Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

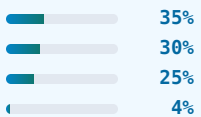
Transcript: **ENST00000268097.10**
cDNA (HGVS): **c.739C>T**
Protein (HGVS): **p.Arg247Trp**
VAF / Read Depth: **VAF 0.56 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#3922 ↗
gnomAD v4 AF: **0.2782%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **21.0** (Raw: 0.629 · Top 0.80%)
Primary Driver: **AlphaMissense**



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PAK3 PAK3 chrX:111,163,700 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a serine-threonine kinase and forms an activated complex with GTP-bound RAS-like (P21), CDC2 and RAC1. This protein may be necessary for dendritic development and for the rapid cytoskeletal reorganization in dendritic spines associated with synaptic plasticity. Defects in this gene are the cause of a non-syndromic form of X-linked intellectual disability. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
chrX:111163700 p.Thr247Pro	MIS, p.Thr247Pro	Heterozygous	Unphased	Not reviewed	Tier1	25.1	–	24.9 (AlphaMissense)	0.48 / 0.72	chrX:111163700 AF: 0.0000%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

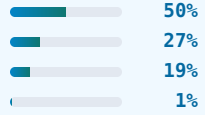
Transcript: **ENST00000372007.10**
cDNA (HGVS): **c.739A>C**
Protein (HGVS): **p.Thr247Pro**
VAF / Read Depth: **VAF 0.27 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0000%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.9** (Raw: 1.106 · Top 0.32%)
Primary Driver: **AlphaMissense**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	–	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy
Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

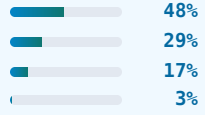
Transcript: **ENST00000294304.12**
cDNA (HGVS): **c.698T>A**
Protein (HGVS): **p.Val233Glu**
VAF / Read Depth: **VAF 0.55 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#866038 ↗
gnomAD v4 AF: **0.0007%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.4** (Raw: 1.189 · Top 0.29%)
Primary Driver: **AlphaMissense**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PARP3 PARP3 chr3:51,944,496 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PARP family. These enzymes modify nuclear proteins by poly-ADP-ribosylation, which is required for DNA repair, regulation of apoptosis, and maintenance of genomic stability. This gene encodes the poly(ADP-ribosyl)transferase 3, which is preferentially localized to the daughter centriole throughout the cell cycle. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs761611493 p.Arg140Gln	MIS, p.Arg140Gln	Heterozygous	Unphased	Not reviewed	Tier1	24.8	–	15.5 (Cactus)	0.40 / 0.14	chr3:51944496 AF: 0.0061%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

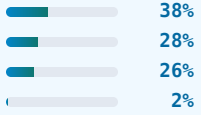
Transcript: ENST00000398755.8
cDNA (HGVS): c.419G>A
Protein (HGVS): p.Arg140Gln
VAF / Read Depth: VAF 0.56 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0061%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 15.5 (Raw: 0.340 · Top 2.83%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

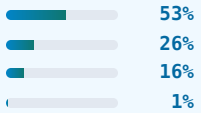
Transcript: ENST00000435607.3
cDNA (HGVS): c.1796A>G
Protein (HGVS): p.His599Arg
VAF / Read Depth: VAF 0.59 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#324537 ↗
gnomAD v4 AF: 0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	—	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4

Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000323670.14
cDNA (HGVS): c.124G>A
Protein (HGVS): p.Gly42Arg
VAF / Read Depth: VAF 0.40 · DP —
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#183298 ↗
gnomAD v4 AF: 0.0040%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.4 (Raw: 1.532 · Top 0.12%)
Primary Driver: AlphaMissense

48%
34%
15%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.3 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	—	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000278550.12
cDNA (HGVS): c.2804C>G
Protein (HGVS): p.Pro935Arg
VAF / Read Depth: VAF 0.44 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#708690 ↗
gnomAD v4 AF: 0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)
Primary Driver: AlphaMissense

40%
37%
13%
4%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs191549326 p.Pro485Ser	MIS, p.Pro485Ser	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier2	24.3	—	24.2 (Cactus)	0.36 / 0.36	chr11:78855981 AF: 0.4409%
rs192353604 c.-264-5528G>C	INT	Heterozygous	Maternal	Not reviewed	Tier3	17.3	—	10.7 (Cactus)	— / —	chr11:79221437 AF: 0.2210%

TTN TTN chr2:178,621,751 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 23.4 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a large abundant protein of striated muscle. The product of this gene is divided into two regions, a N-terminal I-band and a C-terminal A-band. The I-band, which is the elastic part of the molecule, contains two regions of tandem immunoglobulin domains on either side of a PEVK region that is rich in proline, glutamate, valine and lysine. The A-band, which is thought to act as a protein-ruler, contains a mixture of immunoglobulin and fibronectin repeats, and possesses kinase activity. An N-terminal Z-disc region and a C-terminal M-line region bind to the Z-line and M-line of the sarcomere, respectively, so that a single titin molecule spans half the length of a sarcomere. Titin also contains binding sites for muscle associated proteins so it serves as an adhesion template for the assembly of contractile machinery in muscle cells. It has also been identified as a structural protein for chromosomes. Alternative splicing of this gene results in multiple transcript variants. Considerable variability exists in the I-band, the M-line and the Z-disc regions of titin. Variability in the I-band region contributes to the differences in elasticity of different titin isoforms and, therefore, to the differences in elasticity of different muscle types. Mutations in this gene are associated with familial hypertrophic cardiomyopathy 9, and autoantibodies to titin are produced in patients with the autoimmune disease scleroderma. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72677222 c.45083-10A>G	INT	Heterozygous	Unphased	Benign/Likely benign	Tier1	17.8	1.00	26.0 (Splicing)	– / –	chr2:178621751 AF: 1.0145%

1. INHERITANCE & OMIM PATHOLOGY

Dilated cardiomyopathy 1G
Complex / Multifactorial

Autosomal recessive limb-girdle muscular dystrophy type 2J
Autosomal Recessive

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000589042.5
cDNA (HGVS):	c.45083-10A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.49 · DP –
Phased GT:	0/1 (Unphased)

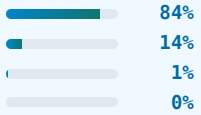
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:1.00 AL:0.78 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#46980 ↗
gnomAD v4 AF:	1.0145%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.0 (Raw: 1.269 · Top 0.25%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs13021201 p.Ser21716Leu	MIS, p.Ser21716Leu	Heterozygous	Maternal	Benign/Likely benign	Tier3	23.4	–	22.8 (Cactus)	0.36 / 0.15	chr2:178584404 AF: 3.6610%
rs33917087 p.Val2823Phe	MIS, p.Val2823Phe	Heterozygous	Maternal	Benign/Likely benign	Tier2	23.0	0.21	22.9 (AlphaMissense)	0.42 / 0.46	chr2:178770234 AF: 1.8566%

GRIA4 GRIA4 chr11:105,611,080 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 23.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Glutamate receptors are the predominant excitatory neurotransmitter receptors in the mammalian brain and are activated in a variety of normal neurophysiologic processes. These receptors are heteromeric protein complexes composed of multiple subunits, arranged to form ligand-gated ion channels. The classification of glutamate receptors is based on their activation by different pharmacologic agonists. The subunit encoded by this gene belongs to a family of AMPA (alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate)-sensitive glutamate receptors, and is subject to RNA editing (AGA->GGA; R->G). Alternative splicing of this gene results in transcript variants encoding different isoforms, which may vary in their signal transduction properties. Some haplotypes of this gene show a positive association with schizophrenia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs150577695 p.Gln28Arg	MIS, p.Gln28Arg	Heterozygous	Paternal	Not reviewed	Tier1	23.0	–	25.8 (AlphaMissense)	0.30 / 0.78	chr11:105611080 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000282499.10
cDNA (HGVS):	c.83A>G
Protein (HGVS):	p.Gln28Arg
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	1|0 (Paternal)

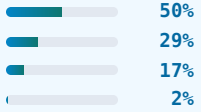
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.245 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB2 GJB2 chr13:20,189,481 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 20.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the gap junction protein family (gap junction beta 2) but is more commonly known as connexin 26. The gap junctions were first characterized by electron microscopy as regionally specialized structures on plasma membranes of contacting adherent cells. These structures were shown to consist of cell-to-cell channels that facilitate the transfer of ions and small molecules between cells. Connexins form hexameric channels in the plasma membrane which regulate passage of ions and small molecules between the cell and its environment or, when joined with another cell, form a dodecameric intercellular gap junction channel. The connexin proteins are grouped into alpha, beta, and gamma subfamilies. Mutations in this gene are responsible for as much as 50% of pre-lingual, recessive deafness. [provided by RefSeq, Mar 2026]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs35887622 p.Met34Thr	MIS, p.Met34Thr	Heterozygous	Unphased	Pathogenic	Tier1	20.9	–	23.6 (Cactus)	0.70 / 0.43	chr13:20189481 AF: 1.2041%

1. INHERITANCE & OMIM PATHOLOGY

Autosomal Recessive Deafness 1A / Dominant 3A

Autosomal Recessive Autosomal Dominant

OMIM #121011 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000382848.5
cDNA (HGVS):	c.101T>C
Protein (HGVS):	p.Met34Thr
VAF / Read Depth:	VAF 0.63 · DP –
Phased GT:	0/1 (Unphased)

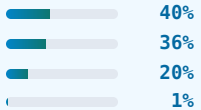
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	reviewed by expert panel
ClinVar VCV:	VCV#17000 ↗
gnomAD v4 AF:	1.2041%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.6 (Raw: 0.892 · Top 0.43%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TRAK1 TRAK1 chr3:42,115,352 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 17.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable GABA receptor binding activity and myosin binding activity. Involved in endosome to lysosome transport. Located in early endosome and mitochondrion. Implicated in developmental and epileptic encephalopathy 68. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs17220321 c.92-10068T>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	17.7	0.89	28.6 (Splicing)	— / —	chr3:42115352 AF: 1.5740%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000327628.10
cDNA (HGVS):	c.92-10068T>A
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.43 · DP —
Phased GT:	0/1 (Unphased)

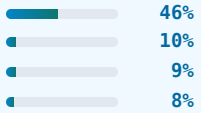
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.89 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	1.5740%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.6 (Raw: 1.480 · Top 0.14%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

5. SpliceAI Deep Learning Splicing Disruption — Top Genes (Δ ≥ 0.10)

Genes exhibiting cryptic splice-site activation or canonical donor/acceptor loss (Δ score ≥ 0.10).

Selected Scope: 23 Genes (23 Tier-1/Pathogenic)
Filtered from 93 matching genes in view

TTN TTN chr2:178,621,751 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 1.00 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a large abundant protein of striated muscle. The product of this gene is divided into two regions, a N-terminal I-band and a C-terminal A-band. The I-band, which is the elastic part of the molecule, contains two regions of tandem immunoglobulin domains on either side of a PEVK region that is rich in proline, glutamate, valine and lysine. The A-band, which is thought to act as a protein-ruler, contains a mixture of immunoglobulin and fibronectin repeats, and possesses kinase activity. An N-terminal Z-disc region and a C-terminal M-line region bind to the Z-line and M-line of the sarcomere, respectively, so that a single titin molecule spans half the length of a sarcomere. Titin also contains binding sites for muscle associated proteins so it serves as an adhesion template for the assembly of contractile machinery in muscle cells. It has also been identified as a structural protein for chromosomes. Alternative splicing of this gene results in multiple transcript variants. Considerable variability exists in the I-band, the M-line and the Z-disc regions of titin. Variability in the I-band region contributes to the differences in elasticity of different titin isoforms and, therefore, to the differences in elasticity of different muscle types. Mutations in this gene are associated with familial hypertrophic cardiomyopathy 9, and autoantibodies to titin are produced in patients with the autoimmune disease scleroderma. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72677222 c.45083-10A>G	INT	Heterozygous	Unphased	Benign/Likely benign	Tier1	17.8	1.00	26.0 (Splicing)	– / –	chr2:178621751 AF: 1.0145%

1. INHERITANCE & OMIM PATHOLOGY

Dilated cardiomyopathy 1G
Complex / Multifactorial

Autosomal recessive limb-girdle muscular dystrophy type 2J
Autosomal Recessive

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000589042.5
cDNA (HGVS):	c.45083-10A>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.49 · DP –
Phased GT:	0/1 (Unphased)

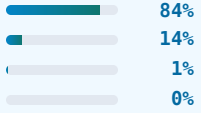
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:1.00 AL:0.78 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#46980 ↗
gnomAD v4 AF:	1.0145%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.0 (Raw: 1.269 · Top 0.25%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs33917087 p.Val2823Phe	MIS, p.Val2823Phe	Heterozygous	Maternal	Benign/Likely benign	Tier2	23.0	0.21	22.9 (AlphaMissense)	0.42 / 0.46	chr2:178770234 AF: 1.8566%

CBLIF CBLIF chr11:59,845,374 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.99 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the cobalamin transport protein family. It encodes a glycoprotein secreted by parietal cells of the gastric mucosa and is required for adequate absorption of vitamin B12. Vitamin B12 is necessary for erythrocyte maturation and mutations in this gene may lead to congenital pernicious anemia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147785187 c.79+1G>A	SPL	Heterozygous	Maternal	Pathogenic	Tier1	32.0	0.99	33.9 (Splicing)	— / —	chr11:59845374 AF: 0.0256%

1. INHERITANCE & OMIM PATHOLOGY

Intrinsic Factor Deficiency / Cobalamin Malabsorption

Autosomal Recessive OMIM #261000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000257248.3
cDNA (HGVS):	c.79+1G>A
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.47 · DP —
Phased GT:	0|1 (Maternal)

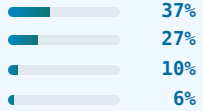
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.04 DL:0.99
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#439755 ↗
gnomAD v4 AF:	0.0256%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.9 (Raw: 1.892 · Top 0.04%)

Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BTN3A2 BTN3A2 chr6:26,370,605 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.98 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the immunoglobulin superfamily, which resides in the juxta-telomeric region of the major histocompatability class 1 locus and is clustered with the other family members on chromosome 6. The encoded protein may be involved in the adaptive immune response. Alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Jun 2013]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs58367598 c.715+2T>G	SPL	Heterozygous	Unphased	Benign	Tier1	28.7	0.98	25.4 (Splicing)	— / —	chr6:26370605 AF: 3.8454%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377708.7
cDNA (HGVS):	c.715+2T>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.64 · DP —
Phased GT:	0/1 (Unphased)

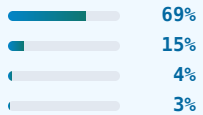
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.01 DL:0.98
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#1238754 ↗
gnomAD v4 AF:	3.8454%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.177 · Top 0.29%)

Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLA2G6 PLA2G6 chr22:38,115,172 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.95 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is an A2 phospholipase, a class of enzyme that catalyzes the release of fatty acids from phospholipids. The encoded protein may play a role in phospholipid remodelling, arachidonic acid release, leukotriene and prostaglandin synthesis, fas-mediated apoptosis, and transmembrane ion flux in glucose-stimulated B-cells. Several transcript variants encoding multiple isoforms have been described, but the full-length nature of only three of them have been determined to date. [provided by RefSeq, Dec 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147795054 c.2034+355G>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	1.7	0.95	25.0 (Splicing)	– / –	chr22:38115172 AF: 0.5672%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000332509.8
cDNA (HGVS):	c.2034+355G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.55 · DP –
Phased GT:	0/1 (Unphased)

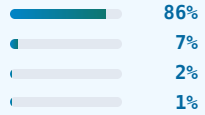
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.95 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.5672%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.0 (Raw: 1.119 · Top 0.32%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

COL23A1 COL23A1 chr5:178,262,215 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.91 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 COL23A1 is a member of the transmembrane collagens, a subfamily of the nonfibrillar collagens that contain a single pass hydrophobic transmembrane domain (Banyard et al., 2003 [PubMed 12644459]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs2973744 c.675+2T>C	SPL	Heterozygous	Unphased	Not reviewed	Tier1	33.0	0.91	26.3 (Splicing)	– / –	chr5:178262215 AF: 0.4032%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000390654.8
cDNA (HGVS):	c.675+2T>C
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.48 · DP –
Phased GT:	0/1 (Unphased)

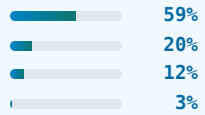
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.02 DL:0.91
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4314750 ↗
gnomAD v4 AF:	0.4032%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TRAK1 TRAK1 chr3:42,115,352 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.89 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable GABA receptor binding activity and myosin binding activity. Involved in endosome to lysosome transport. Located in early endosome and mitochondrion. Implicated in developmental and epileptic encephalopathy 68. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs17220321 c.92-10068T>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	17.7	0.89	28.6 (Splicing)	– / –	chr3:42115352 AF: 1.5740%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000327628.10
cDNA (HGVS):	c.92-10068T>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.43 · DP –
Phased GT:	0/1 (Unphased)

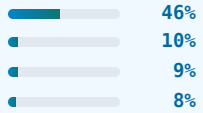
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.89 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	1.5740%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.6 (Raw: 1.480 · Top 0.14%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PKD1L3 PKD1L3 chr16:71,979,984 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.84 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polycystin protein family. The encoded protein contains 11 transmembrane domains, a latrophilin/CL-1-like GPCR proteolytic site (GPS) domain, and a polycystin-1, lipoxygenase, alpha-toxin (PLAT) domain. This protein may function as a component of cation channel pores.[provided by RefSeq, Apr 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72803747 c.1271+23G>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.4	0.84	10.4 (Splicing)	– / –	chr16:71979984 AF: 0.9916%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000620267.2
cDNA (HGVS):	c.1271+23G>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.39 · DP –
Phased GT:	0|1 (Maternal)

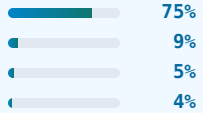
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.84 DL:0.02
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4415201 ↗
gnomAD v4 AF:	0.9916%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 10.4 (Raw: 0.168 · Top 9.03%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NPM1 NPM1 chr5:171,388,520 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.82 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is involved in several cellular processes, including centrosome duplication, protein chaperoning, and cell proliferation. The encoded phosphoprotein shuttles between the nucleolus, nucleus, and cytoplasm, chaperoning ribosomal proteins and core histones from the nucleus to the cytoplasm. This protein is also known to sequester the tumor suppressor ARF in the nucleolus, protecting it from degradation until it is needed. Mutations in this gene are associated with acute myeloid leukemia. Dozens of pseudogenes of this gene have been identified. [provided by RefSeq, Aug 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1770393339 c.58+514T>G	INT	Heterozygous	Unphased	Not reviewed	Tier1	8.7	0.82	19.6 (Splicing)	– / –	chr5:171388520 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000296930.10
cDNA (HGVS):	c.58+514T>G
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.49 · DP –
Phased GT:	0/1 (Unphased)

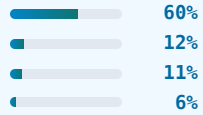
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.01 AL:0.00 DG:0.82 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.6 (Raw: 0.536 · Top 1.10%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

RAD51 RAD51 chr15:40,695,628 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.81 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the RAD51 protein family. RAD51 family members are highly similar to bacterial RecA and Saccharomyces cerevisiae Rad51, and are known to be involved in the homologous recombination and repair of DNA. This protein can interact with the ssDNA-binding protein RPA and RAD52, and it is thought to play roles in homologous pairing and strand transfer of DNA. This protein is also found to interact with BRCA1 and BRCA2, which may be important for the cellular response to DNA damage. BRCA2 is shown to regulate both the intracellular localization and DNA-binding ability of this protein. Loss of these controls following BRCA2 inactivation may be a key event leading to genomic instability and tumorigenesis. Multiple transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Aug 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs5030791 c.-3+203G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	5.5	0.81	19.1 (Splicing)	– / –	chr15:40695628 AF: 1.3014%

1. INHERITANCE & OMIM PATHOLOGY

Breast-Ovarian Cancer Predisposition

Autosomal Dominant OMIM #179570 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000267868.8
cDNA (HGVS):	c.-3+203G>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.60 · DP –
Phased GT:	1|0 (Paternal)

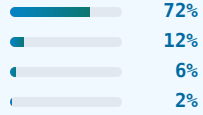
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.81 DL:0.00
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4401961 ↗
gnomAD v4 AF:	1.3014%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.1 (Raw: 0.510 · Top 1.23%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.76 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000304611.13
cDNA (HGVS): c.2362G>A
Protein (HGVS): p.Val788Met
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#556244 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

DNAH7 DNAH7 chr2:196,024,488 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.74 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 DNAH7 is a component of the inner dynein arm of ciliary axonemes (Zhang et al., 2002 [PubMed 11877439]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs555558470 p.Lys228=	SYN, p.Lys228=	Heterozygous	Paternal	Not reviewed	Tier1	12.9	0.74	23.0 (Splicing)	– / –	chr2:196024488 AF: 0.0021%

1. INHERITANCE & OMIM PATHOLOGY

Primary Ciliary Dyskinesia / Respiratory Clearance
Autosomal Recessive OMIM #610061 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

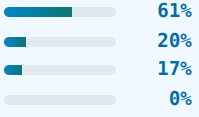
Transcript: ENST00000312428.11
cDNA (HGVS): c.684A>G
Protein (HGVS): p.Lys228=
VAF / Read Depth: VAF 0.65 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.74 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0021%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.0 (Raw: 0.808 · Top 0.50%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

EGFL7 EGFL7 chr9:136,670,809 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.73 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a secreted endothelial cell protein that contains two epidermal growth factor-like domains. The encoded protein may play a role in regulating vasculogenesis. This protein may be involved in the growth and proliferation of tumor cells. Alternate splicing results in multiple transcript variants. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs920684955 c.572-141C>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	0.6	0.73	16.7 (Splicing)	– / –	chr9:136670809 AF: 0.0147%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000308874.12
cDNA (HGVS):	c.572-141C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.58 · DP –
Phased GT:	0|1 (Maternal)

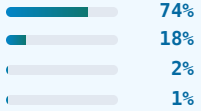
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.73 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0147%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 16.7 (Raw: 0.391 · Top 2.16%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TMEM270 TMEM270 chr7:73,865,329 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.72 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs73142829 p.Gly137Ser	MIS, p.Gly137Ser	Heterozygous	Paternal	Not reviewed	Tier1	0.0	0.72	20.1 (Splicing)	0.02 / 0.08	chr7:73865329 AF: 4.6362%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000320531.3
cDNA (HGVS):	c.409G>A
Protein (HGVS):	p.Gly137Ser
VAF / Read Depth:	VAF 0.47 · DP –
Phased GT:	1|0 (Paternal)

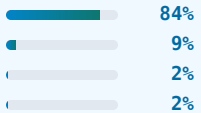
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.72 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4339331 ↗
gnomAD v4 AF:	4.6362%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.1 (Raw: 0.569 · Top 0.97%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

AZIN2 AZIN2 chr1:33,081,925 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.67 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the antizyme inhibitor family, which plays a role in cell growth and proliferation by maintaining polyamine homeostasis within the cell. Antizyme inhibitors are homologs of ornithine decarboxylase (ODC, the key enzyme in polyamine biosynthesis) that have lost the ability to decarboxylase ornithine; however, retain the ability to bind to antizymes. Antizymes negatively regulate intracellular polyamine levels by binding to ODC and targeting it for degradation, as well as by inhibiting polyamine uptake. Antizyme inhibitors function as positive regulators of polyamine levels by sequestering antizymes and neutralizing their effect. This gene encodes antizyme inhibitor 2, the second member of this gene family. Like antizyme inhibitor 1, antizyme inhibitor 2 interacts with all 3 antizymes and stimulates ODC activity and polyamine uptake. However, unlike antizyme inhibitor 1, which is ubiquitously expressed and localized in the nucleus and cytoplasm, antizyme inhibitor 2 is predominantly expressed in the brain and testis and localized in the endoplasmic reticulum-golgi intermediate compartment. Recent studies indicate that antizyme inhibitor 2 is also expressed in specific cell types in ovaries, adrenal glands and pancreas, and in mast cells. The exact function of this gene is not known, however, available data suggest its role in cell growth, spermiogenesis, vesicular trafficking and secretion. Accumulation of antizyme inhibitor 2 has also been observed in brains of patients with Alzheimer's disease. There has been confusion in literature and databases over the nomenclature of this gene, stemming from an earlier report that a human cDNA clone (identical to ODCp/AZIN2) had arginine decarboxylase (ADC) activity (PMID:14738999). Subsequent studies in human and mouse showed that antizyme inhibitor 2 was devoid of arginine decarboxylase activity (PMID:19956990). Alternatively spliced transcript variants have been described for this gene. [provided by RefSeq, Sep 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs190309518 c.-73+113G>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	0.4	0.67	13.2 (Splicing)	— / —	chr1:33081925 AF: 0.0490%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000294517.11
cDNA (HGVS):	c.-73+113G>T
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.40 · DP —
Phased GT:	0/1 (Unphased)

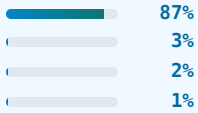
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.06 AL:0.00 DG:0.67 DL:0.00
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4417896 ↗
gnomAD v4 AF:	0.0490%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.2 (Raw: 0.256 · Top 4.79%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

JKAMP JKAMP chr14:59,502,424 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.65 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables ubiquitin protein ligase binding activity. Involved in ERAD pathway. Predicted to be located in endoplasmic reticulum membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs113112586 c.717+1157A>G	INT	Heterozygous	Paternal	Not reviewed	Tier1	13.9	0.65	13.5 (Splicing)	— / —	chr14:59502424 AF: 4.0038%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000616435.5
cDNA (HGVS):	c.717+1157A>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.63 · DP —
Phased GT:	1|0 (Paternal)

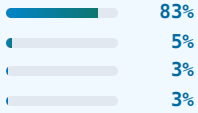
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.40 AL:0.00 DG:0.65 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	4.0038%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.5 (Raw: 0.265 · Top 4.50%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

INPP5K INPP5K chr17:1,508,258 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.63 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with 5-phosphatase activity toward polyphosphate inositol. The protein localizes to the cytosol in regions lacking actin stress fibers. It is thought that this protein may negatively regulate the actin cytoskeleton. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs74437970 c.555-32A>G	INT	Heterozygous	Maternal	Not reviewed	Tier1	7.5	0.63	13.0 (Splicing)	— / —	chr17:1508258 AF: 2.2461%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000421807.7
cDNA (HGVS):	c.555-32A>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.50 · DP —
Phased GT:	0|1 (Maternal)

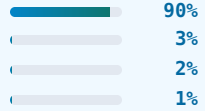
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.63 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4417749 ↗
gnomAD v4 AF:	2.2461%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.0 (Raw: 0.249 · Top 5.02%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.57 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs562989479 c.99+808G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	6.9	0.57	19.2 (Splicing)	— / —	chr1:165828740 AF: 0.3704%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367879.9
cDNA (HGVS):	c.99+808G>T
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.47 · DP —
Phased GT:	1|0 (Paternal)

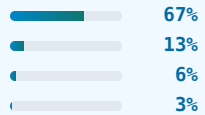
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.57 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.3704%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NEK10 NEK10 chr3:27,344,024 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.56 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein kinase activity. Involved in several processes, including mucociliary clearance; positive regulation of protein phosphorylation; and regulation of ERK1 and ERK2 cascade. Part of protein kinase complex. Implicated in primary ciliary dyskinesia 44. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs113156482 c.362+248G>A	INT	Heterozygous	Maternal	Not reviewed	Tier1	12.8	0.56	20.4 (Splicing)	– / –	chr3:27344024 AF: 1.7310%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000691995.1
cDNA (HGVS):	c.362+248G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.41 · DP –
Phased GT:	0|1 (Maternal)

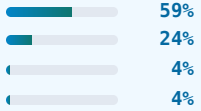
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.56 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	1.7310%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.4 (Raw: 0.584 · Top 0.92%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs1267845859 c.2971-433A>G	INT	Heterozygous	Paternal	Not reviewed	Tier2	0.8	0.47	14.6 (Splicing)	– / –	chr3:27132423 AF: 0.0007%

CNTN1 CNTN1 chr12:40,943,448 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.53 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the immunoglobulin superfamily. It is a glycosylphosphatidylinositol (GPI)-anchored neuronal membrane protein that functions as a cell adhesion molecule. It may play a role in the formation of axon connections in the developing nervous system. Multiple alternatively spliced transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139915282 c.1380-149C>T	INT	Heterozygous	Paternal	Likely benign	Tier1	10.2	0.53	12.1 (Splicing)	– / –	chr12:40943448 AF: 1.8289%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000551295.7
cDNA (HGVS):	c.1380-149C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.52 · DP –
Phased GT:	1|0 (Paternal)

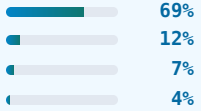
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.53 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#680275 ↗
gnomAD v4 AF:	1.8289%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 12.1 (Raw: 0.221 · Top 6.13%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs146742800 c.2420-1480G>A	INT	Heterozygous	Maternal	Not reviewed	Tier2	6.3	0.22	9.9 (Splicing)	– / –	chr12:41018857 AF: 1.5444%

CYP2E1 CYP2E1 chr10:133,520,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.51 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the cytochrome P450 superfamily of enzymes. The cytochrome P450 proteins are monooxygenases which catalyze many reactions involved in drug metabolism and synthesis of cholesterol, steroids and other lipids. This protein localizes to the endoplasmic reticulum and is induced by ethanol, the diabetic state, and starvation. The enzyme metabolizes both endogenous substrates, such as ethanol, acetone, and acetal, as well as exogenous substrates including benzene, carbon tetrachloride, ethylene glycol, and nitrosamines which are premutagens found in cigarette smoke. Due to its many substrates, this enzyme may be involved in such varied processes as gluconeogenesis, hepatic cirrhosis, diabetes, and cancer. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141625332 c.-118+195C>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	2.5	0.51	1.0 (Cactus)	– / –	chr10:133520755 AF: 1.0596%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000463117.6
cDNA (HGVS):	c.-118+195C>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.43 · DP –
Phased GT:	0/1 (Unphased)

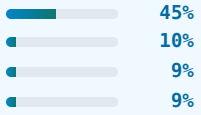
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.51 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	1.0596%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.0 (Raw: -0.042 · Top 80.14%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.22 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000357077.9
cDNA (HGVS):	c.11716C>T
Protein (HGVS):	p.Arg3906Trp
VAF / Read Depth:	VAF 0.52 · DP –
Phased GT:	0/1 (Unphased)

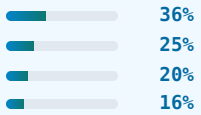
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#18059 ↗
gnomAD v4 AF:	0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.12 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000377403.7
cDNA (HGVS): c.1004C>T
Protein (HGVS): p.Pro335Leu
VAF / Read Depth: VAF 0.43 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.12 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#773105 ↗
gnomAD v4 AF: 0.2420%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.6 (Raw: 1.050 · Top 0.35%)
Primary Driver: Cactus

48%
26%
17%
7%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.11 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000233813.5
cDNA (HGVS): c.578G>A
Protein (HGVS): p.Arg193His
VAF / Read Depth: VAF 0.38 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.11 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0008%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)
Primary Driver: AlphaMissense

39%
39%
15%
3%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

6. AlphaMissense 3D Structural Pathogenicity — Top Missense Genes

Top protein-coding missense variants ranked by DeepMind AlphaFold-derived AlphaMissense structural pathogenicity probability.

Selected Scope: 24 Genes (24 Tier-1/Pathogenic)
Filtered from 334 matching genes in view

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.991 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	–	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000361439.9
cDNA (HGVS):	c.692G>A
Protein (HGVS):	p.Cys231Tyr
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

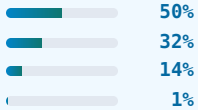
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2358669 ↗
gnomAD v4 AF:	0.0064%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.6 (Raw: 1.597 · Top 0.09%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ELOVL7 ELOVL7 chr5:60,771,922 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.982 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables fatty acid elongase activity. Involved in fatty acid elongation, polyunsaturated fatty acid; fatty acid elongation, saturated fatty acid; and very long-chain fatty acid biosynthetic process. Located in endoplasmic reticulum. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs368309905 p.Ser79Tyr	MIS, p.Ser79Tyr	Heterozygous	Maternal	Not reviewed	Tier1	27.5	–	31.4 (AlphaMissense)	0.69 / 0.98	chr5:60771922 AF: 0.0210%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000508821.6
cDNA (HGVS):	c.236C>A
Protein (HGVS):	p.Ser79Tyr
VAF / Read Depth:	VAF 0.60 · DP –
Phased GT:	0|1 (Maternal)

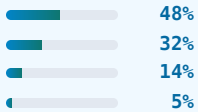
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0210%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 31.4 (Raw: 1.638 · Top 0.07%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.951 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

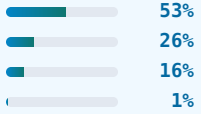
Transcript:	ENST00000435607.3
cDNA (HGVS):	c.1796A>G
Protein (HGVS):	p.His599Arg
VAF / Read Depth:	VAF 0.59 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#324537 ↗
gnomAD v4 AF:	0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.932 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	–	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases
Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder
Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

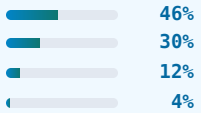
Transcript:	ENST00000373366.3
cDNA (HGVS):	c.223C>T
Protein (HGVS):	p.Arg75Cys
VAF / Read Depth:	VAF 0.37 · DP –
Phased GT:	0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#297191 ↗
gnomAD v4 AF:	0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.921 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	–	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4
Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

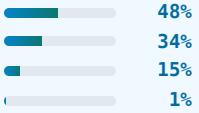
Transcript: ENST00000323670.14
cDNA (HGVS): c.124G>A
Protein (HGVS): p.Gly42Arg
VAF / Read Depth: VAF 0.40 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#183298 ↗
gnomAD v4 AF: 0.0040%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.4 (Raw: 1.532 · Top 0.12%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.818 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	–	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

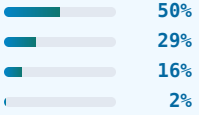
Transcript: ENST00000355112.8
cDNA (HGVS): c.2171T>C
Protein (HGVS): p.Val724Ala
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#567134 ↗
gnomAD v4 AF: 0.0054%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.1 (Raw: 1.282 · Top 0.25%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GRIA4 GRIA4 chr11:105,611,080 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.784 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Glutamate receptors are the predominant excitatory neurotransmitter receptors in the mammalian brain and are activated in a variety of normal neurophysiologic processes. These receptors are heteromeric protein complexes composed of multiple subunits, arranged to form ligand-gated ion channels. The classification of glutamate receptors is based on their activation by different pharmacologic agonists. The subunit encoded by this gene belongs to a family of AMPA (alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate)-sensitive glutamate receptors, and is subject to RNA editing (AGA->GGA; R->G). Alternative splicing of this gene results in transcript variants encoding different isoforms, which may vary in their signal transduction properties. Some haplotypes of this gene show a positive association with schizophrenia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs150577695 p.Gln28Arg	MIS, p.Gln28Arg	Heterozygous	Paternal	Not reviewed	Tier1	23.0	–	25.8 (AlphaMissense)	0.30 / 0.78	chr11:105611080 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

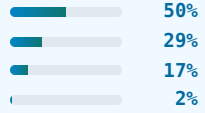
Transcript: ENST00000282499.10
cDNA (HGVS): c.83A>G
Protein (HGVS): p.Gln28Arg
VAF / Read Depth: VAF 0.56 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.245 · Top 0.26%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PTDSS2 PTDSS2 chr11:488,272 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.778 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene catalyzes the conversion of phosphatidylethanolamine to phosphatidylserine, a structural membrane phospholipid that functions in cell signaling, blood coagulation, and apoptosis. The encoded enzyme also has a high affinity for docosahexaenoic acid (DHA) and can use it to make DHA-containing phosphatidylserine. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs908582716 p.Glu232Gly	MIS, p.Glu232Gly	Heterozygous	Unphased	Not reviewed	Tier1	31.0	–	25.8 (AlphaMissense)	0.77 / 0.78	chr11:488272 AF: 0.0096%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

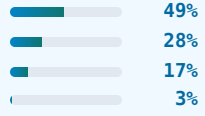
Transcript: ENST00000308020.6
cDNA (HGVS): c.695A>G
Protein (HGVS): p.Glu232Gly
VAF / Read Depth: VAF 0.68 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0096%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.251 · Top 0.26%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.727 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	–	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000294304.12
cDNA (HGVS): c.698T>A
Protein (HGVS): p.Val233Glu
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#866038 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.189 · Top 0.29%)
Primary Driver: AlphaMissense

48%
29%
17%
3%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PAK3 PAK3 chrX:111,163,700 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.717 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a serine-threonine kinase and forms an activated complex with GTP-bound RAS-like (P21), CDC2 and RAC1. This protein may be necessary for dendritic development and for the rapid cytoskeletal reorganization in dendritic spines associated with synaptic plasticity. Defects in this gene are the cause of a non-syndromic form of X-linked intellectual disability. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
chrX:111163700 p.Thr247Pro	MIS, p.Thr247Pro	Heterozygous	Unphased	Not reviewed	Tier1	25.1	–	24.9 (AlphaMissense)	0.48 / 0.72	chrX:111163700 AF: 0.0000%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000372007.10
cDNA (HGVS): c.739A>C
Protein (HGVS): p.Thr247Pro
VAF / Read Depth: VAF 0.27 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0000%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.9 (Raw: 1.106 · Top 0.32%)
Primary Driver: AlphaMissense

50%
27%
19%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.684 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	–	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

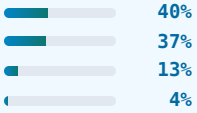
Transcript: ENST00000278550.12
cDNA (HGVS): c.2804C>G
Protein (HGVS): p.Pro935Arg
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#708690 ↗
gnomAD v4 AF: 0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs191549326 p.Pro485Ser	MIS, p.Pro485Ser	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier2	24.3	–	24.2 (Cactus)	0.36 / 0.36	chr11:78855981 AF: 0.4409%

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.673 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

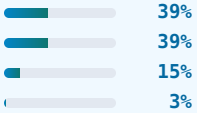
Transcript: ENST00000233813.5
cDNA (HGVS): c.578G>A
Protein (HGVS): p.Arg193His
VAF / Read Depth: VAF 0.38 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.11 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0008%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.616 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000304611.13
cDNA (HGVS): c.2362G>A
Protein (HGVS): p.Val788Met
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#556244 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SLC9A2 SLC9A2 chr2:102,701,098 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.544 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the sodium-hydrogen exchanger (NHE) protein family. These proteins are involved in sodium-ion transport by exchanging intracellular hydrogen ions to external sodium ions and help in the regulation of cell pH and volume. The encoded protein is localized to the apical membrane and is involved in apical absorption of sodium. [provided by RefSeq, Jun 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371605244 p.Arg539Trp	MIS, p.Arg539Trp	Heterozygous	Maternal	Not reviewed	Tier1	26.3	–	22.8 (AlphaMissense)	0.46 / 0.54	chr2:102701098 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

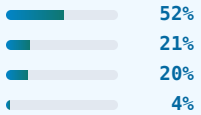
Transcript: ENST00000233969.3
cDNA (HGVS): c.1615C>T
Protein (HGVS): p.Arg539Trp
VAF / Read Depth: VAF 0.51 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.8 (Raw: 0.787 · Top 0.52%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IQCA1 IQCA1 chr2:236,338,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.493 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ATPases Associated with diverse cellular Activities (AAA) superfamily. Members of this superfamily, found in all organisms, participate in a large number of cellular processes and contain the ATPase module consisting of an alpha-beta-alpha core domain and the Walker A and B motifs of the P-loop NTPases. Alternative splicing results in multiple transcript variants that encode different protein isoforms. [provided by RefSeq, Jul 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs115960780 p.Cys693Arg	MIS, p.Cys693Arg	Heterozygous	Paternal	Not reviewed	Tier1	26.2	–	24.0 (AlphaMissense)	0.75 / 0.49	chr2:236338262 AF: 0.6424%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

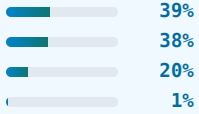
Transcript: ENST00000409907.8
cDNA (HGVS): c.2077T>C
Protein (HGVS): p.Cys693Arg
VAF / Read Depth: VAF 0.56 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.6424%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.0 (Raw: 0.948 · Top 0.40%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB2 GJB2 chr13:20,189,481 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.433 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the gap junction protein family (gap junction beta 2) but is more commonly known as connexin 26. The gap junctions were first characterized by electron microscopy as regionally specialized structures on plasma membranes of contacting adherent cells. These structures were shown to consist of cell-to-cell channels that facilitate the transfer of ions and small molecules between cells. Connexins form hexameric channels in the plasma membrane which regulate passage of ions and small molecules between the cell and its environment or, when joined with another cell, form a dodecameric intercellular gap junction channel. The connexin proteins are grouped into alpha, beta, and gamma subfamilies. Mutations in this gene are responsible for as much as 50% of pre-lingual, recessive deafness. [provided by RefSeq, Mar 2026]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs35887622 p.Met34Thr	MIS, p.Met34Thr	Heterozygous	Unphased	Pathogenic	Tier1	20.9	–	23.6 (Cactus)	0.70 / 0.43	chr13:20189481 AF: 1.2041%

1. INHERITANCE & OMIM PATHOLOGY

Autosomal Recessive Deafness 1A / Dominant 3A

Autosomal Recessive Autosomal Dominant

OMIM #121011 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

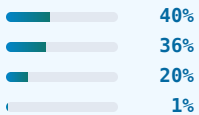
Transcript: ENST00000382848.5
cDNA (HGVS): c.101T>C
Protein (HGVS): p.Met34Thr
VAF / Read Depth: VAF 0.63 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: reviewed by expert panel
ClinVar VCV: VCV#17000 ↗
gnomAD v4 AF: 1.2041%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.6 (Raw: 0.892 · Top 0.43%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.375 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	—	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377801.8
cDNA (HGVS):	c.734C>G
Protein (HGVS):	p.Ser245Cys
VAF / Read Depth:	VAF 0.49 · DP —
Phased GT:	0/1 (Unphased)

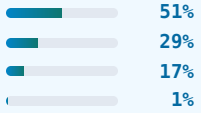
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#3214611 ↗
gnomAD v4 AF:	0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.369 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377403.7
cDNA (HGVS):	c.1004C>T
Protein (HGVS):	p.Pro335Leu
VAF / Read Depth:	VAF 0.43 · DP —
Phased GT:	0/1 (Unphased)

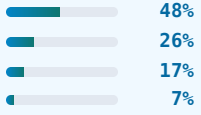
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.12 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#773105 ↗
gnomAD v4 AF:	0.2420%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.6 (Raw: 1.050 · Top 0.35%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ACACB ACACB chr12:109,260,564 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.363 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acetyl-CoA carboxylase (ACC) is a complex multifunctional enzyme system. ACC is a biotin-containing enzyme which catalyzes the carboxylation of acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid synthesis. ACC-beta is thought to control fatty acid oxidation by means of the ability of malonyl-CoA to inhibit carnitine-palmitoyl-CoA transferase I, the rate-limiting step in fatty acid uptake and oxidation by mitochondria. ACC-beta may be involved in the regulation of fatty acid oxidation, rather than fatty acid biosynthesis. [provided by RefSeq, Oct 2022]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139082246 p.Pro2194Leu	MIS, p.Pro2194Leu	Heterozygous	Paternal	Not reviewed	Tier1	25.5	–	24.2 (Cactus)	0.70 / 0.36	chr12:109260564 AF: 0.0037%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

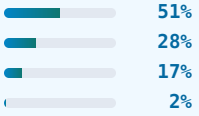
Transcript: **ENST00000338432.12**
cDNA (HGVS): **c.6581C>T**
Protein (HGVS): **p.Pro2194Leu**
VAF / Read Depth: **VAF 0.53 · DP –**
Phased GT: **1|0 (Paternal)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0037%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.2** (Raw: 0.986 · Top 0.38%)
Primary Driver: **Cactus**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.299 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

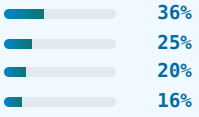
Transcript: **ENST00000357077.9**
cDNA (HGVS): **c.11716C>T**
Protein (HGVS): **p.Arg3906Trp**
VAF / Read Depth: **VAF 0.52 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.22 AL:0.16 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, conflicting classifications**
ClinVar VCV: VCV#18059 ↗
gnomAD v4 AF: **0.1250%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.2** (Raw: 1.146 · Top 0.30%)
Primary Driver: **Splicing**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.296 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	–	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia

Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000268097.10
cDNA (HGVS):	c.739C>T
Protein (HGVS):	p.Arg247Trp
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	0/1 (Unphased)

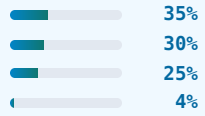
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#3922 ↗
gnomAD v4 AF:	0.2782%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.629 · Top 0.80%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.205 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	–	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367797.9
cDNA (HGVS):	c.883A>G
Protein (HGVS):	p.Thr295Ala
VAF / Read Depth:	VAF 0.34 · DP –
Phased GT:	0|1 (Maternal)

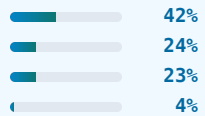
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2884751 ↗
gnomAD v4 AF:	0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs6025 p.Arg534Gln	MIS, p.Arg534Gln	Heterozygous	Unphased	drug response	Tier2	27.9	–	23.6 (Cactus)	– / 0.20	chr1:169549811 AF: 2.1432%

PARP3 PARP3 chr3:51,944,496 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.143 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PARP family. These enzymes modify nuclear proteins by poly-ADP-ribosylation, which is required for DNA repair, regulation of apoptosis, and maintenance of genomic stability. This gene encodes the poly(ADP-ribosyl)transferase 3, which is preferentially localized to the daughter centriole throughout the cell cycle. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs761611493 p.Arg140Gln	MIS, p.Arg140Gln	Heterozygous	Unphased	Not reviewed	Tier1	24.8	–	15.5 (Cactus)	0.40 / 0.14	chr3:51944496 AF: 0.0061%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000398755.8
cDNA (HGVS): c.419G>A
Protein (HGVS): p.Arg140Gln
VAF / Read Depth: VAF 0.56 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0061%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 15.5 (Raw: 0.340 · Top 2.83%)
Primary Driver: Cactus

38%
28%
26%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TMEM270 TMEM270 chr7:73,865,329 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.080 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs73142829 p.Gly137Ser	MIS, p.Gly137Ser	Heterozygous	Paternal	Not reviewed	Tier1	0.0	0.72	20.1 (Splicing)	0.02 / 0.08	chr7:73865329 AF: 4.6362%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000320531.3
cDNA (HGVS): c.409G>A
Protein (HGVS): p.Gly137Ser
VAF / Read Depth: VAF 0.47 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.72 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4339331 ↗
gnomAD v4 AF: 4.6362%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.1 (Raw: 0.569 · Top 0.97%)
Primary Driver: Splicing

84%
9%
2%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

7. REVEL Ensembled Missense Pathogenicity — Top Genes (Score ≥ 0.20)

Top missense variants ranked by Rare Exome Variant Ensemble Learner (REVEL) pathogenicity score.

Selected Scope: 23 Genes (23 Tier-1/Pathogenic)
Filtered from 133 matching genes in view

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.943 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	–	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases
Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder
Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

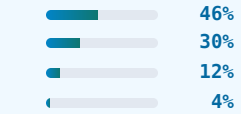
Transcript:	ENST00000373366.3
cDNA (HGVS):	c.223C>T
Protein (HGVS):	p.Arg75Cys
VAF / Read Depth:	VAF 0.37 · DP –
Phased GT:	0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#297191 ↗
gnomAD v4 AF:	0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.937 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	–	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

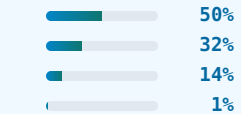
Transcript:	ENST00000361439.9
cDNA (HGVS):	c.692G>A
Protein (HGVS):	p.Cys231Tyr
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2358669 ↗
gnomAD v4 AF:	0.0064%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.6 (Raw: 1.597 · Top 0.09%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.925 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	–	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia
Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000268097.10
cDNA (HGVS):	c.739C>T
Protein (HGVS):	p.Arg247Trp
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#3922 ↗
gnomAD v4 AF:	0.2782%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.629 · Top 0.80%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.925 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000435607.3
cDNA (HGVS):	c.1796A>G
Protein (HGVS):	p.His599Arg
VAF / Read Depth:	VAF 0.59 · DP –
Phased GT:	0/1 (Unphased)

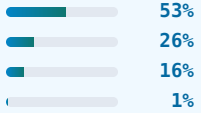
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#324537 ↗
gnomAD v4 AF:	0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.921 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000304611.13
cDNA (HGVS): c.2362G>A
Protein (HGVS): p.Val788Met
VAF / Read Depth: VAF 0.58 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#556244 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.903 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	—	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy
Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

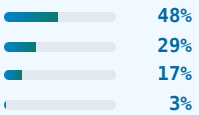
Transcript: ENST00000294304.12
cDNA (HGVS): c.698T>A
Protein (HGVS): p.Val233Glu
VAF / Read Depth: VAF 0.55 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#866038 ↗
gnomAD v4 AF: 0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.4 (Raw: 1.189 · Top 0.29%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.899 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377403.7
cDNA (HGVS):	c.1004C>T
Protein (HGVS):	p.Pro335Leu
VAF / Read Depth:	VAF 0.43 · DP —
Phased GT:	0/1 (Unphased)

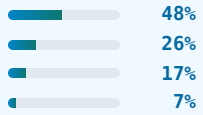
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.12 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#773105 ↗
gnomAD v4 AF:	0.2420%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.6 (Raw: 1.050 · Top 0.35%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.778 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	—	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4

Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000323670.14
cDNA (HGVS):	c.124G>A
Protein (HGVS):	p.Gly42Arg
VAF / Read Depth:	VAF 0.40 · DP —
Phased GT:	0|1 (Maternal)

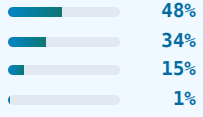
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#183298 ↗
gnomAD v4 AF:	0.0040%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.4 (Raw: 1.532 · Top 0.12%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PTDSS2 PTDSS2 chr11:488,272 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.768 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene catalyzes the conversion of phosphatidylethanolamine to phosphatidylserine, a structural membrane phospholipid that functions in cell signaling, blood coagulation, and apoptosis. The encoded enzyme also has a high affinity for docosahexaenoic acid (DHA) and can use it to make DHA-containing phosphatidylserine. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs908582716 p.Glu232Gly	MIS, p.Glu232Gly	Heterozygous	Unphased	Not reviewed	Tier1	31.0	—	25.8 (AlphaMissense)	0.77 / 0.78	chr11:488272 AF: 0.0096%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000308020.6
cDNA (HGVS):	c.695A>G
Protein (HGVS):	p.Glu232Gly
VAF / Read Depth:	VAF 0.68 · DP —
Phased GT:	0/1 (Unphased)

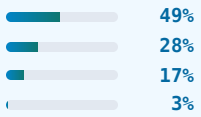
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0096%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.251 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IQCA1 IQCA1 chr2:236,338,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.754 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ATPases Associated with diverse cellular Activities (AAA) superfamily. Members of this superfamily, found in all organisms, participate in a large number of cellular processes and contain the ATPase module consisting of an alpha-beta-alpha core domain and the Walker A and B motifs of the P-loop NTPases. Alternative splicing results in multiple transcript variants that encode different protein isoforms. [provided by RefSeq, Jul 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs115960780 p.Cys693Arg	MIS, p.Cys693Arg	Heterozygous	Paternal	Not reviewed	Tier1	26.2	—	24.0 (AlphaMissense)	0.75 / 0.49	chr2:236338262 AF: 0.6424%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000409907.8
cDNA (HGVS):	c.2077T>C
Protein (HGVS):	p.Cys693Arg
VAF / Read Depth:	VAF 0.56 · DP —
Phased GT:	1|0 (Paternal)

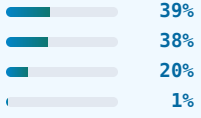
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.6424%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.0 (Raw: 0.948 · Top 0.40%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.722 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000357077.9
cDNA (HGVS): c.11716C>T
Protein (HGVS): p.Arg3906Trp
VAF / Read Depth: VAF 0.52 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#18059 ↗
gnomAD v4 AF: 0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)
Primary Driver: Splicing

36%
25%
20%
16%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB2 GJB2 chr13:20,189,481 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.702 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the gap junction protein family (gap junction beta 2) but is more commonly known as connexin 26. The gap junctions were first characterized by electron microscopy as regionally specialized structures on plasma membranes of contacting adherent cells. These structures were shown to consist of cell-to-cell channels that facilitate the transfer of ions and small molecules between cells. Connexins form hexameric channels in the plasma membrane which regulate passage of ions and small molecules between the cell and its environment or, when joined with another cell, form a dodecameric intercellular gap junction channel. The connexin proteins are grouped into alpha, beta, and gamma subfamilies. Mutations in this gene are responsible for as much as 50% of pre-lingual, recessive deafness. [provided by RefSeq, Mar 2026]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs35887622 p.Met34Thr	MIS, p.Met34Thr	Heterozygous	Unphased	Pathogenic	Tier1	20.9	—	23.6 (Cactus)	0.70 / 0.43	chr13:20189481 AF: 1.2041%

1. INHERITANCE & OMIM PATHOLOGY

Autosomal Recessive Deafness 1A / Dominant 3A
Autosomal Recessive Autosomal Dominant
OMIM #121011 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000382848.5
cDNA (HGVS): c.101T>C
Protein (HGVS): p.Met34Thr
VAF / Read Depth: VAF 0.63 · DP —
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: reviewed by expert panel
ClinVar VCV: VCV#17000 ↗
gnomAD v4 AF: 1.2041%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.6 (Raw: 0.892 · Top 0.43%)
Primary Driver: Cactus

40%
36%
20%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.700 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	–	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000367797.9
cDNA (HGVS): c.883A>G
Protein (HGVS): p.Thr295Ala
VAF / Read Depth: VAF 0.34 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#2884751 ↗
gnomAD v4 AF: 0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)
Primary Driver: Cactus

42%
24%
23%
4%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ACACB ACACB chr12:109,260,564 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.699 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acetyl-CoA carboxylase (ACC) is a complex multifunctional enzyme system. ACC is a biotin-containing enzyme which catalyzes the carboxylation of acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid synthesis. ACC-beta is thought to control fatty acid oxidation by means of the ability of malonyl-CoA to inhibit carnitine-palmitoyl-CoA transferase I, the rate-limiting step in fatty acid uptake and oxidation by mitochondria. ACC-beta may be involved in the regulation of fatty acid oxidation, rather than fatty acid biosynthesis. [provided by RefSeq, Oct 2022]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139082246 p.Pro2194Leu	MIS, p.Pro2194Leu	Heterozygous	Paternal	Not reviewed	Tier1	25.5	–	24.2 (Cactus)	0.70 / 0.36	chr12:109260564 AF: 0.0037%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000338432.12
cDNA (HGVS): c.6581C>T
Protein (HGVS): p.Pro2194Leu
VAF / Read Depth: VAF 0.53 · DP –
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0037%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.2 (Raw: 0.986 · Top 0.38%)
Primary Driver: Cactus

51%
28%
17%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ELOVL7 ELOVL7 chr5:60,771,922 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.695 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables fatty acid elongase activity. Involved in fatty acid elongation, polyunsaturated fatty acid; fatty acid elongation, saturated fatty acid; and very long-chain fatty acid biosynthetic process. Located in endoplasmic reticulum. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs368309905 p.Ser79Tyr	MIS, p.Ser79Tyr	Heterozygous	Maternal	Not reviewed	Tier1	27.5	–	31.4 (AlphaMissense)	0.69 / 0.98	chr5:60771922 AF: 0.0210%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000508821.6
cDNA (HGVS):	c.236C>A
Protein (HGVS):	p.Ser79Tyr
VAF / Read Depth:	VAF 0.60 · DP –
Phased GT:	0|1 (Maternal)

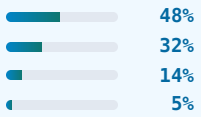
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0210%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 31.4 (Raw: 1.638 · Top 0.07%)

Primary Driver: AlphaMissense



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.493 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	–	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000278550.12
cDNA (HGVS):	c.2804C>G
Protein (HGVS):	p.Pro935Arg
VAF / Read Depth:	VAF 0.44 · DP –
Phased GT:	0/1 (Unphased)

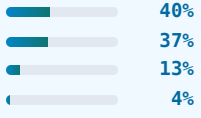
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#708690 ↗
gnomAD v4 AF:	0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)

Primary Driver: AlphaMissense



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs191549326 p.Pro485Ser	MIS, p.Pro485Ser	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier2	24.3	–	24.2 (Cactus)	0.36 / 0.36	chr11:78855981 AF: 0.4409%

PAK3 PAK3 chrX:111,163,700 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.485 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a serine-threonine kinase and forms an activated complex with GTP-bound RAS-like (P21), CDC2 and RAC1. This protein may be necessary for dendritic development and for the rapid cytoskeletal reorganization in dendritic spines associated with synaptic plasticity. Defects in this gene are the cause of a non-syndromic form of X-linked intellectual disability. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
chrX:111163700 p.Thr247Pro	MIS, p.Thr247Pro	Heterozygous	Unphased	Not reviewed	Tier1	25.1	–	24.9 (AlphaMissense)	0.48 / 0.72	chrX:111163700 AF: 0.0000%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

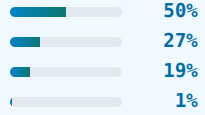
Transcript: **ENST00000372007.10**
cDNA (HGVS): **c.739A>C**
Protein (HGVS): **p.Thr247Pro**
VAF / Read Depth: **VAF 0.27 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0000%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.9** (Raw: 1.106 · Top 0.32%)
Primary Driver: **AlphaMissense**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.473 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	–	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

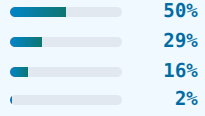
Transcript: **ENST00000355112.8**
cDNA (HGVS): **c.2171T>C**
Protein (HGVS): **p.Val724Ala**
VAF / Read Depth: **VAF 0.44 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#567134 ↗
gnomAD v4 AF: **0.0054%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **26.1** (Raw: 1.282 · Top 0.25%)
Primary Driver: **AlphaMissense**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SLC9A2 SLC9A2 chr2:102,701,098 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.459 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the sodium-hydrogen exchanger (NHE) protein family. These proteins are involved in sodium-ion transport by exchanging intracellular hydrogen ions to external sodium ions and help in the regulation of cell pH and volume. The encoded protein is localized to the apical membrane and is involved in apical absorption of sodium. [provided by RefSeq, Jun 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371605244 p.Arg539Trp	MIS, p.Arg539Trp	Heterozygous	Maternal	Not reviewed	Tier1	26.3	–	22.8 (AlphaMissense)	0.46 / 0.54	chr2:102701098 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000233969.3
cDNA (HGVS):	c.1615C>T
Protein (HGVS):	p.Arg539Trp
VAF / Read Depth:	VAF 0.51 · DP –
Phased GT:	0|1 (Maternal)

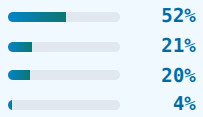
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.8 (Raw: 0.787 · Top 0.52%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.438 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000233813.5
cDNA (HGVS):	c.578G>A
Protein (HGVS):	p.Arg193His
VAF / Read Depth:	VAF 0.38 · DP –
Phased GT:	0/1 (Unphased)

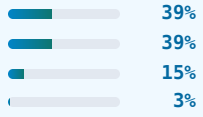
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.11 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0008%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.417 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	—	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377801.8
cDNA (HGVS):	c.734C>G
Protein (HGVS):	p.Ser245Cys
VAF / Read Depth:	VAF 0.49 · DP —
Phased GT:	0/1 (Unphased)

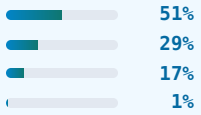
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#3214611 ↗
gnomAD v4 AF:	0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PARP3 PARP3 chr3:51,944,496 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.401 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PARP family. These enzymes modify nuclear proteins by poly-ADP-ribosylation, which is required for DNA repair, regulation of apoptosis, and maintenance of genomic stability. This gene encodes the poly(ADP-ribosyl)transferase 3, which is preferentially localized to the daughter centriole throughout the cell cycle. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs761611493 p.Arg140Gln	MIS, p.Arg140Gln	Heterozygous	Unphased	Not reviewed	Tier1	24.8	—	15.5 (Cactus)	0.40 / 0.14	chr3:51944496 AF: 0.0061%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000398755.8
cDNA (HGVS):	c.419G>A
Protein (HGVS):	p.Arg140Gln
VAF / Read Depth:	VAF 0.56 · DP —
Phased GT:	0/1 (Unphased)

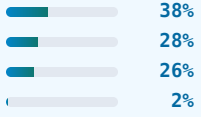
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0061%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 15.5 (Raw: 0.340 · Top 2.83%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GRIA4 GRIA4 chr11:105,611,080 Multisystem 🔥 Tier-1 / Pathogenic Gene REVEL: 0.298 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Glutamate receptors are the predominant excitatory neurotransmitter receptors in the mammalian brain and are activated in a variety of normal neurophysiologic processes. These receptors are heteromeric protein complexes composed of multiple subunits, arranged to form ligand-gated ion channels. The classification of glutamate receptors is based on their activation by different pharmacologic agonists. The subunit encoded by this gene belongs to a family of AMPA (alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate)-sensitive glutamate receptors, and is subject to RNA editing (AGA->GGA; R->G). Alternative splicing of this gene results in transcript variants encoding different isoforms, which may vary in their signal transduction properties. Some haplotypes of this gene show a positive association with schizophrenia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs150577695 p.Gln28Arg	MIS, p.Gln28Arg	Heterozygous	Paternal	Not reviewed	Tier1	23.0	–	25.8 (AlphaMissense)	0.30 / 0.78	chr11:105611080 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000282499.10
cDNA (HGVS):	c.83A>G
Protein (HGVS):	p.Gln28Arg
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	1|0 (Paternal)

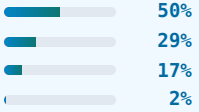
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.245 · Top 0.26%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

8. Rare & Private Variants (gnomAD AF ≤ 1%) — Tier-1 & Top High-Impact Genes

Rare coding and regulatory variants (gnomAD v4 allele frequency ≤ 1.0%), prioritizing Tier-1 / Pathogenic and high-CADD/AVI rare alleles.

Selected Scope: 32 Genes (32 Tier-1/Pathogenic)
Filtered from 627 matching genes in view

PEX6 PEX6 chr6:42,966,044 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0007% · CADD 35.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the AAA (ATPases associated with diverse cellular activities) family of ATPases. This member is a predominantly cytoplasmic protein, which plays a direct role in peroxisomal protein import and is required for PTS1 (peroxisomal targeting signal 1, a C-terminal tripeptide of the sequence ser-lys-leu) receptor activity. Mutations in this gene cause peroxisome biogenesis disorders of complementation group 4 and complementation group 6. Several transcript variants encoding different isoforms have been found for this gene. [provided by RefSeq, Oct 2015]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs267608240 p.Val788Met	MIS, p.Val788Met	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	35.0	0.76	32.3 (Cactus)	0.92 / 0.62	chr6:42966044 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Peroxisome biogenesis disorder 4A (Zellweger)
Complex / Multifactorial OMIM #614862; 614863 ↗

Peroxisome biogenesis disorder 4B
Complex / Multifactorial OMIM #614862; 614863 ↗

Zellweger spectrum disorders
Complex / Multifactorial OMIM #614862; 614863 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

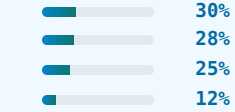
Transcript:	ENST00000304611.13
cDNA (HGVS):	c.2362G>A
Protein (HGVS):	p.Val788Met
VAF / Read Depth:	VAF 0.58 · DP —
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.07 DL:0.76
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#556244 ↗
gnomAD v4 AF:	0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.3 (Raw: 1.716 · Top 0.06%)
Primary Driver: Cactus



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

COL23A1 COL23A1 chr5:178,262,215 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0000% · CADD 33.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 COL23A1 is a member of the transmembrane collagens, a subfamily of the nonfibrillar collagens that contain a single pass hydrophobic transmembrane domain (Banyard et al., 2003 [PubMed 12644459]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs2973744 c.675+2T>C	SPL	Heterozygous	Unphased	Not reviewed	Tier1	33.0	0.91	26.3 (Splicing)	– / –	chr5:178262215 AF: 0.4032%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

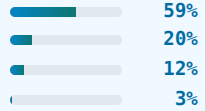
Transcript: **ENST00000390654.8**
cDNA (HGVS): **c.675+2T>C**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.48 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.02 DL:0.91**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4314750 ↗
gnomAD v4 AF: **0.4032%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs5873597 c.1297-277_1297-276del	INT	Heterozygous	Unknown	Not reviewed	Tier3	–	–	–	– / –	chr5:178246735 AF: 0.0000%

IGFBP5 IGFBP5 chr2:216,678,221 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0008% · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables insulin-like growth factor I binding activity and receptor ligand activity. Involved in several processes, including cellular response to cAMP; regulation of smooth muscle cell migration; and regulation of smooth muscle cell proliferation. Part of insulin-like growth factor ternary complex. Biomarker of pulmonary fibrosis. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs755706987 p.Arg193His	MIS, p.Arg193His	Heterozygous	Unphased	Not reviewed	Tier1	32.0	0.11	26.4 (AlphaMissense)	0.44 / 0.67	chr2:216678221 AF: 0.0008%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

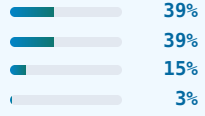
Transcript: **ENST00000233813.5**
cDNA (HGVS): **c.578G>A**
Protein (HGVS): **p.Arg193His**
VAF / Read Depth: **VAF 0.38 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.11 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0008%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.4 (Raw: 1.325 · Top 0.23%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

CBLIF CBLIF chr11:59,845,374 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0256% · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the cobalamin transport protein family. It encodes a glycoprotein secreted by parietal cells of the gastric mucosa and is required for adequate absorption of vitamin B12. Vitamin B12 is necessary for erythrocyte maturation and mutations in this gene may lead to congenital pernicious anemia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147785187 c.79+1G>A	SPL	Heterozygous	Maternal	Pathogenic	Tier1	32.0	0.99	33.9 (Splicing)	– / –	chr11:59845374 AF: 0.0256%

1. INHERITANCE & OMIM PATHOLOGY

Intrinsic Factor Deficiency / Cobalamin Malabsorption

Autosomal Recessive OMIM #261000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

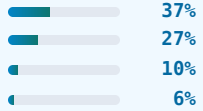
Transcript: ENST00000257248.3
cDNA (HGVS): c.79+1G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.47 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.04 DL:0.99
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#439755 ↗
gnomAD v4 AF: 0.0256%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.9 (Raw: 1.892 · Top 0.04%)
Primary Driver: Splicing



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PTDSS2 PTDSS2 chr11:488,272 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0096% · CADD 31.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene catalyzes the conversion of phosphatidylethanolamine to phosphatidylserine, a structural membrane phospholipid that functions in cell signaling, blood coagulation, and apoptosis. The encoded enzyme also has a high affinity for docosahexaenoic acid (DHA) and can use it to make DHA-containing phosphatidylserine. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs908582716 p.Glu232Gly	MIS, p.Glu232Gly	Heterozygous	Unphased	Not reviewed	Tier1	31.0	–	25.8 (AlphaMissense)	0.77 / 0.78	chr11:488272 AF: 0.0096%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

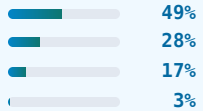
Transcript: ENST00000308020.6
cDNA (HGVS): c.695A>G
Protein (HGVS): p.Glu232Gly
VAF / Read Depth: VAF 0.68 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0096%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.251 · Top 0.26%)
Primary Driver: AlphaMissense



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs142613557 c.367+914G>A	INT	Heterozygous	Unphased	Not reviewed	Tier2	0.9	0.24	0.4 (Cactus)	– / –	chr11:474891 AF: 0.6829%

PLCXD3 PLCXD3 chr5:41,381,904 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0013% · CADD 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable phosphoric diester hydrolase activity. Predicted to be involved in lipid catabolic process and signal transduction. Predicted to be located in cytoplasm. Predicted to be active in glutamatergic synapse. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1210486304 p.Ser245Cys	MIS, p.Ser245Cys	Heterozygous	Unphased	Uncertain significance	Tier1	27.6	—	24.1 (Cactus)	0.42 / 0.38	chr5:41381904 AF: 0.0013%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377801.8
cDNA (HGVS):	c.734C>G
Protein (HGVS):	p.Ser245Cys
VAF / Read Depth:	VAF 0.49 · DP —
Phased GT:	0/1 (Unphased)

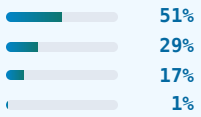
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#3214611 ↗
gnomAD v4 AF:	0.0013%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.1 (Raw: 0.974 · Top 0.39%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ANK2 ANK2 chr4:113,373,306 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.1250% · CADD 27.7 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the ankyrin family of proteins that link the integral membrane proteins to the underlying spectrin-actin cytoskeleton. Ankyrins play key roles in activities such as cell motility, activation, proliferation, contact and the maintenance of specialized membrane domains. Most ankyrins are typically composed of three structural domains: an amino-terminal domain containing multiple ankyrin repeats; a central region with a highly conserved spectrin binding domain; and a carboxy-terminal regulatory domain which is the least conserved and subject to variation. The protein encoded by this gene is required for targeting and stability of Na/Ca exchanger 1 in cardiomyocytes. Mutations in this gene cause long QT syndrome 4 and cardiac arrhythmia syndrome. Multiple transcript variants encoding different isoforms have been described. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121912706 p.Arg3906Trp	MIS, p.Arg3906Trp	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	27.7	0.22	25.2 (Splicing)	0.72 / 0.30	chr4:113373306 AF: 0.1250%

1. INHERITANCE & OMIM PATHOLOGY

ANK2-related disorder
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiovascular phenotype
Complex / Multifactorial OMIM #600919; 115000 ↗

Cardiac arrhythmia
Complex / Multifactorial OMIM #600919; 115000 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000357077.9
cDNA (HGVS):	c.11716C>T
Protein (HGVS):	p.Arg3906Trp
VAF / Read Depth:	VAF 0.52 · DP —
Phased GT:	0/1 (Unphased)

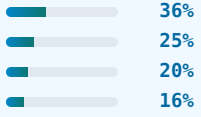
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.22 AL:0.16 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#18059 ↗
gnomAD v4 AF:	0.1250%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.2 (Raw: 1.146 · Top 0.30%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

rs183491249 c.85-40398G>A	INT	Heterozygous	Maternal	Not reviewed	Tier3	5.2	—	4.2 (Cactus)	— / —	chr4:113134018 AF: 0.4297%

BLM BLM chr15:90,765,392 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0054% · CADD 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The Bloom syndrome is an autosomal recessive disorder characterized by growth deficiency, microcephaly and immunodeficiency among others. It is caused by homozygous or compound heterozygous mutation in the gene encoding DNA helicase RecQ protein on chromosome 15q26. This Bloom-associated helicase unwinds a variety of DNA substrates including Holliday junction, and is involved in several pathways contributing to the maintenance of genome stability. Identification of pathogenic Bloom variants is required for heterozygote testing in at-risk families. [provided by RefSeq, May 2020]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs750687788 p.Val724Ala	MIS, p.Val724Ala	Heterozygous	Unphased	Uncertain significance	Tier1	27.5	–	26.1 (AlphaMissense)	0.47 / 0.82	chr15:90765392 AF: 0.0054%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary cancer-predisposing syndrome
Complex / Multifactorial OMIM #210900 ↗

Bloom syndrome
Complex / Multifactorial OMIM #210900 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

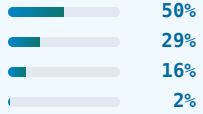
Transcript: ENST00000355112.8
cDNA (HGVS): c.2171T>C
Protein (HGVS): p.Val724Ala
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#567134 ↗
gnomAD v4 AF: 0.0054%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.1 (Raw: 1.282 · Top 0.25%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ELOVL7 ELOVL7 chr5:60,771,922 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0210% · CADD 27.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables fatty acid elongase activity. Involved in fatty acid elongation, polyunsaturated fatty acid; fatty acid elongation, saturated fatty acid; and very long-chain fatty acid biosynthetic process. Located in endoplasmic reticulum. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs368309905 p.Ser79Tyr	MIS, p.Ser79Tyr	Heterozygous	Maternal	Not reviewed	Tier1	27.5	–	31.4 (AlphaMissense)	0.69 / 0.98	chr5:60771922 AF: 0.0210%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

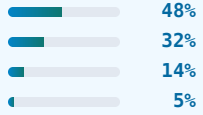
Transcript: ENST00000508821.6
cDNA (HGVS): c.236C>A
Protein (HGVS): p.Ser79Tyr
VAF / Read Depth: VAF 0.60 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0210%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 31.4 (Raw: 1.638 · Top 0.07%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

SLC9A2 SLC9A2 chr2:102,701,098 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0002% · CADD 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the sodium-hydrogen exchanger (NHE) protein family. These proteins are involved in sodium-ion transport by exchanging intracellular hydrogen ions to external sodium ions and help in the regulation of cell pH and volume. The encoded protein is localized to the apical membrane and is involved in apical absorption of sodium. [provided by RefSeq, Jun 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371605244 p.Arg539Trp	MIS, p.Arg539Trp	Heterozygous	Maternal	Not reviewed	Tier1	26.3	–	22.8 (AlphaMissense)	0.46 / 0.54	chr2:102701098 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

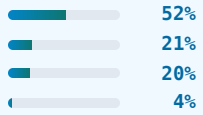
Transcript: ENST00000233969.3
cDNA (HGVS): c.1615C>T
Protein (HGVS): p.Arg539Trp
VAF / Read Depth: VAF 0.51 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.8 (Raw: 0.787 · Top 0.52%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GJB3 GJB3 chr1:34,784,985 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0441% · CADD 26.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the connexin gene family. The encoded protein is a component of gap junctions, which are composed of arrays of intercellular channels that provide a route for the diffusion of low molecular weight materials from cell to cell. Mutations in this gene can cause non-syndromic deafness or erythrokeratodermia variabilis, a skin disorder. Alternative splicing results in multiple transcript variants encoding the same protein. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs370476720 p.Arg75Cys	MIS, p.Arg75Cys	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	26.2	–	32.4 (AlphaMissense)	0.94 / 0.93	chr1:34784985 AF: 0.0441%

1. INHERITANCE & OMIM PATHOLOGY

Inborn genetic diseases
Complex / Multifactorial OMIM #133200 ↗

GJB3-related disorder
Complex / Multifactorial OMIM #133200 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

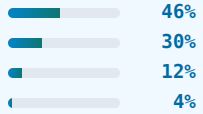
Transcript: ENST00000373366.3
cDNA (HGVS): c.223C>T
Protein (HGVS): p.Arg75Cys
VAF / Read Depth: VAF 0.37 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#297191 ↗
gnomAD v4 AF: 0.0441%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 32.4 (Raw: 1.725 · Top 0.06%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

IQCA1 IQCA1 chr2:236,338,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.4781% · CADD 26.2 | Showing 2 of 2 variants | OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ATPases Associated with diverse cellular Activities (AAA) superfamily. Members of this superfamily, found in all organisms, participate in a large number of cellular processes and contain the ATPase module consisting of an alpha-beta-alpha core domain and the Walker A and B motifs of the P-loop NTPases. Alternative splicing results in multiple transcript variants that encode different protein isoforms. [provided by RefSeq, Jul 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs115960780 p.Cys693Arg	MIS, p.Cys693Arg	Heterozygous	Paternal	Not reviewed	Tier1	26.2	–	24.0 (AlphaMissense)	0.75 / 0.49	chr2:236338262 AF: 0.6424%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000409907.8
cDNA (HGVS):	c.2077T>C
Protein (HGVS):	p.Cys693Arg
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	1|0 (Paternal)

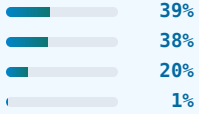
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.6424%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.0 (Raw: 0.948 · Top 0.40%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1553608365 c.1870-2866_1870-2865insTATATATATAGA	INT	Homozygous	N/A	Not reviewed	Tier3	–	–	–	– / –	chr2:236347509 AF: 0.4781%

NSMCE1 NSMCE1 chr16:27,225,755 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0064% · CADD 25.6 | Showing 1 of 1 variants | OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables protein dimerization activity and ubiquitin protein ligase activity. Involved in DNA damage response. Located in nucleoplasm. Part of Smc5-Smc6 complex. [provided by Alliance of Genome Resources, Jul 2025]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs774697446 p.Cys231Tyr	MIS, p.Cys231Tyr	Heterozygous	Paternal	Uncertain significance	Tier1	25.6	–	30.6 (AlphaMissense)	0.94 / 0.99	chr16:27225755 AF: 0.0064%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000361439.9
cDNA (HGVS):	c.692G>A
Protein (HGVS):	p.Cys231Tyr
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

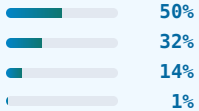
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2358669 ↗
gnomAD v4 AF:	0.0064%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.6 (Raw: 1.597 · Top 0.09%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

ACACB ACACB chr12:109,260,564 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0037% · CADD 25.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acetyl-CoA carboxylase (ACC) is a complex multifunctional enzyme system. ACC is a biotin-containing enzyme which catalyzes the carboxylation of acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid synthesis. ACC-beta is thought to control fatty acid oxidation by means of the ability of malonyl-CoA to inhibit carnitine-palmitoyl-CoA transferase I, the rate-limiting step in fatty acid uptake and oxidation by mitochondria. ACC-beta may be involved in the regulation of fatty acid oxidation, rather than fatty acid biosynthesis. [provided by RefSeq, Oct 2022]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs139082246 p.Pro2194Leu	MIS, p.Pro2194Leu	Heterozygous	Paternal	Not reviewed	Tier1	25.5	–	24.2 (Cactus)	0.70 / 0.36	chr12:109260564 AF: 0.0037%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000338432.12
cDNA (HGVS):	c.6581C>T
Protein (HGVS):	p.Pro2194Leu
VAF / Read Depth:	VAF 0.53 · DP –
Phased GT:	1|0 (Paternal)

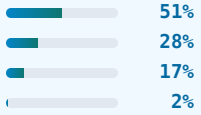
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0037%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.2 (Raw: 0.986 · Top 0.38%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

H6PD H6PD chr1:9,262,317 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.2420% · CADD 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 There are 2 forms of glucose-6-phosphate dehydrogenase. G form is X-linked and H form, encoded by this gene, is autosomally linked. This H form shows activity with other hexose-6-phosphates, especially galactose-6-phosphate, whereas the G form is specific for glucose-6-phosphate. Both forms are present in most tissues, but H form is not found in red cells. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs140631516 p.Pro335Leu	MIS, p.Pro335Leu	Heterozygous	Unphased	Likely benign	Tier1	25.7	0.12	24.6 (Cactus)	0.90 / 0.37	chr1:9262317 AF: 0.2420%

1. INHERITANCE & OMIM PATHOLOGY

H6PD-related disorder

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000377403.7
cDNA (HGVS):	c.1004C>T
Protein (HGVS):	p.Pro335Leu
VAF / Read Depth:	VAF 0.43 · DP –
Phased GT:	0/1 (Unphased)

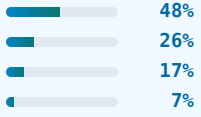
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.12 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#773105 ↗
gnomAD v4 AF:	0.2420%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.6 (Raw: 1.050 · Top 0.35%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PAK3 PAK3 chrX:111,163,700 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0000% · CADD 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a serine-threonine kinase and forms an activated complex with GTP-bound RAS-like (P21), CDC2 and RAC1. This protein may be necessary for dendritic development and for the rapid cytoskeletal reorganization in dendritic spines associated with synaptic plasticity. Defects in this gene are the cause of a non-syndromic form of X-linked intellectual disability. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
chrX:111163700 p.Thr247Pro	MIS, p.Thr247Pro	Heterozygous	Unphased	Not reviewed	Tier1	25.1	–	24.9 (AlphaMissense)	0.48 / 0.72	chrX:111163700 AF: 0.0000%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

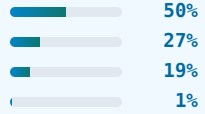
Transcript: **ENST00000372007.10**
cDNA (HGVS): **c.739A>C**
Protein (HGVS): **p.Thr247Pro**
VAF / Read Depth: **VAF 0.27 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0000%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.9** (Raw: 1.106 · Top 0.32%)
Primary Driver: **AlphaMissense**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

LRP5 LRP5 chr11:68,363,758 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0007% · CADD 24.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a transmembrane low-density lipoprotein receptor that binds and internalizes ligands in the process of receptor-mediated endocytosis. This protein also acts as a co-receptor with Frizzled protein family members for transducing signals by Wnt proteins and was originally cloned on the basis of its association with type 1 diabetes mellitus in humans. This protein plays a key role in skeletal homeostasis and many bone density related diseases are caused by mutations in this gene. Mutations in this gene also cause familial exudative vitreoretinopathy. Alternative splicing results in multiple transcript variants. [provided by RefSeq, May 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1177484229 p.Val233Glu	MIS, p.Val233Glu	Heterozygous	Unphased	Uncertain significance	Tier1	24.9	–	25.4 (AlphaMissense)	0.90 / 0.73	chr11:68363758 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Retinal dystrophy
Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

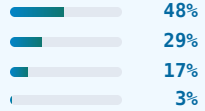
Transcript: **ENST00000294304.12**
cDNA (HGVS): **c.698T>A**
Protein (HGVS): **p.Val233Glu**
VAF / Read Depth: **VAF 0.55 · DP –**
Phased GT: **0/1 (Unphased)**

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#866038 ↗
gnomAD v4 AF: **0.0007%**

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.4** (Raw: 1.189 · Top 0.29%)
Primary Driver: **AlphaMissense**



Ref vs. Alt Track Predictions ↗ LitVar2 ↗

HEXA HEXA chr15:72,350,584 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.2782% · CADD 25.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the glycosyl hydrolase 20 family of proteins. The encoded preproprotein is proteolytically processed to generate the alpha subunit of the lysosomal enzyme beta-hexosaminidase. This enzyme, together with the cofactor GM2 activator protein, catalyzes the degradation of the ganglioside GM2, and other molecules containing terminal N-acetyl hexosamines. Mutations in this gene lead to an accumulation of GM2 ganglioside in neurons, the underlying cause of neurodegenerative disorders termed the GM2 gangliosidoses, including Tay-Sachs disease (GM2-gangliosidosis type I). Alternative splicing results in multiple transcript variants, at least one of which encodes a preproprotein that is proteolytically processed. [provided by RefSeq, Jan 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs121907970 p.Arg247Trp	MIS, p.Arg247Trp	Heterozygous	Unphased	Benign/Likely benign|other	Tier1	25.1	–	21.0 (AlphaMissense)	0.93 / 0.30	chr15:72350584 AF: 0.2782%

1. INHERITANCE & OMIM PATHOLOGY

Hereditary ataxia

Complex / Multifactorial OMIM #272800 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000268097.10
cDNA (HGVS):	c.739C>T
Protein (HGVS):	p.Arg247Trp
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, multiple submitters, no conflicts
ClinVar VCV:	VCV#3922 ↗
gnomAD v4 AF:	0.2782%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.629 · Top 0.80%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PARP3 PARP3 chr3:51,944,496 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0061% · CADD 24.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PARP family. These enzymes modify nuclear proteins by poly-ADP-ribosylation, which is required for DNA repair, regulation of apoptosis, and maintenance of genomic stability. This gene encodes the poly(ADP-ribosyl)transferase 3, which is preferentially localized to the daughter centriole throughout the cell cycle. Alternatively spliced transcript variants encoding different isoforms have been identified. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs761611493 p.Arg140Gln	MIS, p.Arg140Gln	Heterozygous	Unphased	Not reviewed	Tier1	24.8	–	15.5 (Cactus)	0.40 / 0.14	chr3:51944496 AF: 0.0061%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000398755.8
cDNA (HGVS):	c.419G>A
Protein (HGVS):	p.Arg140Gln
VAF / Read Depth:	VAF 0.56 · DP –
Phased GT:	0/1 (Unphased)

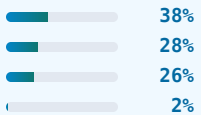
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0061%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 15.5 (Raw: 0.340 · Top 2.83%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

F5 F5 chr1:169,556,715 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0004% · CADD 24.7 | Showing 1 of 1 variants | OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an essential cofactor of the blood coagulation cascade. This factor circulates in plasma, and is converted to the active form by the release of the activation peptide by thrombin during coagulation. This generates a heavy chain and a light chain which are held together by calcium ions. The activated protein is a cofactor that participates with activated coagulation factor X to activate prothrombin to thrombin. Defects in this gene result in either an autosomal recessive hemorrhagic diathesis or an autosomal dominant form of thrombophilia, which is known as activated protein C resistance. [provided by RefSeq, Oct 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs371760153 p.Thr295Ala	MIS, p.Thr295Ala	Heterozygous	Maternal	Uncertain significance	Tier1	24.7	–	21.0 (Cactus)	0.70 / 0.20	chr1:169556715 AF: 0.0004%

1. INHERITANCE & OMIM PATHOLOGY

Congenital factor V deficiency

Complex / Multifactorial

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367797.9
cDNA (HGVS):	c.883A>G
Protein (HGVS):	p.Thr295Ala
VAF / Read Depth:	VAF 0.34 · DP –
Phased GT:	0|1 (Maternal)

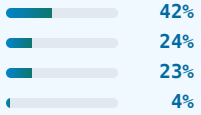
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, single submitter
ClinVar VCV:	VCV#2884751 ↗
gnomAD v4 AF:	0.0004%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.0 (Raw: 0.630 · Top 0.80%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

C19orf12 C19orf12 chr19:29,708,290 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0040% · CADD 24.3 | Showing 1 of 1 variants | OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a small transmembrane protein. Mutations in this gene are a cause of neurodegeneration with brain iron accumulation-4 (NBIA4), but the specific function of the encoded protein is unknown. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Dec 2011]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs200133991 p.Gly42Arg	MIS, p.Gly42Arg	Heterozygous	Maternal	Conflicting classifications of pathogenicity	Tier1	24.3	–	29.4 (AlphaMissense)	0.78 / 0.92	chr19:29708290 AF: 0.0040%

1. INHERITANCE & OMIM PATHOLOGY

Neurodegeneration with Brain Iron Accumulation 4

Autosomal Recessive OMIM #614297 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000323670.14
cDNA (HGVS):	c.124G>A
Protein (HGVS):	p.Gly42Arg
VAF / Read Depth:	VAF 0.40 · DP –
Phased GT:	0|1 (Maternal)

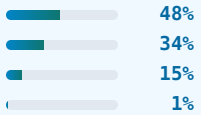
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided, conflicting classifications
ClinVar VCV:	VCV#183298 ↗
gnomAD v4 AF:	0.0040%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.4 (Raw: 1.532 · Top 0.12%)

Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

TENM4 TENM4 chr11:78,738,523 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.1990% · CADD 24.3 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene plays a role in establishing proper neuronal connectivity during development. Defects in this gene have been associated with hereditary essential tremor-5. [provided by RefSeq, Oct 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs141706152 p.Pro935Arg	MIS, p.Pro935Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.0	–	26.8 (AlphaMissense)	0.49 / 0.68	chr11:78738523 AF: 0.5497%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

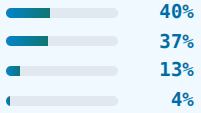
Transcript: ENST00000278550.12
cDNA (HGVS): c.2804C>G
Protein (HGVS): p.Pro935Arg
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#708690 ↗
gnomAD v4 AF: 0.5497%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.8 (Raw: 1.365 · Top 0.21%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs181299543 c.-264-22173G>C	INT	Heterozygous	Paternal	Not reviewed	Tier3	17.0	–	13.3 (Cactus)	– / –	chr11:79238082 AF: 0.1990%
rs192353604 c.-264-5528G>C	INT	Heterozygous	Maternal	Not reviewed	Tier3	17.3	–	10.7 (Cactus)	– / –	chr11:79221437 AF: 0.2210%

SCN4A SCN4A chr17:63,961,242 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.4126% · CADD 24.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Voltage-gated sodium channels are transmembrane glycoprotein complexes composed of a large alpha subunit with 24 transmembrane domains and one or more regulatory beta subunits. They are responsible for the generation and propagation of action potentials in neurons and muscle. This gene encodes one member of the sodium channel alpha subunit gene family. It is expressed in skeletal muscle, and mutations in this gene have been linked to several myotonia and periodic paralysis disorders. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs187401185 p.His599Arg	MIS, p.His599Arg	Heterozygous	Unphased	Conflicting classifications of pathogenicity	Tier1	24.4	–	27.8 (AlphaMissense)	0.93 / 0.95	chr17:63961242 AF: 0.4126%

1. INHERITANCE & OMIM PATHOLOGY

Paramyotonia congenita of Von Eulenburg
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Hyperkalemic periodic paralysis
Complex / Multifactorial OMIM #168300; 613345; 614198; 608390; 170500 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

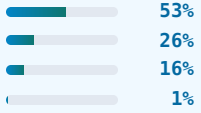
Transcript: ENST00000435607.3
cDNA (HGVS): c.1796A>G
Protein (HGVS): p.His599Arg
VAF / Read Depth: VAF 0.59 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#324537 ↗
gnomAD v4 AF: 0.4126%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

GRIA4 GRIA4 chr11:105,611,080 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0002% · CADD 23.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Glutamate receptors are the predominant excitatory neurotransmitter receptors in the mammalian brain and are activated in a variety of normal neurophysiologic processes. These receptors are heteromeric protein complexes composed of multiple subunits, arranged to form ligand-gated ion channels. The classification of glutamate receptors is based on their activation by different pharmacologic agonists. The subunit encoded by this gene belongs to a family of AMPA (alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate)-sensitive glutamate receptors, and is subject to RNA editing (AGA->GGA; R->G). Alternative splicing of this gene results in transcript variants encoding different isoforms, which may vary in their signal transduction properties. Some haplotypes of this gene show a positive association with schizophrenia. [provided by RefSeq, Jul 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs150577695 p.Gln28Arg	MIS, p.Gln28Arg	Heterozygous	Paternal	Not reviewed	Tier1	23.0	—	25.8 (AlphaMissense)	0.30 / 0.78	chr11:105611080 AF: 0.0002%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

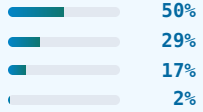
Transcript: ENST00000282499.10
cDNA (HGVS): c.83A>G
Protein (HGVS): p.Gln28Arg
VAF / Read Depth: VAF 0.56 · DP —
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0002%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.8 (Raw: 1.245 · Top 0.26%)
Primary Driver: AlphaMissense



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

DNAH7 DNAH7 chr2:196,024,488 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0021% · CADD 12.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 DNAH7 is a component of the inner dynein arm of ciliary axonemes (Zhang et al., 2002 [PubMed 11877439]).[supplied by OMIM, Mar 2008]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs555558470 p.Lys228=	SYN, p.Lys228=	Heterozygous	Paternal	Not reviewed	Tier1	12.9	0.74	23.0 (Splicing)	— / —	chr2:196024488 AF: 0.0021%

1. INHERITANCE & OMIM PATHOLOGY

Primary Ciliary Dyskinesia / Respiratory Clearance
Autosomal Recessive OMIM #610061 ↗

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

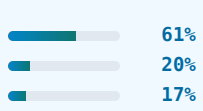
Transcript: ENST00000312428.11
cDNA (HGVS): c.684A>G
Protein (HGVS): p.Lys228=
VAF / Read Depth: VAF 0.65 · DP —
Phased GT: 1|0 (Paternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.74 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0021%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.0 (Raw: 0.808 · Top 0.50%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

NPM1 NPM1 chr5:171,388,520 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0007% · CADD 8.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is involved in several cellular processes, including centrosome duplication, protein chaperoning, and cell proliferation. The encoded phosphoprotein shuttles between the nucleolus, nucleus, and cytoplasm, chaperoning ribosomal proteins and core histones from the nucleus to the cytoplasm. This protein is also known to sequester the tumor suppressor ARF in the nucleolus, protecting it from degradation until it is needed. Mutations in this gene are associated with acute myeloid leukemia. Dozens of pseudogenes of this gene have been identified. [provided by RefSeq, Aug 2017]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs1770393339 c.58+514T>G	INT	Heterozygous	Unphased	Not reviewed	Tier1	8.7	0.82	19.6 (Splicing)	— / —	chr5:171388520 AF: 0.0007%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000296930.10
cDNA (HGVS):	c.58+514T>G
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.49 · DP —
Phased GT:	0/1 (Unphased)

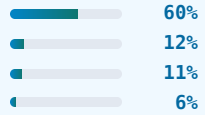
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.01 AL:0.00 DG:0.82 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0007%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.6 (Raw: 0.536 · Top 1.10%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.3704% · CADD 6.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs562989479 c.99+808G>T	INT	Heterozygous	Paternal	Not reviewed	Tier1	6.9	0.57	19.2 (Splicing)	— / —	chr1:165828740 AF: 0.3704%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000367879.9
cDNA (HGVS):	c.99+808G>T
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.47 · DP —
Phased GT:	1|0 (Paternal)

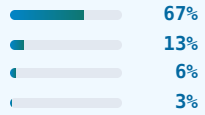
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.57 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.3704%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLB1 PLB1 chr2:28,461,598 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0012% · CADD 3.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a membrane-associated phospholipase that displays lysophospholipase and phospholipase A2 activities through removal of sn-1 and sn-2 fatty acids of glycerophospholipids. In addition, it displays lipase and retinyl ester hydrolase activities. The encoded protein is highly conserved and is composed of a large, glycosylated extracellular domain composed of four tandem homologous domains, followed by a hydrophobic segment that anchors the enzyme to the membrane and a short C-terminal cytoplasmic tail. This gene has been identified as a candidate rheumatoid arthritis risk gene. [provided by RefSeq, Jul 2016]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs56116810 c.-114+4401G>A	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.5	–	3.6 (Cactus)	– / –	chr2:28461598 AF: 0.0012%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000416713.5
cDNA (HGVS):	c.-114+4401G>A
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.50 · DP –
Phased GT:	0|1 (Maternal)

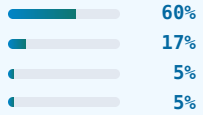
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review:	criteria provided
gnomAD v4 AF:	0.0012%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 3.6 (Raw: 0.009 · Top 43.90%)

Primary Driver: Cactus



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PKD1L3 PKD1L3 chr16:71,979,984 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.9916% · CADD 3.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polycystin protein family. The encoded protein contains 11 transmembrane domains, a latrophilin/CL-1-like GPCR proteolytic site (GPS) domain, and a polycystin-1, lipoxygenase, alpha-toxin (PLAT) domain. This protein may function as a component of cation channel pores.[provided by RefSeq, Apr 2009]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs72803747 c.1271+23G>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	3.4	0.84	10.4 (Splicing)	– / –	chr16:71979984 AF: 0.9916%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000620267.2
cDNA (HGVS):	c.1271+23G>T
Protein (HGVS):	–
VAF / Read Depth:	VAF 0.39 · DP –
Phased GT:	0|1 (Maternal)

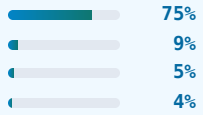
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.00 AL:0.00 DG:0.84 DL:0.02
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4415201 ↗
gnomAD v4 AF:	0.9916%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 10.4 (Raw: 0.168 · Top 9.03%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

PLA2G6 PLA2G6 chr22:38,115,172 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.5672% · CADD 1.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is an A2 phospholipase, a class of enzyme that catalyzes the release of fatty acids from phospholipids. The encoded protein may play a role in phospholipid remodelling, arachidonic acid release, leukotriene and prostaglandin synthesis, fas-mediated apoptosis, and transmembrane ion flux in glucose-stimulated B-cells. Several transcript variants encoding multiple isoforms have been described, but the full-length nature of only three of them have been determined to date. [provided by RefSeq, Dec 2010]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs147795054 c.2034+355G>A	INT	Heterozygous	Unphased	Not reviewed	Tier1	1.7	0.95	25.0 (Splicing)	– / –	chr22:38115172 AF: 0.5672%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

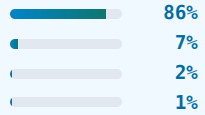
Transcript: ENST00000332509.8
cDNA (HGVS): c.2034+355G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.95 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.5672%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.0 (Raw: 1.119 · Top 0.32%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

EGFL7 EGFL7 chr9:136,670,809 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0147% · CADD 0.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a secreted endothelial cell protein that contains two epidermal growth factor-like domains. The encoded protein may play a role in regulating vasculogenesis. This protein may be involved in the growth and proliferation of tumor cells. Alternate splicing results in multiple transcript variants. [provided by RefSeq, Feb 2012]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs920684955 c.572-141C>T	INT	Heterozygous	Maternal	Not reviewed	Tier1	0.6	0.73	16.7 (Splicing)	– / –	chr9:136670809 AF: 0.0147%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

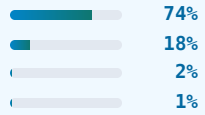
Transcript: ENST00000308874.12
cDNA (HGVS): c.572-141C>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0|1 (Maternal)

3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.73 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0147%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 16.7 (Raw: 0.391 · Top 2.16%)
Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

AZIN2 AZIN2 chr1:33,081,925 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0490% · CADD 0.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the antizyme inhibitor family, which plays a role in cell growth and proliferation by maintaining polyamine homeostasis within the cell. Antizyme inhibitors are homologs of ornithine decarboxylase (ODC, the key enzyme in polyamine biosynthesis) that have lost the ability to decarboxylase ornithine; however, retain the ability to bind to antizymes. Antizymes negatively regulate intracellular polyamine levels by binding to ODC and targeting it for degradation, as well as by inhibiting polyamine uptake. Antizyme inhibitors function as positive regulators of polyamine levels by sequestering antizymes and neutralizing their effect. This gene encodes antizyme inhibitor 2, the second member of this gene family. Like antizyme inhibitor 1, antizyme inhibitor 2 interacts with all 3 antizymes and stimulates ODC activity and polyamine uptake. However, unlike antizyme inhibitor 1, which is ubiquitously expressed and localized in the nucleus and cytoplasm, antizyme inhibitor 2 is predominantly expressed in the brain and testis and localized in the endoplasmic reticulum-golgi intermediate compartment. Recent studies indicate that antizyme inhibitor 2 is also expressed in specific cell types in ovaries, adrenal glands and pancreas, and in mast cells. The exact function of this gene is not known, however, available data suggest its role in cell growth, spermiogenesis, vesicular trafficking and secretion. Accumulation of antizyme inhibitor 2 has also been observed in brains of patients with Alzheimer's disease. There has been confusion in literature and databases over the nomenclature of this gene, stemming from an earlier report that a human cDNA clone (identical to ODCp/AZIN2) had arginine decarboxylase (ADC) activity (PMID:14738999). Subsequent studies in human and mouse showed that antizyme inhibitor 2 was devoid of arginine decarboxylase activity (PMID:19956990). Alternatively spliced transcript variants have been described for this gene. [provided by RefSeq, Sep 2014]

VARIANT / HGVS	CONSEQUENCE	ZYGOSITY / DP	PHASE	CLINVAR	TIER	CADD	SPLICEAI	ALPHAGENOME AVI	REVEL / AM	COORDINATE & AF
rs190309518 c.-73+113G>T	INT	Heterozygous	Unphased	Not reviewed	Tier1	0.4	0.67	13.2 (Splicing)	— / —	chr1:33081925 AF: 0.0490%

1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

2. TRANSCRIPT, PHASING & DEPTH

Transcript:	ENST00000294517.11
cDNA (HGVS):	c.-73+113G>T
Protein (HGVS):	—
VAF / Read Depth:	VAF 0.40 · DP —
Phased GT:	0/1 (Unphased)

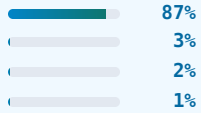
3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ:	AG:0.06 AL:0.00 DG:0.67 DL:0.00
ClinVar Review:	criteria provided
ClinVar VCV:	VCV#4417896 ↗
gnomAD v4 AF:	0.0490%

4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.2 (Raw: 0.256 · Top 4.79%)

Primary Driver: Splicing



🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗